

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\127-CA-CB.pvdb

Date of Analysis : Sep 9, 2026 9:26am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 127-CA-CB

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Vessel Design Summary:

Step: 32

9:26am

Sep 9, 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2110.000 mm ID
Vessel Design Length, Tangent to Tangent 22802.70 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 150 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 2210.000 KPa.
Channel Side Design Pressure 1000.000 KPa.
Shop Shell Side Test Pressure 4002.499 KPa.
Field Shell Side Test Pressure 2873.000 KPa.
Shop Channel Side Test Pressure 1556.888 KPa.
Field Channel Side Test Pressure 1300.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SB-338 1	...	...	R50250	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flg	1023.309	103.42	1132.000	3.0000	N/A	No
LHS Ch Flg	1020.878	103.42	1662.000	3.0000	N/A	No
LHS Ch Shell	1020.692	103.42	1413.000	1.0000	N/A	No
LSH TS Flang	1020.878	103.42	1662.000	3.0000	N/A	No
Main Shell	2230.712	0.00	2686.000	3.0000	N/A	No
RHS CH Flang 1	1020.878	103.42	1662.000	3.0000	N/A	No
Rhs Ch Shell	1020.692	103.42	1631.000	1.0000	N/A	No
RHS CH Flang 2	1020.878	103.42	1662.000	3.0000	N/A	No
Rhs Blind Flg	1023.309	103.42	1132.000	3.0000	N/A	No

Liquid Level: 2378.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



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Vessel Design Summary:

Step: 32 9:26am Sep 9,2026

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	122.2	119.0	127.0	119.0	112.2	63.6	1.00	1.00
Body Flg	385.2	263.0	211.0	203.0	196.5	196.5	1.00	1.00
Cylinder	3385.2	3000.0	16.0	14.0	10.2	8.9	0.85	0.85
Body Flg	3651.3	263.0	211.0	203.0	196.5	196.5	1.00	1.00
Cylinder	19031.3	15260.0	24.0	24.0	20.3	...	1.00	1.00
Body Flg	19417.5	263.0	211.0	203.0	196.5	196.5	1.00	1.00
Cylinder	22417.5	3000.0	16.0	16.0	10.2	8.9	0.85	0.85
Body Flg	22683.7	263.0	211.0	203.0	196.5	196.5	1.00	1.00
Body Flg	22802.7	119.0	127.0	119.0	112.2	63.6	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-Tube Inlet	WRC-297	4.372	Shell	Failed
N1-Shell Inlet	WRC-297	4.372	Shell	Failed
N2-Shell Outle	WRC-297	4.372	Shell	Failed
N4-Tube Outlet	WRC-297	3.309	Shell	Failed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	425.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	1760.000	mm
Wear Pad Width	680.000	mm
Wear Pad Thickness	16.000	mm
Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	3000.000	mm

Baseplate Length	2070.100	mm
Baseplate Thickness	50.000	mm
Baseplate Width	450.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	14.000	mm
Web Thickness	14.000	mm
Height of Center Web	695.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



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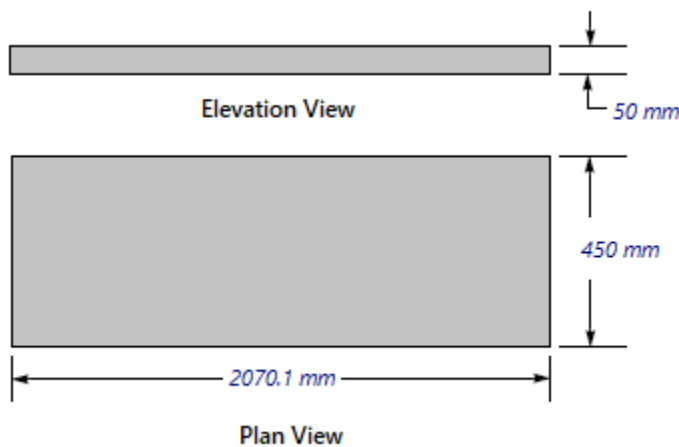
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Vessel Design Summary:

Step: 32

9:26am

Sep 9, 2026

**Summary of Maximum Saddle Loads, Operating Case, Un-Factored:**

Maximum Vertical Saddle Load	105516.90	Kgf
Maximum Transverse Saddle Shear Load	6227.01	Kgf
Maximum Longitudinal Saddle Shear Load	10229.61	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	88580.69	Kgf
Maximum Transverse Saddle Shear Load	697.87	Kgf
Maximum Longitudinal Saddle Shear Load	226.30	Kgf

Weights:

Fabricated - Bare w/o removable internals	101036.5	kg.
Shop Test - Fabricated + water (full)	172220.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	101036.5	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	101036.5	kg.
Empty - Fab. + intls. + details + wghts.	104851.3	kg.
Operating - Empty + operating liquid (no ca)	176154.6	kg.
Field Test - Empty weight + test medium (full)	176035.5	kg.

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Nozzle Summary:

Step: 30 9:26am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-Tube Inlet	1000	OK	...	OK	8.69	OK	Passed
N1-Shell Inlet	2210	...	...	OK	11.33	OK	Passed
N2-Shell Outlet	2210	...	...	OK	11.33	OK	Passed
N4-Tube Outlet	1000	OK	...	OK	8.85	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2210 Nozzle : N1-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 1000 Nozzle : N3-Tube Inlet [Tubeside]

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3-Tube Inlet	1885.175	0.000	2500.000
60	N1-Shell Inlet	5771.350	0.000	1162.198
60	N2-Shell Outlet	17041.350	180.000	1162.198
80	N4-Tube Outlet	20917.520	180.000	2500.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 29 9:26am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Shell Inlet	24.016 in	Actual	WNF	24.016	17.450	...	...	419.05	300
N2-Shell Outlet	24.016 in	Actual	WNF	24.016	17.450	...	...	419.05	300
N3-Tube Inlet	62.992 in	Actual	WNF	62.992	30.000	2300.00	16.00	981.23	150
N4-Tube Outlet	62.992 in	Actual	WNF	62.992	30.000	2300.00	16.00	983.23	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Shell In	SA-266 2	22.000	10.000	...	...	...
N2-Shell Ou	SA-266 2	22.000	10.000	...	...	...
N3-Tube Inl	SA-516 70	14.000	10.000	12.000	16.000	...
N4-Tube Out	SA-516 70	16.000	10.000	12.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Shell Inlet	5771.350	0.0	350.00	0.00	Main Shell
N2-Shell Outlet	17041.350	180.0	350.00	0.00	Main Shell
N3-Tube Inlet	1885.175	0.0	600.00	0.00	LHS Ch Shell
N4-Tube Outlet	20917.520	180.0	600.00	0.00	Rhs Ch Shell

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Bill of Material:

Step: 28 9:26am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 127.0mm THK X 2385.0mm OD	SA-266 2
4	WELD NECK FLANGE: 211.0mm THK X 2385.0mm OD	SA-266 2
2	CYLINDER: 16.0mm THK X 2110.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 24.0mm THK X 2110.0mm ID X 15260.0mm	SA-516 70
2	CLADDING: 3000.000mm X 3.000mm THK	...
29	BAFFLES: 2120.0mm DIA X 10.0mm THK	SA-516 70
12	TIE RODS: 16mm DIA X 14459mm	SA-193 B7M
2	TUBESHEET: 2385.0mm X 120.0mm THK	SA-266 2
3704	TUBES: 15500.0mm X 25.4mm DIA X 2.1mm THK	SB-338 1
2	SADDLE: 425.0mm X 120 DEG	SA-516 70
1	INSULATION: 15260mm X 100mm THK	Hot Insulation
4	GASKET: 2186.0mm OD X 2142.0mm ID	...
224	BODY FLANGE BOLTS: 41.3mm DIA	SA-193 B7
336	NUTS FOR BODY FLANGE BOLTS: 41.3mm DIA	...
1	NAMEPLATE	...

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MDMT Summary:

Step: 31 9:26am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell [8]	B	-2	-12	-29	0.815	24.000	1.00	Yes
Left Saddle f[24]	B	-14	-24	-29	0.815	16.000	1.00	Yes
RH fixed sadd[24]	B	-14	-24	-29	0.815	16.000	1.00	Yes
N1-Shell Inlet[1]	B	-2	-29	-29	0.815	24.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
N2-Shell Outle[1]	B	-2	-29	-29	0.822	24.000	1.00	Yes
Nozzle Flg [5]	A	-18	-48					
Tubesheet: SS[13]	B	4	-25		0.529	30.000	1.00	Yes
Warmest MDMT:		4	-12					
LHS Blind Flg[11]	B	4	-6		0.823	29.750	1.00	No
LHS Ch Flg [11]	B	-31	-48		0.482	16.000	1.00	No
LSH TS Flang [11]	B	-31	-48		0.482	16.000	1.00	No
RHS CH Flang [11]	B	-31	-48		0.482	16.000	1.00	No
RHS CH Flang [11]	B	-31	-48		0.482	16.000	1.00	No
Rhs Blind Flg[11]	B	4	-6		0.823	29.750	1.00	No
LHS Ch Shell [8]	B	-35	-48		0.592	14.000	0.85	No
Rhs Ch Shell [8]	B	-31	-48		0.513	16.000	0.85	No
N3-Tube Inlet [1]	B	-31	-48		0.591	16.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N4-Tube Outlet[1]	B	-31	-48		0.523	16.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
Tubesheet: CS[14]	B	-13	-42		0.529	30.000	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		4	-6					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-25.0	-12.0	Pass
Channel/Tube	-6.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material



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MDMT Summary:

Step: 31 9:26am Sep 9, 2026

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:26am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2210	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	150.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-3	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	2210	KPa.
User defined MAWP, Channel Side	999.99	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 9:26am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed [V]	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	17500	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration [Ss]	0.147	
Max. Eff. Ground Acceleration [S]	0.065	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio z/h	0.000	
Amplification Factor Ap	1.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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FileName : 127-CA-CB

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Input Echo:

Step: 1 9:26am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9	
Element From Node	10	
Element To Node	20	
Element Type	Flange	
Description	LHS Blind Flg	
Distance "FROM" to "TO"	119	mm
Flange Outside Diameter	2385	mm
Element Thickness	119	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	127	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2378	mm
Liquid Density	0.0009984	kg./cm ³

Element#	2/9
Element From Node	20



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Input Echo:

Step: 1 9:26am Sep 9,2026

Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	263 mm
Flange Inside Diameter	2110 mm
Element Thickness	203 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	211 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	Yes
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2130 mm
Liquid Density	0.0009984 kg./cm ³

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	3000 mm
Inside Diameter	2110 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	Yes
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2110 mm
Liquid Density	0.0009984 kg./cm ³



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Input Echo:

Step: 1 9:26am Sep 9,2026

Element From Node	30
Detail Type	Lining
Detail ID	Titanium clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	N3-Tube Inlet
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	62.99 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	644.79 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element#	4/9
Element From Node	40
Element To Node	60
Element Type	Flange
Description	LSH TS Flang
Distance "FROM" to "TO"	263 mm
Flange Inside Diameter	2110 mm
Element Thickness	203 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	211 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	Yes

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2130 mm
Liquid Density	0.0009984 kg./cm ³

Element#	5/9
Element From Node	60
Element To Node	70



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Input Echo:

Step: 1 9:26am Sep 9,2026

Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	15260 mm
Inside Diameter	2110 mm
Element Thickness	24 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2210 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	232 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Left Saddle flo
Dist. from "FROM" Node / Offset dist	3000 mm
Width of Saddle	425 mm
Height of Saddle at Bottom	1760 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	680 mm
Thickness of Wear Plate	16 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Saddle
Detail ID	RH fixed saddle
Dist. from "FROM" Node / Offset dist	12240 mm
Width of Saddle	425 mm
Height of Saddle at Bottom	1760 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	680 mm
Thickness of Wear Plate	16 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
-------------------	----



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Input Echo:

Step: 1 9:26am Sep 9,2026

Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2110 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	15260 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	N1-Shell Inlet
Dist. from "FROM" Node / Offset dist	2000 mm
Nozzle Diameter	24.02 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	282.97 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	60
Detail Type	Nozzle
Detail ID	N2-Shell Outlet
Dist. from "FROM" Node / Offset dist	13270 mm
Nozzle Diameter	24.02 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	282.97 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	3000 mm
Miscellaneous Weight	800 Kg
Offset from Element Centerline	1752 mm

Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	12216 mm
Miscellaneous Weight	800 Kg
Offset from Element Centerline	1752 mm

Element#	6/9
Element From Node	70
Element To Node	80
Element Type	Flange
Description	RHS CH Flang 1
Distance "FROM" to "TO"	263 mm
Flange Inside Diameter	2110 mm
Element Thickness	203 mm
Internal Corrosion Allowance	3 mm



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Input Echo:

Step: 1 9:26am Sep 9, 2026

Nominal Thickness	211	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	Yes	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2130	mm
Liquid Density	0.0009984	kg./cm ³

Element#	7/9	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Rhs Ch Shell	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	2110	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	242.6	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	Yes	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm



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Input Echo:

Step: 1 9:26am Sep 9, 2026

Height/Length of Liquid	2110	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	80	
Detail Type	Lining	
Detail ID	Titanium Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N4-Tube Outlet	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	62.99	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	846.7	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element#	8/9	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	RHS CH Flang 2	
Distance "FROM" to "TO"	263	mm
Flange Inside Diameter	2110	mm
Element Thickness	203	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	211	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	Yes	

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 90	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2130	mm
Liquid Density	0.0009984	kg./cm ³



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Input Echo:

Step: 1 9:26am Sep 9,2026

Element#	9/9
Element From Node	100
Element To Node	110
Element Type	Flange
Description	Rhs Blind Flg
Distance "FROM" to "TO"	119 mm
Flange Outside Diameter	2385 mm
Element Thickness	119 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	127 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	100
Detail Type	Liquid
Detail ID	Liquid: 100
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2378 mm
Liquid Density	0.0009984 kg./cm ³

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XY Coordinate Calculations:

Step: 2 9:26am

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flg		122.175	...	119	...
LHS Ch Flg		385.175	...	263	...
LHS Ch Shell		3385.17	...	3000	...
LSH TS Flang		3651.35	...	263	...
Main Shell		19031.3	...	15260	...
RHS CH Flang 1		19417.5	...	263	...
Rhs Ch Shell		22417.5	...	3000	...
RHS CH Flang 2		22683.7	...	263	...
Rhs Blind Flg		22802.7	...	119	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: New Flange

Flng: 13

9:26am

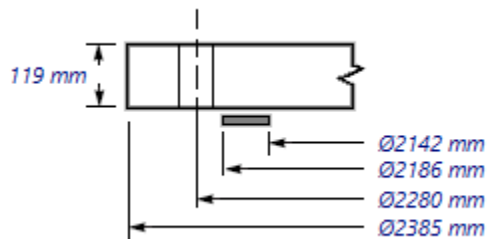
Sep 9, 2026

Flange Input Data Values

Description: New Flange :

LHS Blind Flg

Flange Type		Blind	
Design Pressure	P	1023.31	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2385.000	mm
Flange Thickness	t	119.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	7162.831	mm
Diameter of Bolt Circle	C	2280.000	mm
Nominal Bolt Diameter	a	41.2750	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2197.000	mm
Flange Face Inside Diameter	Fid	2110.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2186.000	mm
Gasket Inside Diameter	Gi	2142.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	2169.285 mm

Basic Flange and Bolt Loads:



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Flg Calc [Int P]: New Flange

Flng: 13

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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2169^2 \cdot 1023$$

$$= 385679.438 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2169 \cdot 3 \cdot 1023$$

$$= 35661.898 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(385679 + 35662 + 0, 0)$$

$$= 421341.312 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69 \cdot 8.36 \cdot \pi \cdot 2169 + 0 \cdot 0 \cdot 0$$

$$= 400456.562 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((421341 + 0 + 4 \cdot 0 / 2169) / 172, 400457 / 172)$$

$$= 240.238 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	240.238	606.966	
Radial Distance between Bolts and Edge:	41.275	52.500	
Circ. Spacing between the Bolts:	88.900	127.841	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 607 \cdot 172 / (69 \cdot \pi (2186 + 2142))$$

$$= 11.136 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 607$$

$$= 1064527.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 421341.31 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2280 - 2169) / 2$$

$$= 55.3576 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 421341 \cdot 55.4 \cdot 1 / (116^2 \cdot 2169) + 0.3 \cdot 1 \cdot 1023 \cdot 2169^2 / 116^2$$

$$= 122.26 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1064527 \cdot 55.4 \cdot 1 / (116^2 \cdot 2169)$$



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Flg Calc [Int P]: New Flange

Flng: 13

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= 37.62 N./mm²

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b

= 421341/607

= 68.08 N./mm²

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)

= (400457/607)

= 64.70 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	122.26	138.00	37.62	138.00
Bolting	68.08	172.00	64.70	172.00

Reqd. Blind Flange Thickness at Center 112.184 mm

Reqd. Blind Flange Thickness at Gasket 63.564 mm

Estimated M.A.W.P. (Operating) 1155 KPa.

Estimated Finished Weight of Flange at given Thk. 4120.4 kg.

Estimated Unfinished Weight of Forging at given Thk 4120.4 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.82, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	116.17	138.00	35.74	138.00
Bolting	68.08	172.00	64.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 109.195 mm

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Flg Calc [Int P]: New Flange

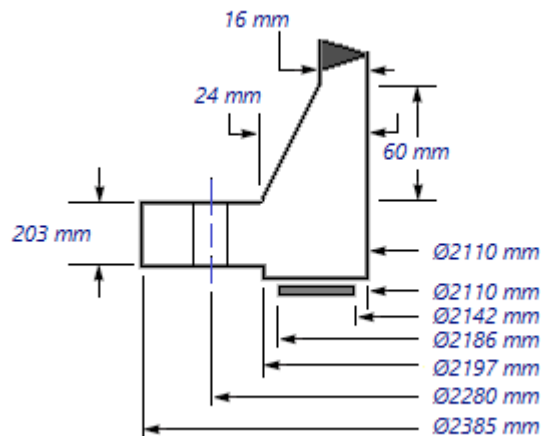
Flng: 14 9:26am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	1020.88	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2110.000	mm
Flange Outside Diameter	A	2385.000	mm
Flange Thickness	t	203.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	24.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2280.000	mm
Nominal Bolt Diameter	a	41.2750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2197.000	mm
Flange Face Inside Diameter	Fid	2110.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2186.000	mm
Gasket Inside Diameter	Gi	2142.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1021 \cdot (2110/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1021) + Ca$$

$$= 10.8622 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1058 + 0.6 \cdot 13)$$

$$= 1683.114 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1055 + 0.6 \cdot 16)$$

$$= 2073.860 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	200.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2116.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	21.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	61.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2169.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2169^2 \cdot 1021$$

$$= 384763.219 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2169 \cdot 3 \cdot 1021$$

$$= 35577.184 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2116^2 \cdot 1021$$

$$= 366093.250 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 384763 - 366093$$

$$= 18669.949 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(384763 + 35577 + 0, 0)$$

$$= 420340.406 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2169 + 0 \cdot 0$$

$$= 400456.531 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((420340 + 0 + 4 \cdot 0 / 2169) / 172, 400457 / 172)$$

$$= 239.668 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 41.3 + 6 \cdot 200/(3 + 0.5)$$

$$= 425.407 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2280 \cdot \sin(\pi / 56)$$

$$= 127.841 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{128/(2 \cdot 41.3 + 200)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	239.668	606.966	
Radial Distance between Hub and Bolts:	53.975	61.000	
Radial Distance between Bolts and Edge:	41.275	52.500	
Circ. Spacing between the Bolts:	88.900	127.841	425.407

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 607 \cdot 172 / (68.9 \cdot \pi (2186 + 2142))$$

$$= 11.136 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 607$$

$$= 1064527.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 420340 - 384763$$

$$= 35577.18 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2280 - 2169) / 2$$

$$= 55.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 (C - B) / 2 + hg$$

$$= 0.5 (2280 - 2116) / 2 + 55.4$$

$$= 68.6787 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2280 - 2116 - 21) / 2$$

$$= 71.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	366093.	71.4999	1.0000	26176.
Face Pressure, Mt	18670.	68.6787	1.0000	1282.
Gasket Load, Mg	35577.	55.3576	1.0000	1970.
Gasket Seating, Matm	1064527.	55.3576	1.0000	58931.

Total Moment for Operation, Mo 29428. Kg-m.
Total Moment for Gasket seating, Mg 58931. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 165.855 mm
Hub Ratio, h/h0 = HL / H0 0.362
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.615

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.865
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.152

Stress factor from Table 4.16.4 [K]:

= A / B
= 2385/2116
= 1.127

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 (K² • log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 (1.13² • log10(1.13))) / (1.13² - 1)) / (1.13 - 1)
= 16.238

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / ((1.0472 + 1.9448 • 1.13²) • (1.13 - 1))
= 1.867

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / (1.36136(1.13² - 1) • (1.13 - 1))
= 17.844

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.13² + 1) / (1.13² - 1)
= 8.396

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 17.8 • 13² • 166 / 0.33
= 1527723.125 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.87 / 166
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³ / d
= (200 • 0.0052 + 1) / 1.87 + 200³ / 1527723
= 6.331

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.15 • 29428 / 2116) / (6.33 • 21²)
= 56.27 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.15 \cdot 58931 / 2116) / (6.33 \cdot 21^2)$$

$$= 112.68 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 29428 / 2116) / (6.33 \cdot 200^2)$$

$$= 1.29 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 58931 / 2116) / (6.33 \cdot 200^2)$$

$$= 2.58 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (16.2 \cdot 29428 / (200^2 \cdot 2116)) - 8.4 \cdot 1.29$$

$$= 44.55 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (16.2 \cdot 58931 / (200^2 \cdot 2116)) - 8.4 \cdot 2.58$$

$$= 89.22 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (56.3 + \max(1.29, 44.6)) / 2$$

$$= 50.41 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (113 + \max(2.58, 89.2)) / 2$$

$$= 100.95 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 420340 / 607$$

$$= 67.91 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (400457 / 607)$$

$$= 64.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	56.27	207.00	112.68	207.00
Radial Flange	1.29	138.00	2.58	138.00
Tangential Flange	44.55	138.00	89.22	138.00
Maximum Average	50.41	138.00	100.95	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Required Flange Thickness [Rigidity] 196.545 mm

Estimated M.A.W.P. (Operating) 1683 KPa.

Estimated Finished Weight of Flange at given Thk. 1589.7 kg.

Estimated Unfinished Weight of Forging at given Thk 1979.0 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	452.91	499.99	0.906	452.91	499.99	0.906
Tangential Flange	St	357.75	374.99	0.954	357.75	374.99	0.954
Tangential Flange	Sto	152.91	250.00	0.612	152.91	250.00	0.612
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	368.09	749.99	0.491	368.09	749.99	0.491
Combined	Sh + Sto	605.82	749.99	0.808	605.82	749.99	0.808
Radial Flange	Sr	10.34	499.99	0.021	10.34	499.99	0.021
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.56/1.35)

= 1.1547

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)

= max(0.25; 0.51/1.32) if h/h0 < 0.35, else max (0.25; 0.3/0.76)

= 0.3898

In this case, the hub ratio h/h0, 0.3618 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	689.6 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	10.8387 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.660 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2186^2 - 2142^2)$$

$$= 1495.7 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 1496 / (56 \cdot 10.8)$$

$$= 512.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(513; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 690)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 508 \geq 140 (1496 / (56 \cdot 10.8))$$

$$\therefore 507.5 \geq 345.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 508 \geq (97 \cdot 1496 + \pi/4 \cdot 1021 \cdot 2142^2) / (0.7 \cdot 10.8 \cdot 56)$$

$$\therefore 507.5 \geq 428.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 508 \leq 275 (1496 / (56 \cdot 10.8))$$

$$\therefore 507.5 \leq 677.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 508 \leq 690 (1/0.66)$$



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:: 507.5 ≤ 1045.3 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • φb

= 508 • 0.26 • 10.8 • 41.3

= 601.9 Kg-m.

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 58931 / 1 • 9807 • 0.33 / (5.82 • 201334 • 13² • 166 • 0.3)

= 1.000 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 29428 / 1 • 9807 • 0.33 / (5.82 • 197847 • 13² • 166 • 0.3)

= 0.508 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 58931 / 1 • 9807 • 0.33 / (6.33 • 201334 • 13² • 166 • 0.3)

= 0.920 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 29428 / 1 • 9807 • 0.33 / (6.33 • 197847 • 13² • 166 • 0.3)

= 0.467 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.48, Temperature Reduction per Fig. UCS 66.1 = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	55.74	207.00	109.49	207.00
Radial Flange	1.64	138.00	3.21	138.00
Tangential Flange	40.27	138.00	79.11	138.00
Maximum Average	48.00	138.00	94.30	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 185.776 mm

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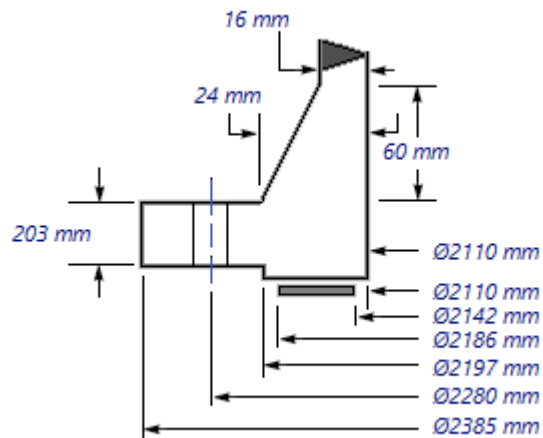
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Flange Input Data Values

Description: New Flange :

LSH TS Flang

Flange Type	Integral Weld Neck		
Design Pressure	P	1020.88	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2110.000	mm
Flange Outside Diameter	A	2385.000	mm
Flange Thickness	t	203.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	24.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2280.000	mm
Nominal Bolt Diameter	a	41.2750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2197.000	mm
Flange Face Inside Diameter	Fid	2110.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2186.000	mm
Gasket Inside Diameter	Gi	2142.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1021 \cdot (2110/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1021) + Ca$$

$$= 10.8622 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1058 + 0.6 \cdot 13)$$

$$= 1683.114 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1055 + 0.6 \cdot 16)$$

$$= 2073.860 \text{ KPa.}$$

Corroded Flange Thickness, tc = T-ci	200.000 mm
Corroded Flange ID, Bcor = B+2*ci	2116.000 mm
Corroded Large Hub, g1Cor = g1-ci	21.000 mm
Corroded Small Hub, g0Cor = go-ci	13.000 mm
Code R Dimension, R = ((C-Bcor)/2) - g1cor	61.000 mm

Gasket Contact Width, N = (Go - Gi)/2	22.000 mm
Basic Gasket Width, bo = N/2	11.000 mm
Effective Gasket Width, b = Cb*sqrt(bo)	8.358 mm
Gasket Reaction Diameter, G = Go - 2 * b	2169.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2169^2 \cdot 1021$$

$$= 384763.219 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2169 \cdot 3 \cdot 1021$$

$$= 35577.184 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 2116^2 \cdot 1021$$

$$= 366093.250 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 384763 - 366093$$

$$= 18669.949 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(384763 + 35577 + 0, 0)$$

$$= 420340.406 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2169 + 0 \cdot 0$$

$$= 400456.531 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((420340 + 0 + 4 \cdot 0 / 2169) / 172, 400457 / 172)$$

$$= 239.668 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 41.3 + 6 \cdot 200/(3 + 0.5)$$

$$= 425.407 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2280 \cdot \sin(\pi / 56)$$

$$= 127.841 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{128/(2 \cdot 41.3 + 200)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	239.668	606.966	
Radial Distance between Hub and Bolts:	53.975	61.000	
Radial Distance between Bolts and Edge:	41.275	52.500	
Circ. Spacing between the Bolts:	88.900	127.841	425.407

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 607 \cdot 172 / (68.9 \cdot \pi (2186 + 2142))$$

$$= 11.136 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 607$$

$$= 1064527.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 420340 - 384763$$

$$= 35577.18 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2280 - 2169) / 2$$

$$= 55.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2280 - 2116) / 2 + 55.4)$$

$$= 68.6787 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2280 - 2116 - 21) / 2$$

$$= 71.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	366093.	71.4999	1.0000	26176.
Face Pressure, Mt	18670.	68.6787	1.0000	1282.
Gasket Load, Mg	35577.	55.3576	1.0000	1970.
Gasket Seating, Matm	1064527.	55.3576	1.0000	58931.

Total Moment for Operation, Mo 29428. Kg-m.
Total Moment for Gasket seating, Mg 58931. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 165.855 mm
Hub Ratio, h/h0 = HL / H0 0.362
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.615

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.865
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.152

Stress factor from Table 4.16.4 [K]:

= A / B
= 2385/2116
= 1.127

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.13² • log10(1.13))) / (1.13² - 1))/(1.13 - 1)
= 16.238

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / ((1.0472 + 1.9448 • 1.13²) • (1.13 - 1))
= 1.867

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / (1.36136(1.13² - 1) • (1.13 - 1))
= 17.844

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.13² + 1) / (1.13² - 1)
= 8.396

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 17.8 • 13² • 166 / 0.33
= 1527723.125 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.87/166
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (200 • 0.0052 + 1) / 1.87 + 200³/1527723
= 6.331

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.15•29428/2116)/(6.33•21²)
= 56.27 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.15 \cdot 58931 / 2116) / (6.33 \cdot 21^2)$$

$$= 112.68 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 29428 / 2116) / (6.33 \cdot 200^2)$$

$$= 1.29 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 58931 / 2116) / (6.33 \cdot 200^2)$$

$$= 2.58 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (16.2 \cdot 29428 / (200^2 \cdot 2116)) - 8.4 \cdot 1.29$$

$$= 44.55 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (16.2 \cdot 58931 / (200^2 \cdot 2116)) - 8.4 \cdot 2.58$$

$$= 89.22 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (56.3 + \max(1.29, 44.6)) / 2$$

$$= 50.41 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (113 + \max(2.58, 89.2)) / 2$$

$$= 100.95 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 420340 / 607$$

$$= 67.91 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (400457 / 607)$$

$$= 64.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	56.27	207.00	112.68	207.00
Radial Flange	1.29	138.00	2.58	138.00
Tangential Flange	44.55	138.00	89.22	138.00
Maximum Average	50.41	138.00	100.95	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Required Flange Thickness [Rigidity] 196.545 mm

Estimated M.A.W.P. (Operating) 1683 KPa.

Estimated Finished Weight of Flange at given Thk. 1589.7 kg.

Estimated Unfinished Weight of Forging at given Thk 1979.0 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	452.91	499.99	0.906	452.91	499.99	0.906
Tangential Flange	St	357.75	374.99	0.954	357.75	374.99	0.954
Tangential Flange	Sto	152.91	250.00	0.612	152.91	250.00	0.612
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	368.09	749.99	0.491	368.09	749.99	0.491
Combined	Sh + Sto	605.82	749.99	0.808	605.82	749.99	0.808
Radial Flange	Sr	10.34	499.99	0.021	10.34	499.99	0.021
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.56/1.35)

= 1.1547

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.51/1.32) if $h/h_0 < 0.35$, else max (0.25; 0.3/0.76)

= 0.3898

In this case, the hub ratio h/h_0 , 0.3618 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	689.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	10.8387 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.660 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2186^2 - 2142^2)$$

$$= 1495.7 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207 \cdot 1496 / (56 \cdot 10.8)$$

$$= 510.1 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(510; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 690)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 508 \geq 140 (1496 / (56 \cdot 10.8))$$

$$\therefore 507.5 \geq 345.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 508 \geq (97 \cdot 1496 + \pi/4 \cdot 1021 \cdot 2142^2) / (0.7 \cdot 10.8 \cdot 56)$$

$$\therefore 507.5 \geq 428.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 508 \leq 276 (1496 / (56 \cdot 10.8))$$

$$\therefore 507.5 \leq 680.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 508 \leq 690 (1/0.66)$$



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:: 507.5 ≤ 1045.3 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • φb

= 508 • 0.2 • 10.8 • 41.3

= 463.0 Kg-m.

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 58931 / 1 • 9807 • 0.33 / (5.82 • 201334 • 13² • 166 • 0.3)

= 1.000 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 29428 / 1 • 9807 • 0.33 / (5.82 • 197847 • 13² • 166 • 0.3)

= 0.508 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 58931 / 1 • 9807 • 0.33 / (6.33 • 201334 • 13² • 166 • 0.3)

= 0.920 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 29428 / 1 • 9807 • 0.33 / (6.33 • 197847 • 13² • 166 • 0.3)

= 0.467 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.48, Temperature Reduction per Fig. UCS 66.1 = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	55.74	207.00	109.49	207.00
Radial Flange	1.64	138.00	3.21	138.00
Tangential Flange	40.27	138.00	79.11	138.00
Maximum Average	48.00	138.00	94.30	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 185.776 mm

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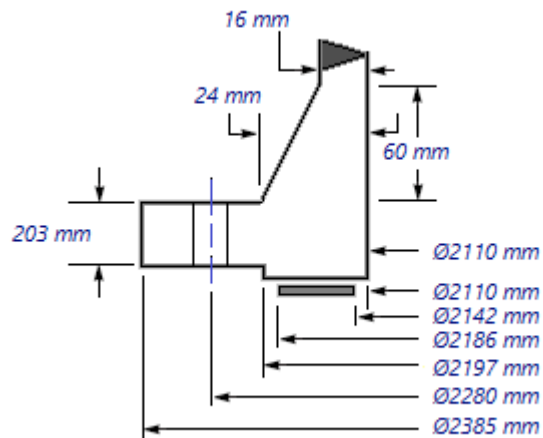
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Flange Input Data Values

Description: RHS TS Flang :

RHS CH Flang 1

Flange Type	Integral Weld Neck		
Design Pressure	P	1020.88	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2110.000	mm
Flange Outside Diameter	A	2385.000	mm
Flange Thickness	t	203.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	24.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2280.000	mm
Nominal Bolt Diameter	a	41.2750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2197.000	mm
Flange Face Inside Diameter	Fid	2110.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2186.000	mm
Gasket Inside Diameter	Gi	2142.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1021 \cdot (2110/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1021) + Ca$$

$$= 10.8622 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1058 + 0.6 \cdot 13)$$

$$= 1683.114 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1055 + 0.6 \cdot 16)$$

$$= 2073.860 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	200.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2116.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	21.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	61.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $bo = N/2$	11.000 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2169.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2169^2 \cdot 1021$$

$$= 384763.219 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2169 \cdot 3 \cdot 1021$$

$$= 35577.184 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2116^2 \cdot 1021$$

$$= 366093.250 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 384763 - 366093$$

$$= 18669.949 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(384763 + 35577 + 0, 0)$$

$$= 420340.406 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69 \cdot 8.36 \cdot \pi \cdot 2169 + 0 \cdot 0 \cdot 0$$

$$= 400456.562 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((420340 + 0 + 4 \cdot 0/2169)/172, 400457/172)$$

$$= 239.668 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 41.3 + 6 \cdot 200/(3 + 0.5)$$

$$= 425.407 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2280 \cdot \sin(\pi / 56)$$

$$= 127.841 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{128/(2 \cdot 41.3 + 200)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	239.668	606.966	
Radial Distance between Hub and Bolts:	53.975	61.000	
Radial Distance between Bolts and Edge:	41.275	52.500	
Circ. Spacing between the Bolts:	88.900	127.841	425.407

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 607 \cdot 172 / (69 \cdot \pi (2186 + 2142))$$

$$= 11.136 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 607$$

$$= 1064527.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 420340 - 384763$$

$$= 35577.18 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2280 - 2169) / 2$$

$$= 55.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 (C - B) / 2 + hg$$

$$= 0.5 (2280 - 2116) / 2 + 55.4$$

$$= 68.6787 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2280 - 2116 - 21) / 2$$

$$= 71.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	366093.	71.4999	1.0000	26176.
Face Pressure, Mt	18670.	68.6787	1.0000	1282.
Gasket Load, Mg	35577.	55.3576	1.0000	1970.
Gasket Seating, Matm	1064527.	55.3576	1.0000	58931.

Total Moment for Operation, Mo 29428. Kg-m.
Total Moment for Gasket seating, Mg 58931. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 165.855 mm
Hub Ratio, h/h0 = HL / H0 0.362
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.615

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.865
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.152

Stress factor from Table 4.16.4 [K]:

= A / B
= 2385/2116
= 1.127

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.13² • log10(1.13))) / (1.13² - 1)) / (1.13 - 1)
= 16.238

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / ((1.0472 + 1.9448 • 1.13²) • (1.13 - 1))
= 1.867

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / (1.36136(1.13² - 1) • (1.13 - 1))
= 17.844

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.13² + 1) / (1.13² - 1)
= 8.396

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 17.8 • 13² • 166 / 0.33
= 1527723.125 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.87/166
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (200 • 0.0052 + 1) / 1.87 + 200³/1527723
= 6.331

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.15•29428/2116)/(6.33•21²)
= 56.27 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.15 \cdot 58931 / 2116) / (6.33 \cdot 21^2)$$

$$= 112.68 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 29428 / 2116) / (6.33 \cdot 200^2)$$

$$= 1.29 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 58931 / 2116) / (6.33 \cdot 200^2)$$

$$= 2.58 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (16.2 \cdot 29428 / (200^2 \cdot 2116)) - 8.4 \cdot 1.29$$

$$= 44.55 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (16.2 \cdot 58931 / (200^2 \cdot 2116)) - 8.4 \cdot 2.58$$

$$= 89.22 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (56.3 + \max(1.29, 44.6)) / 2$$

$$= 50.41 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (113 + \max(2.58, 89.2)) / 2$$

$$= 100.95 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 420340 / 607$$

$$= 67.91 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (400457 / 607)$$

$$= 64.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	56.27	207.00	112.68	207.00
Radial Flange	1.29	138.00	2.58	138.00
Tangential Flange	44.55	138.00	89.22	138.00
Maximum Average	50.41	138.00	100.95	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Required Flange Thickness [Rigidity] 196.545 mm

Estimated M.A.W.P. (Operating) 1683 KPa.

Estimated Finished Weight of Flange at given Thk. 1589.7 kg.

Estimated Unfinished Weight of Forging at given Thk 1979.0 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	452.91	499.99	0.906	452.91	499.99	0.906
Tangential Flange	St	357.75	374.99	0.954	357.75	374.99	0.954
Tangential Flange	Sto	152.91	250.00	0.612	152.91	250.00	0.612
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	368.09	749.99	0.491	368.09	749.99	0.491
Combined	Sh + Sto	605.82	749.99	0.808	605.82	749.99	0.808
Radial Flange	Sr	10.34	499.99	0.021	10.34	499.99	0.021
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.56/1.35)

= 1.1547

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.51/1.32) if $h/h_0 < 0.35$, else max (0.25; 0.3/0.76)

= 0.3898

In this case, the hub ratio h/h_0 , 0.3618 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	689.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	10.8387 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.660 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2186^2 - 2142^2)$$

$$= 1495.7 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1496 / (56 \cdot 10.8)$$

$$= 510.1 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(510; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 690)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 508 \geq 140 (1496 / (56 \cdot 10.8))$$

$$\therefore 507.5 \geq 345.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 508 \geq (97 \cdot 1496 + \pi/4 \cdot 1021 \cdot 2142^2) / (0.7 \cdot 10.8 \cdot 56)$$

$$\therefore 507.5 \geq 428.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 508 \leq 276 (1496 / (56 \cdot 10.8))$$

$$\therefore 507.5 \leq 680.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 508 \leq 690 (1/0.66)$$



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:: 507.5 ≤ 1045.3 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi_b$$

$$= 508 \cdot 0.2 \cdot 10.8 \cdot 41.3$$

$$= 463.0 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58931 / 1 \cdot 9807 \cdot 0.33 / (5.82 \cdot 201334 \cdot 13^2 \cdot 166 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 29428 / 1 \cdot 9807 \cdot 0.33 / (5.82 \cdot 197847 \cdot 13^2 \cdot 166 \cdot 0.3)$$

$$= 0.508 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58931 / 1 \cdot 9807 \cdot 0.33 / (6.33 \cdot 201334 \cdot 13^2 \cdot 166 \cdot 0.3)$$

$$= 0.920 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 29428 / 1 \cdot 9807 \cdot 0.33 / (6.33 \cdot 197847 \cdot 13^2 \cdot 166 \cdot 0.3)$$

$$= 0.467 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.48, Temperature Reduction per Fig. UCS 66.1 = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	55.74	207.00	109.49	207.00
Radial Flange	1.64	138.00	3.21	138.00
Tangential Flange	40.27	138.00	79.11	138.00
Maximum Average	48.00	138.00	94.30	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 185.776 mm

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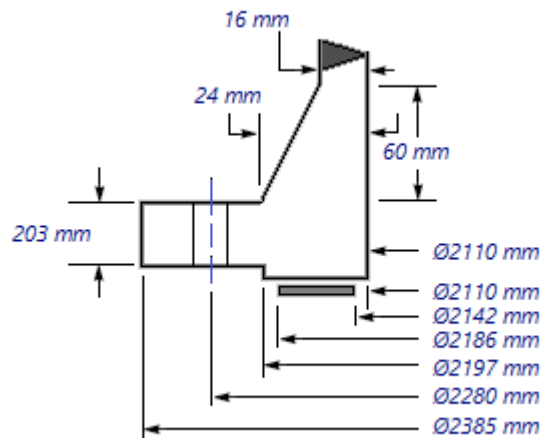
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Flange Input Data Values

Description: New Flange :

RHS CH Flang 2

Flange Type	Integral Weld Neck		
Design Pressure	P	1020.88	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2110.000	mm
Flange Outside Diameter	A	2385.000	mm
Flange Thickness	t	203.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	24.0000	mm
Length of Hub	h	60.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2280.000	mm
Nominal Bolt Diameter	a	41.2750	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2197.000	mm
Flange Face Inside Diameter	Fid	2110.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2186.000	mm
Gasket Inside Diameter	Gi	2142.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1021 \cdot (2110/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1021) + Ca$$

$$= 10.8622 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1058 + 0.6 \cdot 13)$$

$$= 1683.114 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1055 + 0.6 \cdot 16)$$

$$= 2073.860 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	200.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2116.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	21.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	61.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $bo = N/2$	11.000 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2169.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2169^2 \cdot 1021$$

$$= 384763.219 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2169 \cdot 3 \cdot 1021$$

$$= 35577.184 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2116^2 \cdot 1021$$

$$= 366093.250 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 384763 - 366093$$

$$= 18669.949 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(384763 + 35577 + 0, 0)$$

$$= 420340.406 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69 \cdot 8.36 \cdot \pi \cdot 2169 + 0 \cdot 0 \cdot 0$$

$$= 400456.562 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((420340 + 0 + 4 \cdot 0/2169)/172, 400457/172)$$

$$= 239.668 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 41.3 + 6 \cdot 200/(3 + 0.5)$$

$$= 425.407 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2280 \cdot \sin(\pi / 56)$$

$$= 127.841 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{128/(2 \cdot 41.3 + 200)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	239.668	606.966	
Radial Distance between Hub and Bolts:	53.975	61.000	
Radial Distance between Bolts and Edge:	41.275	52.500	
Circ. Spacing between the Bolts:	88.900	127.841	425.407

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 607 \cdot 172 / (69 \cdot \pi (2186 + 2142))$$

$$= 11.136 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 607$$

$$= 1064527.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 420340 - 384763$$

$$= 35577.18 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2280 - 2169) / 2$$

$$= 55.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 (C - B) / 2 + hg$$

$$= 0.5 (2280 - 2116) / 2 + 55.4$$

$$= 68.6787 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2280 - 2116 - 21) / 2$$

$$= 71.4999 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	366093.	71.4999	1.0000	26176.
Face Pressure, Mt	18670.	68.6787	1.0000	1282.
Gasket Load, Mg	35577.	55.3576	1.0000	1970.
Gasket Seating, Matm	1064527.	55.3576	1.0000	58931.

Total Moment for Operation, Mo 29428. Kg-m.
Total Moment for Gasket seating, Mg 58931. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 165.855 mm
Hub Ratio, h/h0 = HL / H0 0.362
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.615

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.865
Factor V per 4.16.5 0.327
Factor f per 4.16.5 1.152

Stress factor from Table 4.16.4 [K]:

= A / B
= 2385/2116
= 1.127

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.13² • log10(1.13))) / (1.13² - 1))/(1.13 - 1)
= 16.238

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / ((1.0472 + 1.9448 • 1.13²) • (1.13 - 1))
= 1.867

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.13²(1 + 8.55246 • log10(1.13)) - 1) / (1.36136(1.13² - 1) • (1.13 - 1))
= 17.844

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.13² + 1) / (1.13² - 1)
= 8.396

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 17.8 • 13² • 166 / 0.33
= 1527723.125 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.87/166
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (200 • 0.0052 + 1) / 1.87 + 200³/1527723
= 6.331

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.15•29428/2116)/(6.33•21²)
= 56.27 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.15 \cdot 58931 / 2116) / (6.33 \cdot 21^2)$$

$$= 112.68 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 29428 / 2116) / (6.33 \cdot 200^2)$$

$$= 1.29 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.39 \cdot 58931 / 2116) / (6.33 \cdot 200^2)$$

$$= 2.58 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (16.2 \cdot 29428 / (200^2 \cdot 2116)) - 8.4 \cdot 1.29$$

$$= 44.55 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (16.2 \cdot 58931 / (200^2 \cdot 2116)) - 8.4 \cdot 2.58$$

$$= 89.22 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (56.3 + \max(1.29, 44.6)) / 2$$

$$= 50.41 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (113 + \max(2.58, 89.2)) / 2$$

$$= 100.95 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 420340 / 607$$

$$= 67.91 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (400457 / 607)$$

$$= 64.70 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	56.27	207.00	112.68	207.00
Radial Flange	1.29	138.00	2.58	138.00
Tangential Flange	44.55	138.00	89.22	138.00
Maximum Average	50.41	138.00	100.95	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Required Flange Thickness [Rigidity] 196.545 mm

Estimated M.A.W.P. (Operating) 1683 KPa.

Estimated Finished Weight of Flange at given Thk. 1589.7 kg.

Estimated Unfinished Weight of Forging at given Thk 1979.0 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	452.91	499.99	0.906	452.91	499.99	0.906
Tangential Flange	St	357.75	374.99	0.954	357.75	374.99	0.954
Tangential Flange	Sto	152.91	250.00	0.612	152.91	250.00	0.612
Combined	Sh/f + St	749.99	749.99	1.000	749.99	749.99	1.000
Combined	Sr + St	368.09	749.99	0.491	368.09	749.99	0.491
Combined	Sh + Sto	605.82	749.99	0.808	605.82	749.99	0.808
Radial Flange	Sr	10.34	499.99	0.021	10.34	499.99	0.021
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.56/1.35)

= 1.1547

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)

= max(0.25; 0.51/1.32) if h/h0 < 0.35, else max (0.25; 0.3/0.76)

= 0.3898

In this case, the hub ratio h/h0, 0.3618 was ≥ 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	689.6 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	10.8387 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.660 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2186^2 - 2142^2)$$

$$= 1495.7 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 1496 / (56 \cdot 10.8)$$

$$= 512.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(513; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 690)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 508 \geq 140 (1496 / (56 \cdot 10.8))$$

$$:: 507.5 \geq 345.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 508 \geq (97 \cdot 1496 + \pi/4 \cdot 1021 \cdot 2142^2) / (0.7 \cdot 10.8 \cdot 56)$$

$$:: 507.5 \geq 428.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 508 \leq 275 (1496 / (56 \cdot 10.8))$$

$$:: 507.5 \leq 677.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 508 \leq 690 (1/0.66)$$



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:: 507.5 ≤ 1045.3 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

= Sbsel • K • Abs • φb

= 508 • 0.26 • 10.8 • 41.3

= 601.9 Kg-m.

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb ≥ SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 58931 / 1 • 9807 • 0.33 / (5.82 • 201334 • 13² • 166 • 0.3)

= 1.000 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 29428 / 1 • 9807 • 0.33 / (5.82 • 197847 • 13² • 166 • 0.3)

= 0.508 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

= 52.14 • Ma / Bsc • Cnvfac • V / (λ • Eamb • go² • ho • Ki)

= 52.14 • 58931 / 1 • 9807 • 0.33 / (6.33 • 201334 • 13² • 166 • 0.3)

= 0.920 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

= 52.14 • Mo / Bsc • Cnvfac • V / (λ • Eop • goc² • ho • Ki)

= 52.14 • 29428 / 1 • 9807 • 0.33 / (6.33 • 197847 • 13² • 166 • 0.3)

= 0.467 (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.48, Temperature Reduction per Fig. UCS 66.1 = 35 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	55.74	207.00	109.49	207.00
Radial Flange	1.64	138.00	3.21	138.00
Tangential Flange	40.27	138.00	79.11	138.00
Maximum Average	48.00	138.00	94.30	138.00
Bolting	67.91	172.00	64.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 185.776 mm

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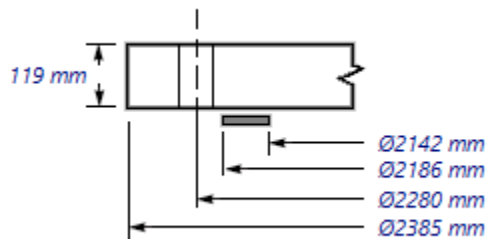
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Flange Input Data Values

Description: New Flange :

Rhs Blind Flg

Flange Type		Blind	
Design Pressure	P	1023.31	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2385.000	mm
Flange Thickness	t	119.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	7162.831	mm
Diameter of Bolt Circle	C	2280.000	mm
Nominal Bolt Diameter	a	41.2750	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2197.000	mm
Flange Face Inside Diameter	Fid	2110.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2186.000	mm
Gasket Inside Diameter	Gi	2142.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	2169.285 mm

Basic Flange and Bolt Loads:



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Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2169^2 \cdot 1023$$

$$= 385679.438 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2169 \cdot 3 \cdot 1023$$

$$= 35661.898 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(385679 + 35662 + 0, 0)$$

$$= 421341.312 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 69 \cdot 8.36 \cdot \pi \cdot 2169 + 0 \cdot 0 \cdot 0$$

$$= 400456.562 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((421341 + 0 + 4 \cdot 0 / 2169) / 172, 400457 / 172)$$

$$= 240.238 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	240.238	606.966	
Radial Distance between Bolts and Edge:	41.275	52.500	
Circ. Spacing between the Bolts:	88.900	127.841	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 607 \cdot 172 / (69 \cdot \pi (2186 + 2142))$$

$$= 11.136 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 607$$

$$= 1064527.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 421341.31 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2280 - 2169) / 2$$

$$= 55.3576 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 421341 \cdot 55.4 \cdot 1 / (116^2 \cdot 2169) + 0.3 \cdot 1 \cdot 1023 \cdot 2169^2 / 116^2$$

$$= 122.26 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1064527 \cdot 55.4 \cdot 1 / (116^2 \cdot 2169)$$



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= 37.62 N./mm²

Bolt Stress, Operating [BS_o]:

= W_{m1} / A_b

= 421341/607

= 68.08 N./mm²

Bolt Stress, Seating [BS_a]:

= (W_{m2} / A_b)

= (400457/607)

= 64.70 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	122.26	138.00	37.62	138.00
Bolting	68.08	172.00	64.70	172.00

Reqd. Blind Flange Thickness at Center 112.184 mm

Reqd. Blind Flange Thickness at Gasket 63.564 mm

Estimated M.A.W.P. (Operating) 1155 KPa.

Estimated Finished Weight of Flange at given Thk. 4120.4 kg.

Estimated Unfinished Weight of Forging at given Thk 4120.4 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.82, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -6 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	116.17	138.00	35.74	138.00
Bolting	68.08	172.00	64.70	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 109.195 mm

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Internal Pressure Calculations: Step: 9 9:26am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flg		1023.3	127	3	2385	138	1
LHS Ch Flg		1020.9	211	3	2110	138	2
LHS Ch Shell		1020.7	16	1	2110	117.3	3
LSH TS Flang		1020.9	211	3	2110	138	4
Main Shell		2230.7	24	3	2110	138	5
RHS CH Flang 1		1020.9	211	3	2110	138	6
Rhs Ch Shell		1020.7	16	1	2110	117.3	7
RHS CH Flang 2		1020.9	211	3	2110	138	8
Rhs Blind Flg		1023.3	127	3	2385	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flg		1000	1132	1216	119	112.184	1
LHS Ch Flg		1000	1662	2074	203	196.545	2
LHS Ch Shell		1000	1413	1544	14	10.2378	3
LSH TS Flang		1000	1662	2074	203	196.545	4
Main Shell		2210	2686	3097	24	20.2709	5
RHS CH Flang 1		1000	1662	2074	203	196.545	6
Rhs Ch Shell		1000	1631	1763	16	10.2378	7
RHS CH Flang 2		1000	1662	2074	203	196.545	8
Rhs Blind Flg		1000	1132	1216	119	112.184	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 2210 KPa.
 Shell Side MAPnc = 3097 KPa.
 Channel Side MAWP = 1000 KPa.
 Channel Side MAPnc = 1216 KPa.

Elements Suitable for Design Internal Pressure.

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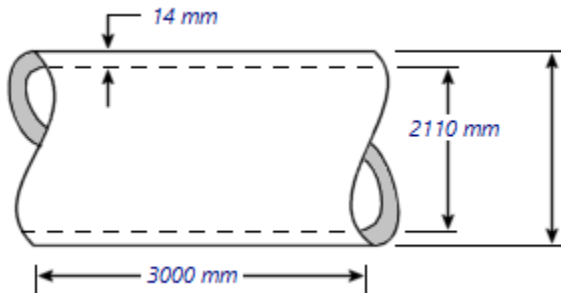
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Element: LHS Ch Shell

Step: 10 9:26am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1020.692 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	3000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2110.000 mm	Actual Stress	98.3 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	201878.8 cm ²	SG of Contents	1.000
Element Volume	10491894.0 cm ³	Weight of Contents	10485.5 Kg
Empty Weight	2484.8 Kg	Operating Weight	12970.2 Kg
		Comp. All. Ope. Stress	116.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1021 \cdot 1056) / (138 \cdot 0.85 - 0.6 \cdot 1021) \\
 &= 9.2378 + 1.0000 = 10.2378 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 20.692 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 13) / (1056 + 0.6 \cdot 13) \\
 &= 1433 - 20.7 = 1413 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 14) / (1055 + 0.6 \cdot 14) \\
 &= 1544 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1021 \cdot (1056 + 0.6 \cdot 13)) / (0.85 \cdot 13) \\
 &= 98.269 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS Ch Shell

Step: 10

9:26am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.753 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14$, $t_r = 9.05$, $c = 1$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.59$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -35 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1000.000 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-35 °C
Stress/Thk. Ratio	0.592	Temperature Difference	13 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	2138.00	3000.00	164.46	1.4032	0.0004488	44.87

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 44.9) / (3 \cdot 164) = 364 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.891	2138.00	3000.00	270.95	1.4032	0.0002102	21.02

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21) / (3 \cdot 271) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	2138.00	9857.32	164.46	4.6105	0.0001277	12.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 12.8) / (3 \cdot 164) = 103 \text{ KPa.}$$



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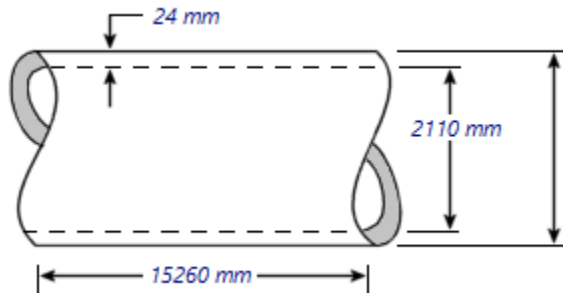
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Element: Main Shell

Step: 11 9:26am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	2230.712 KPa.	Int. Design Temperature	150 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	15260.001 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2110.000 mm	Actual Stress	113.7 N./mm ²
Spec. Min. Thk t,e, or sa	24.0000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	1034560.6 cm ²	SG of Contents	1.000
Element Volume	24722994.0 cm ³	Weight of Contents	24707.9 Kg
Empty Weight	19030.1 Kg	Operating Weight	43738.0 Kg
		Comp. All. Ope. Stress	129.2 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2231 \cdot 1058) / (138 \cdot 1 - 0.6 \cdot 2231) \\
 &= 17.2709 + 3.0000 = 20.2709 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 20.712 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 21) / (1058 + 0.6 \cdot 21) \\
 &= 2707 - 20.7 = 2686 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 24) / (1055 + 0.6 \cdot 24) \\
 &= 3097 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2231 \cdot (1058 + 0.6 \cdot 21)) / (1 \cdot 21) \\
 &= 113.730 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell

Step: 11

9:26am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.125 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 24$, $t_r = 17.1$, $c = 3$ mm, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.81$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was not specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2210.000 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	24.000 mm	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.815	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-12 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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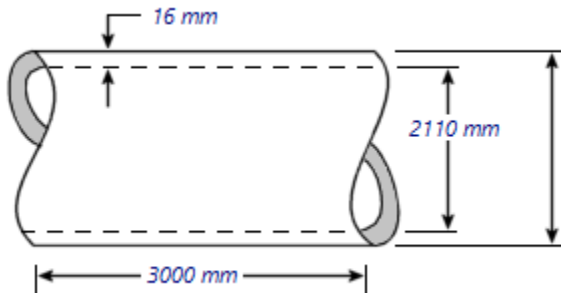
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Element: Rhs Ch Shell

Step: 12 9:26am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Rhs Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1020.692 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	3000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2110.000 mm	Actual Stress	85.3 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	201878.8 cm ²	SG of Contents	1.000
Element Volume	10491894.0 cm ³	Weight of Contents	10485.5 Kg
Empty Weight	2484.8 Kg	Operating Weight	12970.2 Kg
		Comp. All. Ope. Stress	116.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1021 \cdot 1056) / (138 \cdot 0.85 - 0.6 \cdot 1021) \\
 &= 9.2378 + 1.0000 = 10.2378 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 20.692 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 15) / (1056 + 0.6 \cdot 15) \\
 &= 1652 - 20.7 = 1631 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 16) / (1055 + 0.6 \cdot 16) \\
 &= 1763 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1021 \cdot (1056 + 0.6 \cdot 15)) / (0.85 \cdot 15) \\
 &= 85.263 \text{ N./mm}^2
 \end{aligned}$$



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Element: Rhs Ch Shell

Step: 12

9:26am

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.753 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16$, $t_r = 9.05$, $c = 1$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.51$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1000.000 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-31 °C
Stress/Thk. Ratio	0.513	Temperature Difference	18 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Rhs Ch Shell

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2142.00	3000.00	142.80	1.4006	0.0005512	55.10

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 55.1) / (3 \cdot 143) = 514 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.899	2142.00	3000.00	271.17	1.4006	0.0002104	21.04

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21) / (3 \cdot 271) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2142.00	13474.93	142.80	6.2908	0.0001109	11.08

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 11.1) / (3 \cdot 143) = 103 \text{ KPa.}$$



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Shop Pressure Test Results: Step: 13 9:26am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2873.000 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2873.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	4002.499 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2431.000 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3160.300 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3160.300 KPa.
App 27-4	= M.A.W.P.	2210.000 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1300.000 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1556.888 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1100.000 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1000.000 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	141.0	234.0	0.602	1577.57
Main Shell	179.3	234.0	0.766	4023.18
Rhs Ch Shell	123.5	234.0	0.528	1577.57

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	138.00	1.000
N3-Tube Inlet	Pad	138.00	138.00	1.000
N1-Shell Inlet	Nozzle	138.00	138.00	1.000
N2-Shell Outlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flg	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LSH TS Flang	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS CH Flang 1	138.00	138.00	1.000
Rhs Ch Shell	138.00	138.00	1.000
RHS CH Flang 2	138.00	138.00	1.000
Rhs Blind Flg	138.00	138.00	1.000



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Shop Pressure Test Results:

Step: 13

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Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	68.90	45.30	1.521
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	41.44	234.00	0.177
N1-Shell Inlet	68.71	225.00	0.305
N2-Shell Outlet	68.71	225.00	0.305
N4-Tube Outlet	41.44	234.00	0.177

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 14 9:26am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2873.000 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2873.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	4002.499 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2431.000 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3160.300 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3160.300 KPa.
App 27-4	= M.A.W.P.	2210.000 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1300.000 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1556.888 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1100.000 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1000.000 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 2210 • 1

= 2873.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1000 • 1

= 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	127.2	234.0	0.543	1320.68
Main Shell	147.5	234.0	0.630	2893.68
Rhs Ch Shell	110.3	234.0	0.471	1320.68

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	138.00	1.000
N3-Tube Inlet	Pad	138.00	138.00	1.000
N1-Shell Inlet	Nozzle	138.00	138.00	1.000
N2-Shell Outlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
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Field Pressure Test Results: Step: 14 9:26am Sep 9, 2026

LHS Blind Flg	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LSH TS Flang	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS CH Flang 1	138.00	138.00	1.000
Rhs Ch Shell	138.00	138.00	1.000
RHS CH Flang 2	138.00	138.00	1.000
Rhs Blind Flg	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	68.90	45.30	1.521
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	34.69	234.00	0.148
N1-Shell Inlet	49.42	225.00	0.220
N2-Shell Outlet	49.42	225.00	0.220
N4-Tube Outlet	34.69	234.00	0.148

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		2873.00 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2110.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2110.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3704
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	15500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		15260.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SB-338 1
Tube Material UNS Number		R50250
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		68.90 N./mm ²
Tube Allowable Stress At Ambient		68.90 N./mm ²
Tube Yield Stress At design Temperature	Syt	138.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1029.310 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	31602.9 cm ²



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Perimeter of Tube Layout Cp 6619.8 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 120.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 2385.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? Yes
Is the Tubesheet Flat on Tubeside? No
Step 1 Diameter on the Tubeside dt1 2194.00 mm
Step 1 Depth on the Tubeside ht1 9.00 mm
Step 2 Diameter on the Tubeside dt2 2127.00 mm
Step 2 Depth on the Tubeside ht2 7.00 mm
Is the Tubesheet Flat on Shellside? Yes
Calculated Tubesheet Diameter per Div 2, 4.18.4 2385.00 mm

Note: Tubesheet diameter is now: 2385.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 520.000 mm
Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Channel
Flange Outside Diameter A 2385.000 mm
Flange Inside Diameter B 2110.000 mm
Flange Face Outside Diameter Fod 2197.000 mm
Flange Face Inside Diameter Fid 2110.000 mm
Gasket Outside Diameter Go 2186.000 mm
Gasket Inside Diameter Gi 2142.000 mm
Small end Hub thk. g0 16.0000 mm
Large end Hub thk. g1 24.0000 mm
Gasket Factor, m 3.00
Gasket Design Seating Stress y 68.95 N./mm²
Flange Facing Sketch Code Sketch 1a
Gasket Thickness tg 3.1750 mm
Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 2280.000 mm
Nominal Bolt Diameter dB 41.2750 mm
Thread Series TEMA Thread Series



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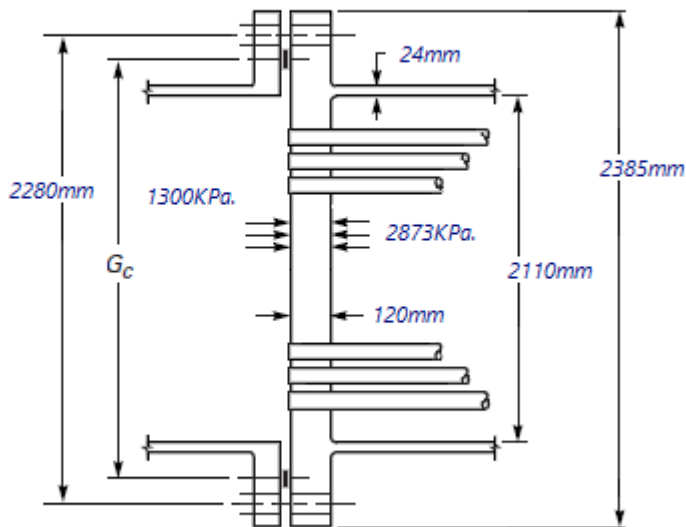
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Number of Bolts	n	56
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with		Shell
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	104.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-5 Titanium Gr 1,2,3,7,11,12,16 and 17
Tubes - TM-5 R50250

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000082793 /°C
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Elastic Mod. at Design Temperature	21.1 °C	0.10687E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.10687E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.10687E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.10687E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1300 \cdot (25.4/2 - 0)) / (68.9 \cdot 1 + 0.4 \cdot 1300)$$

$$= 0.2378 + 0.0000 = 0.2378 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart NFT-3 at 21.11 °C
Elastic Modulus for Material 106866296.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	15260.00	12.04	50.0000	0.0075908	62.32
EMAWP = (4*B)/(3*(D/T)) = (4 *62.3)/(3 *12) = 6903 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.0271	25.40	15260.00	24.73	50.0000	0.0017988	53.29
EMAWP = (4*B)/(3*(D/T)) = (4 *53.3)/(3 *24.7) = 2873 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.0271 mm
Allowable Internal Pressure at Corroded thickness	12261.13 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	6902.53 KPa.
Required External Design Pressure	2873.00 KPa.
Required Thickness due to Shell Side pressure	1.0271 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	22.000 mm
Basic Gasket Width,	b0 = N / 2	11.000 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	8.355 mm
Gasket Reaction Diameter,	G = Go-2.0*b	2169.291 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:



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Tube expansion depth ratio [p]:

= l_{tx} / h (modified for corrosion if present)

= 0/120

= 0.0000 (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$

= $\max(25.4 - 2 \cdot 2.11 (106866296/202701232) \cdot$

$(68.9/138) \cdot (0); 25.4 - 2 \cdot 2.11)$

= 25.4000 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 1029 + 25.4$

= 2084.020 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(31.8 - 25.4) / 31.8$

= 0.200

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= 2084/2

= 1042.0100 mm

Effective tube pitch [p*]:

= $p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

= $31.8 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2084 \cdot 31.8) / (\pi \cdot 2084^2)) }$

= 31.7500 mm

Effective ligament efficiency for bending [μ*]:

= $(p* - d*) / p*$

= $(31.8 - 25.4) / 31.8$

= 0.2000

Ratio [ps]:

= as / ao

= 1055/1042

= 1.0125

Ratio [pc]:

= ac / ao

= 1085/1042

= 1.0409

Compute parameter [xt]:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

= $1 - 3704 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1042))^2$

= 0.6174

Determine parameter [xs]:

= $1 - Nt(dt / (2 \cdot ao))^2$

= $1 - 3704 (25.4 / (2 \cdot 1042))^2$

= 0.4498

Determine the value [h'g]:

= $\max((hg - CATC); 0)$ (For pressure only cases)

= $\max((0 - 0); 0)$

= 0.000 mm

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:



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$$\begin{aligned}
 &= \pi \cdot ts (Ds + ts) Es / L \\
 &= \pi \cdot 24 (2110 + 24) 202701232 / 15260 \\
 &= 217948.5312 \text{ Kgf/mm}
 \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned}
 &= \pi \cdot tt (Dt - tt) Et / L \\
 &= \pi \cdot 2.11 (25.4 - 2.11) 106866296 / 15260 \\
 &= 110.2515 \text{ Kgf/mm}
 \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned}
 &= Ks / (Nt \cdot Kt) \\
 &= 217949 / (3704 \cdot 110) \\
 &= 0.53370
 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned}
 &= 1 / (1 + Ks/Kj) \\
 &= 1 / (1 + 217949/0) \\
 &= 1.0000 (= 1 \text{ if No Exp. Jt.})
 \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts) \\
 &= ((12(1 - 0.3^2))^{1/4}) / ((2110 + 24)24) \\
 &= 0.0080 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [ks]:

$$\begin{aligned}
 &= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2)) \\
 &= 0.008 \cdot 202701232 \cdot 24^3 / (6(1 - 0.3^2)) \\
 &= 4122401536.0000 \text{ KPa} \cdot \text{mm}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(2110 + 24) \cdot 4.12E+09 / 120^3 \cdot (1 + 120 \cdot 0.008 + (120 \cdot 0.008)^2 / 2) \\
 &= 74179400.0000 \text{ KPa}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts))) \\
 &= 2110^2 / (4 \cdot 202701232 \cdot 24) \cdot (1 - 2110 \cdot 0.3 / (2(2110 + 24))) \\
 &= 0.19485E+00 \text{ mm/N} \cdot \text{mm}^2
 \end{aligned}$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 3.779528$; $\mu^* = 0.200000$; $E^* = 31489462 \text{ KPa}$.

$E^*/E = 0.155349$; $\nu^* = 0.458572$

Note: As h/p (3.780) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$\begin{aligned}
 &= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4} \\
 &= ((24(1 - 0.46^2)3704 \cdot 106866296 \cdot 2.11 \\
 &\quad (25.4 - 2.11)1042^2) / (31489462 \cdot 15260 \cdot 120^3))^{1/4} \\
 &= 4.6858
 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned}
 Z_d &= 0.014611; & Z_v &= 0.044605; & Z_m &= 0.301300 \\
 Z_a &= 0.162509E+02; & Z_w &= 0.044605
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned}
 &= A / D_o \\
 &= 2385 / 2084 \\
 &= 1.1444
 \end{aligned}$$



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Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 31489462 \cdot (74179400 + 0 + 202701232 \cdot \ln(1.14))$$

$$= 1.7456$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.46) 1.75$$

$$= 2.5461$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.01 - 1 - 2.55 \cdot 0.045) / (1 + 2.55 \cdot 0.3)$$

$$= -0.057213$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.015 + -0.057 \cdot 0.045) / 2 \cdot 4.69^4$$

$$= 2.9068$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.045 + -0.057 \cdot 0.3) / 2 \cdot 4.69^4$$

$$= 6.5967$$

Compute parameter [U]:

$$= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.045 + (1.01 - 1) 0.3) 4.69^4 / (1 + 2.55 \cdot 0.3)$$

$$= 13.1933$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (2169 - 2280) / 2084 = -0.053$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 4.12E+09 \cdot 0.008 \cdot 0.19484627 (1 + 120 \cdot 0.008)$$

$$= 12829.8428 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s$$

$$= 1042^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 12830$$

$$= -12744.9463 \text{ mm}^2$$

Parameter [ω_c]:

$$= \psi_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.04 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 120) - 0.000 (0/120 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= a o^2 [(\psi_c^2 + 1) \cdot (\psi_c - 1) / 4 - (\psi_s - 1) / 2] - \omega_c$$

$$= 1042^2 [(1.04^2 + 1) \cdot (1.04 - 1) / 4 - (1.01 - 1) / 2] - 0$$

$$= 16372.8418 \text{ mm}^2$$

Pressure [P^*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$



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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 2873 \{ 0.45 + 2(1 - 0.45) 0.32 + [2/0.53 \cdot (2110/2084)^2] 0.3 - [(1.01^2 - 1)/(1 \cdot 0.53)] - [(1 - 1)/(2 \cdot 1 \cdot 0.53)] [(0^2 - (2110)^2)/2084^2] \}$$

$$= 5479.7949 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.62 + 2(1 - 0.62) 0.32 + 1/(1 \cdot 0.53)] 1300$$

$$= 3556.7725 \text{ KPa.}$$

Pressure [Pγ]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 3704 \cdot 110 \cdot 0 / (3.14 \cdot 1042^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0.053 \cdot 13.2 \cdot 535253 / (2 \cdot \pi \cdot 1042^2)$$

$$= 539.2316 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(13.2/1042^2) \cdot (-12745 \cdot 2873 - 16373 \cdot 1300)$$

$$= 703.5521 \text{ KPa.}$$

Calculate the pressure [Pω]:

$$= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c)$$

$$= (13.2/1042^2) \cdot (12830 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P\gamma + P\omega + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.53 / (1 + 1 \cdot 0.53(2.91 + (1.01 - 1)6.6)) \cdot (5480 - 3557 + 0 + 0 + 539 + 704)$$

$$= 651.0352 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-12745 \cdot 2873 - 16373 \cdot 1300) - (12830 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 535253 \cdot -0.053 / (2 \cdot \pi)] / (1 + 2.55 \cdot 0.3)$$

$$= -5902.1758 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.057 + 2 \cdot -5902 / (651 \cdot 1042^2)$$

$$= -0.22097$$

Fm Value from Table 4.18.3 = 0.110484

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$



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$$= (1.5 \cdot 0.11 / 0.2) \cdot (2 \cdot 1042 / (120 - 0))^2 \cdot 651$$

$$= 162.7171 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.081 / 0.2) \cdot (2 \cdot 1042 / (69.5 - 0))^2 \cdot 657$
 $= 356.2411 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 69.5 + 0 + 0$
 $= 69.5366 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$
 $= (2 \cdot 0.2 \cdot 120 / 1042) \cdot 190$
 $= 8.7522 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 3.8100 mm
 Tubesheet Shear Stress = 86.4317 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(69.5; 3.81)$
 $= 69.5366 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = -0.097, F_{tmax} = 1.47$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((2873 \cdot 0.45 - 1300 \cdot 0.62) - (651) \cdot -0.097) / (0.62 - 0.45)$
 $= 3.2974 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((2873 \cdot 0.45 - 1300 \cdot 0.62) - (651) \cdot 1.47) / (0.62 - 0.45)$
 $= -2.8063 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 3.3 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= \beta T \cdot S_{y,t} = 131.0976 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.015 + -0.22 \cdot 0.045)4.69^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2)$



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ASME TS Calc: Pressure Test

Case: 1 9:26a Sep 9, 2026

Determine the factor [rt]:

$$= \left(\sqrt{\left(\frac{dt^2}{4} + \left(\frac{dt}{2} - 2 \cdot \frac{tt^2}{4} \right)^2} \right) \right) / 4$$

$$= \left(\sqrt{\left(\frac{25.4^2}{4} + \left(\frac{25.4}{2} - 2 \cdot \frac{2.11^2}{4} \right)^2} \right) \right) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot Et}{Syt} \right)}$$

$$= \sqrt{\left(\frac{2 \cdot \pi^2 \cdot 106866296}{138} \right)}$$

$$= 163.6864$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 1 \cdot 520/8.27$$

$$= 62.8932$$

Buckling Allowable Stress [Stb]:

$$= (Sy/t/Fs) \cdot (1 - Ft/(2 \cdot Ct))$$

$$= (138/2) \cdot (1 - 62.9/(2 \cdot 164))$$

$$= 131.098 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 3.3 \cdot 1.54$$

$$= 51.91 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 131$$

$$= 2063.802 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 131$$

$$= 2019.6411 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 131/131$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 131$$

$$= 1306.8267 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot d_o}{2} \right)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{\left(\frac{0.75 \cdot 25.4}{2} \right)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm



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ASME TS Calc: Pressure Test

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Tube-Tubesheet Jt allowable, 2064 is >= tube strength 2064 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 260 = 370 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1042^2 / ((2110 + 24) 24) \cdot [651 + (1.01^2 - 1)(2873 - 1300)] + 1055^2 \cdot 1300 / ((2110 + 24) 24)$$

$$= 42.8927 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 24 mm, the minimum Shell length adjacent to the tubesheet should be 405.060 mm

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	162.7 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	13.0 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	3.3 <=	131.1 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.8 <=	-131.1 N./mm ²	Pass
Maximum Force on any one Tube	51.9 <=	2063.8 Kgf	Pass
Axial Membrane Stress in Shell	42.9 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	69.537	120.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	10.346 40.017	97	356 11 190	Ps+Pt-Th	D1 Ok
D2uc	55.841 ...	122	356 24 190	Ps+Pt-Th	D2 Ok
D3uc	69.537 ...	163	356 13 190	Ps+Pt-Th	D3 Ok
Max:	69.5366 40.017 mm	0.457	0.128 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:



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ASME TS Calc: Pressure Test

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With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					---- Shell Band Stress ----					Pass
Case#	Ten	Allwd	Cmp	Allwd	Ten	Allwd	Cmp	Allwd	Fail	
D1uc	16	370	...	...	...	...	...	...	...	Ok
D2uc	27	370	...	...	...	...	...	...	...	Ok
D3uc	43	370	...	...	...	...	...	...	...	Ok
Max RATIO	0.116		...	...	...	...	...	...	...	

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	9	131	-8	-131	149	2063	...	370	...	...	Ok
D2uc	11	131	-11	-131	173	2063	...	370	...	...	Ok
D3uc	3	131	-3	-131	52	2063	...	370	...	...	Ok
Max RATIO	0.084		0.084		0.084		...	...	...	...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	69.537	120.000	Ok
Tubesheet thickness flanged extension :	40.017	104.000	Ok
Tube thickness :	1.027	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (24.000) adj. to tubesheet: 405.060 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Compresses by : -0.606 mm
 Due to Pressure + Thermal Shell Compresses by : -0.606 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1163	...	...	...	...	...
2	-0.6057	...	...	...	...	...
3	-0.4894	...	...	...	...	...



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ASME TS Calc: Pressure Test

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4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculations: Step: 15 9:26am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	116	No Calc	No Calc
20	30	No Calc	...	200	No Calc	No Calc
30	40	3000	2138	13	0.00044885	44.8748
40	60	No Calc	...	200	No Calc	No Calc
60	70	15260	2158	21	0.00015706	15.6921
70	80	No Calc	...	200	No Calc	No Calc
80	90	3000	2142	15	0.00055116	55.1031
90	100	No Calc	...	200	No Calc	No Calc
100	110	No Calc	...	116	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	119	63.5762	103.419	No Calc
20	30	203	196.545	103.419	No Calc
30	40	14	8.89062	103.419	364
40	60	203	196.545	103.419	No Calc
60	70	24	No Calc	...	204
70	80	203	196.545	103.419	No Calc
80	90	16	8.89911	103.419	514
90	100	203	196.545	103.419	No Calc
100	110	119	63.5762	103.419	No Calc

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	3000	9857.32	No Calc	No Calc
40	60	No Calc	No Calc	No Calc	No Calc
60	70	15260	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	3000	13474.9	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 16

9:26am

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	4397.44	...	4397.44	...	439.744
20	30	1649.92	951970	1608.09	953166	164.992
30	40	2484.75	10491894	2330.55	10511794	248.475
40	60	1649.92	951970	1608.09	953166	164.992
60	70	19030.1	24722994	16674.8	25026940	1903.01
70	80	1649.92	951970	1608.09	953166	164.992
80	90	2484.75	10491894	2330.55	10511794	248.475
90	100	1649.92	951970	1608.09	953166	164.992
100	110	4397.44	...	4397.44	...	439.744
Total		39394	49514668.00	36563	49863184.00	3939



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	59.5	3.49993	...	Liquid: 10
20	Liqd	951.39	131.5	-9.99997	...	Liquid: 20
30	Liqd	10485.5	1500	...	...	Liquid: 30
30	Lini	477.159	1500	...	...	Titanium clad
30	Nozl	709.268	1500	1855	...	N3-Tube Inlet
40	Liqd	951.39	131.5	-9.99997	...	Liquid: 40
60	Sadl	1090.82	3000	1387.5	...	Left Saddle flo
60	Sadl	1090.82	12240	1387.5	...	RH fixed saddle
60	Liqd	24707.9	7630	...	...	Liquid: 60
60	Insl	3814.87	7630	...	...	Insulation
60	Nozl	311.271	2000	1360	...	N1-Shell Inlet
60	Nozl	311.271	13270	-1360	...	N2-Shell Outlet
60	Wght	800	3000	1752	...	Fireproofing we
60	Wght	800	12216	-1752	...	Fireproofing we
70	Liqd	951.39	131.5	-9.99997	...	Liquid: 70
80	Liqd	10485.5	1500	...	...	Liquid: 80
80	Lini	477.159	1500	...	...	Titinium Clad
80	Nozl	931.37	1500	-1855	...	N4-Tube Outlet
90	Liqd	951.39	131.5	-9.99997	...	Liquid: 90
100	Liqd	...	59.5	3.49993	...	Liquid: 100
40	FTsh	2650.43	326.175	...	...	Fixed Tubesheet
40	Tube	43990.3	8016.17	...	...	Tubes
40	RTsh	2650.43	15706.2	...	...	Tubesheet
60	Baf1	103.862	520	...	...	Baffles
60	Baf1	103.862	1040	...	...	Baffles
60	Baf1	103.862	1560	...	...	Baffles
60	Baf1	103.862	2080	...	...	Baffles
60	Baf1	103.862	2600	...	...	Baffles
60	Baf1	103.862	3120	...	...	Baffles
60	Baf1	103.862	3640	...	...	Baffles
60	Baf1	103.862	4160	...	...	Baffles
60	Baf1	103.862	4680	...	...	Baffles
60	Baf1	103.862	5200	...	...	Baffles
60	Baf1	103.862	5720	...	...	Baffles
60	Baf1	103.862	6240	...	...	Baffles



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Element and Detail Weights: Step: 16 9:26am Sep 9,2026

60 Baf1	103.862	6760	...	...	Baffles
60 Baf1	103.862	7280	...	...	Baffles
60 Baf1	103.862	7800	...	...	Baffles
60 Baf1	103.862	8320	...	...	Baffles
60 Baf1	103.862	8840	...	...	Baffles
60 Baf1	103.862	9360	...	...	Baffles
60 Baf1	103.862	9880	...	...	Baffles
60 Baf1	103.862	10400	...	...	Baffles
60 Baf1	103.862	10920	...	...	Baffles
60 Baf1	103.862	11440	...	...	Baffles
60 Baf1	103.862	11960	...	...	Baffles
60 Baf1	103.862	12480	...	...	Baffles
60 Baf1	103.862	13000	...	...	Baffles
60 Baf1	103.862	13520	...	...	Baffles
60 Baf1	103.862	14040	...	...	Baffles
60 Baf1	103.862	14560	...	...	Baffles
60 Baf1	104.499	15080	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	2181.6
Liquid	49484.5
Insulation	3814.9
Cladding & Overlay	954.3
Nozzles	2263.2
Weights	1600.0
Exchanger Components	52303.8
Liquid in Tubes	20218.8

Sum of the Detail Weights 132821.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	43333.5	43333.5	43333.5	43333.5	43333.5	43333.5
Saddles	2181.6	2181.6	2181.6	2181.6	2181.6	2181.6
Cladding	954.3	954.3	954.3	954.3	954.3	954.3
Nozzles	2263.2	2263.2	2263.2	2263.2	2263.2	2263.2
Exchanger	52303.8	52303.8	52303.8	52303.8	52303.8	52303.8
Insulation	...	...	...	...	3814.9	3814.9 *
Test Weights	...	1600.0	...	...	...	...
Ope Weights	...	...	...	...	...	1600.0
Ope. Liquid	...	...	...	...	...	49484.5
Tube Ope Lqd	...	...	...	...	...	20218.8
Test Liquid	...	49365.3	...	...	...	...
Tube Tst Lqd	...	20218.9	...	...	...	...
Totals	101036.5	172220.7	101036.5	101036.5	104851.3	176154.6

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	101036.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	172220.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	101036.5 kg.



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Element and Detail Weights: Step: 16 9:26am Sep 9,2026

Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	101036.5 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	104851.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	176154.6 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	173040.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	176035.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	20231230.0 cm ³
Weight of Ope Liq in tubes :	20218.8 kg.
Weight of Water in tubes :	20218.9 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	54190.9	...
20	30	29585.8	17963.9
30	40	201879	198863
40	60	29585.8	17963.9
60	70	1034560	1011549
70	80	29585.8	17963.9
80	90	201879	198863
90	100	29585.8	17963.9
100	110	54190.9	...

Total	1665043.2	1481130.6 cm ²
Total	1792.2	1594.3 ft ²

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Nozzle Flange MAWP:

Step: 17

9:26am

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
N1-Shell Inlet	4510.0	5100.0	150	300	GR 1.1	...	...	...	...
N2-Shell Outle	4510.0	5100.0	150	300	GR 1.1	...	...	...	...
N4-Tube Outlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4510.037 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1820.434 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation:

Step: 18 9:26am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	17500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.2 \cdot 0.92)/(1 + 1.7 \cdot 3.4 \cdot 0.2)))$$

$$= \min(0.85, 0.89)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.617
Structure Height to Diameter ratio	10.075

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1760 + 17500 = 19260 \text{ mm}$$

$$= 63.2 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (63.2 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (63.2/900)^{2/9.5}$$

$$= 1.15$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.15 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.62 • 3.67

= 30.8 lbs. [14] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flg	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
LHS Ch Flg	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
LHS Ch Shell	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
LSH TS Flang	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
Main Shell	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
RHS CH Flang 1	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
Rhs Ch Shell	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
RHS CH Flang 2	19260.0	0.000	0.000	0.000	1.149	1.000	78.120
Rhs Blind Flg	19260.0	0.000	0.000	0.000	1.149	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fireproofing	0.00	21012.00	78.12	0.00
Fireproofing	0.00	17500.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	19260	2862	3405.78	78.12	15.3509
20	30	19260	2532	6659.16	78.12	30.0149
30	40	19260	2565.6	76968	78.12	346.919
40	60	19260	2532	6659.16	78.12	30.0149
60	70	19260	2829.6	431797	78.12	1946.24
70	80	19260	2532	6659.16	78.12	30.0149
80	90	19260	2570.4	77112	78.12	347.568
90	100	19260	2532	6659.16	78.12	30.0149
100	110	19260	2862	3405.78	78.12	15.3509

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.



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Wind Load Calculation:

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Earthquake Load Calculation: Step: 19 9:26am Sep 9,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hnx$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(9.690.75)$
 $= 0.110$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/33)$
 $= 0.030$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.11, 0.03) \text{ per 12.8.2}$
 $= 0.030$
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 $= 0.3 \cdot Sds \cdot W \cdot I$
 $= 0.3 \cdot 0.16 \cdot 173973 \cdot 1.25$
 $= 10229.611$ Kgf

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.16 \cdot 173973$
 $= 5455.8$ Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1192.5	15815.7	1026.84



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Earthquake Load Calculation:

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20	30	1055	15815.7	908.438
30	40	1055	15815.7	908.438
40	60	1055	15815.7	908.438
60	Sad1	1055	15815.7	908.438
Sad1	70	1055	15815.7	908.438
60	70	1055	15815.7	908.438
70	80	1055	15815.7	908.438
80	90	1055	15815.7	908.438
90	100	1055	15815.7	908.438
100	110	1192.5	15815.7	1026.84

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 20 9:26am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	11391.350 mm
Center of Gravity of the Liquid	11401.471 mm
Center of Gravity of the Insulation	11401.351 mm
Center of Gravity of the Lining	11401.348 mm
Center of Gravity of the Nozzles	12336.614 mm
Center of Gravity of the Added Weights (Operating)	11379.351 mm
Center of Gravity of the Tubesheet(s)	11401.349 mm
Center of Gravity of the Tubes	11401.349 mm
Center of Gravity of the Baffles	11572.894 mm
Center of Gravity of Bare Shell New and Cold	11401.837 mm
Center of Gravity of Bare Shell Corroded	11401.870 mm
Vessel CG in the Operating Condition	11416.395 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	11427.406 mm
Vessel CG in the Test Condition	11416.786 mm

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Saddle Calcs: Road Transport:

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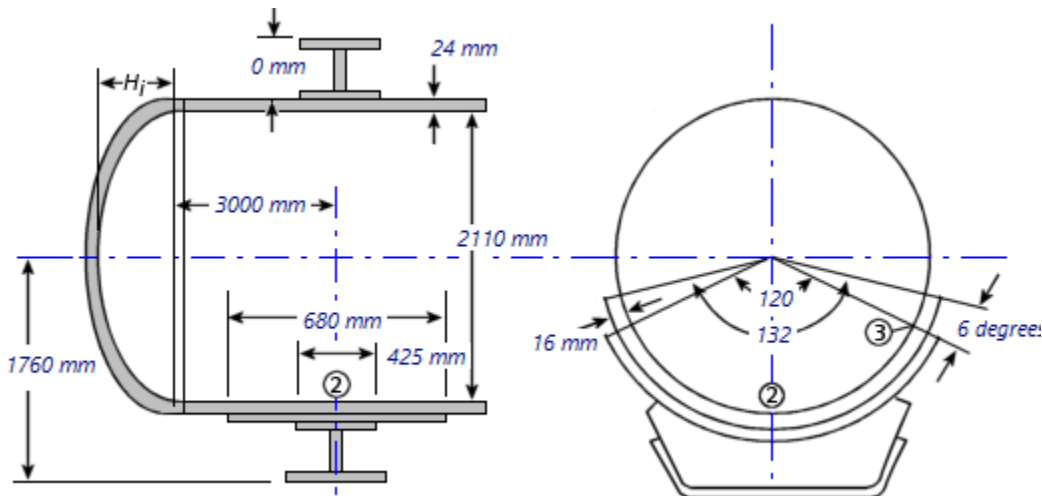
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1067.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	15260.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	680.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		57263.34	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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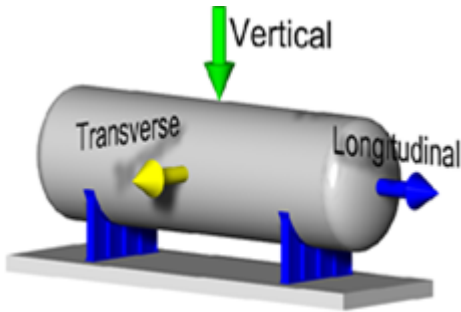
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Saddle Calcs: Road Transport:

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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-5.58	-234.00
Long. Stress at Top of Midspan	5.58	138.00
Long. Stress at Bottom of Midspan	5.58	138.00
Long. Stress at Top of Saddles	33.90	138.00
Long. Stress at Bottom of Saddles	-18.79	-234.00
Long. Stress at Bottom of Saddles	18.79	138.00
Tangential Shear in Shell	15.58	110.40
Circ. Stress at Horn of Saddle	33.03	207.00
Circ. Compressive Stress in Shell	1.58	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 667 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 5662.6 \text{ lbf} \\
 &= 2568.545 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 7.08^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 401.3 \text{ lbf} \\
 &= 182.0 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 98855 \cdot 1 + 2569 \\
 &= 12454.0 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 98855 \cdot 1 + 182 \\
 &= 10067.5 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 12454/2 \cdot 1760/1897 \\
 &= 5778.5 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 10068 \cdot 1760/9240 \\
 &= 1917.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 49033 \cdot 0.05
 \end{aligned}$$



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= 2451.657 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 49033 + 2452 + max(1918, 5779)

= 57263.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 58354.15 Kgf

Transverse Shear Load 6227.01 Kgf

Longitudinal Shear Load 10067.51 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)

= min(2(2110/2 + 24 + 16)sin(120/2), 2•1067)

= 1896.596 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -57263•3000[1-(1-3000/15260+(1067²-0²)/

(2•3000•15260))/(1+(4•0)/(3•15260))]

= -31637.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 57263•15260/4((1+2(1067²-0²)/(15260²))/(1+(4•0)/

(3•15260))-4•3000/15260)

= 48806.7 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)

= 0 • 1067/(2•24) - 48807/(π•1067²•24)

= -5.58 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 0 • 1067/(2 • 24) + 48807/(π • 1067² • 24)

= 5.58 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)

= 0•1067/(2•24)--31637/(0.11•π•1067²•24)

= 33.90 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 0•1067/(2•24)+-31637/(0.19•π•1067²•24)

= -18.79 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 57263 (15260 - 2 • 3000)/(15260 + (4 • 0/3))

= 34748.3 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 34748 / (1067 \cdot 24)$$

$$= 15.58 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1067 \cdot 24)}$$

$$= 124.819 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(1067 \cdot 24)}, 2 \cdot 3000)$$

$$= 674.64 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 57263 \cdot 0.1 / (675(24 + 1 \cdot 16))$$

$$= -1.58 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -57263 / (4(24 + 1 \cdot 16)675) -$$

$$3 \cdot 0.053 \cdot 57263 / (2(24 + 1 \cdot 16)^2)$$

$$= -33.03 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	695.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1055 \cdot 24)} + 680)24^3) / 12$$

$$= 106.932 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 107 + 223(294 - 12)^2$$

$$= 177517.078 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	928.2	24.0	12.0	222.8	267330.6	0.178E+06
Wearplate	680.0	16.0	32.0	108.8	348160.0	0.748E+05
Web	14.0	615.0	347.5	86.1	2991973.5	0.296E+05
BasePlate	450.0	50.0	680.0	225.0	15299993.0	0.335E+06
Totals	...	...	...	642.7	18907458.0	0.617E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 18907458/643$$

$$= 294.199 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 57263$$

$$= 11654.3301 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 2.7219 \text{ N./mm}^2$

Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1055 \cdot \sin(120/2) / 1.05$$

$$= 887.522 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 10343.6768 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 4.8345 \text{ N./mm}^2$

Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(57263 + 1091) \cdot 450) / (4 \cdot 2070 \cdot 173)}$$

$$= 23.201 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1055 + 24 + 16)$$

$$= 1896.596 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1897 / (5 - 1)$$

$$= 474.149 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 474 \cdot 450 / 2$$

$$= 1066.835 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 58354 / (2070 \cdot 450)$$

$$= 6.264 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1067 \cdot 6.26$$

$$= 6682.903 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 57.5 + 33.2$$

$$= 90.730 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 6683/90.7$$

$$= 7.223 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	425.0	...	59.5	...	0.896E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 425 - 14$$

$$= 411.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/90.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(8956/90.7)}$$

$$= 99.353 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1760 - \cos(120/2) (1055 + 24 + 16) - 50$$

$$= 1162.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1162/99.4$$

$$= 11.701$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 10068 / (2 \cdot 2070) \cdot 474 \cdot 1162/2$$

$$= 670.173 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 152.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 7.22/152 + (670 \cdot 213/89559920) / 173$$



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= 0.137

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	411.0	0.0	57.5	0.0	0.896E+04
Web	474.1	14.0	0.0	66.4	0.0	10.8
Totals	...	...	...	123.9	...	0.897E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/124

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 425/2

= 212.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1760 - cos(1095/2)(50 + 0 + 0) - 0

= 615.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{8967/124}$

= 85.063 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 615/85.1

= 7.230

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 10068/(2 • 2070)

= 2.432 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 2.43 • 474 • 615

= 709.087 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 709/421954

= 16.480 N./mm²

Compressive Allowable, KL/r < Cc (7.23 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.23)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (7.23) / (8 \cdot 123) - (7.23^3) / (8 \cdot 123^3))$

= 153.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 10.6/154 + (709 • 213/8967)/173

= 0.164

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Saddle Calcs: Road Transport: Step: 21 9:26am Sep 9,2026

Vessel Mean Radius	Rm	1067.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	15260.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	680.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		58091.30	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-5.66	-234.00
Long. Stress at Top of Midspan		5.66	138.00
Long. Stress at Bottom of Midspan		5.66	138.00
Long. Stress at Top of Saddles		34.39	138.00
Long. Stress at Bottom of Saddles		-19.06	-234.00
Long. Stress at Bottom of Saddles		19.06	138.00
Tangential Shear in Shell		15.80	110.40
Circ. Stress at Horn of Saddle		33.51	207.00
Circ. Compressive Stress in Shell		1.60	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 667 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 5662.6 \text{ lbf} \\
 &= 2568.545 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 7.08^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 401.3 \text{ lbf} \\
 &= 182.0 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 98855 \cdot 1 + 2569
 \end{aligned}$$



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Saddle Calcs: Road Transport: Step: 21 9:26am Sep 9, 2026

= 12454.0 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$

= $0.1 \cdot 98855 \cdot 1 + 182$

= 10067.5 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$

= $12454 / 2 \cdot 1760 / 1897$

= 5778.5 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$

= $10068 \cdot 1760 / 9240$

= 1917.6 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$

= $49822 \cdot 0.05$

= 2491.084 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= $49822 + 2491 + \max(1918, 5779)$

= 58091.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 59182.12 Kgf

Transverse Shear Load 6227.01 Kgf

Longitudinal Shear Load 10067.51 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$

= $\min(2(2110/2 + 24 + 16) \sin(120/2), 2 \cdot 1067)$

= 1896.596 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$

= $-58091 \cdot 3000 [1 - (1 - 3000/15260 + (1067^2 - 0^2)/$

$(2 \cdot 3000 \cdot 15260) / (1 + (4 \cdot 0)/(3 \cdot 15260))]$

= -32094.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/3L) - 4a/L)$

= $58091 \cdot 15260/4 ((1 + 2(1067^2 - 0^2)/(15260^2)) / (1 + (4 \cdot 0)/$

$(3 \cdot 15260)) - 4 \cdot 3000/15260)$

= 49512.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$

= $0 \cdot 1067 / (2 \cdot 24) - 49512 / (\pi \cdot 1067^2 \cdot 24)$

= -5.66 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1067 / (2 \cdot 24) + 49512 / (\pi \cdot 1067^2 \cdot 24)$$

$$= 5.66 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1067 / (2 \cdot 24) - 32095 / (0.11 \cdot \pi \cdot 1067^2 \cdot 24)$$

$$= 34.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1067 / (2 \cdot 24) + 32095 / (0.19 \cdot \pi \cdot 1067^2 \cdot 24)$$

$$= -19.06 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 58091 (15260 - 2 \cdot 3000) / (15260 + (4 \cdot 0 / 3))$$

$$= 35250.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 35251 / (1067 \cdot 24)$$

$$= 15.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1067 \cdot 24)}$$

$$= 124.819 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(1067 \cdot 24)}, 2 \cdot 3000)$$

$$= 674.64 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 58091 \cdot 0.1 / (675(24 + 1 \cdot 16))$$

$$= -1.60 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -58091 / (4(24 + 1 \cdot 16)675) -$$

$$3 \cdot 0.053 \cdot 58091 / (2(24 + 1 \cdot 16)^2)$$

$$= -33.51 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	695.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1055 \cdot 24)} + 680)24^3)/12$$

$$= 106.932 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 107 + 223(294 - 12)^2$$

$$= 177517.078 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	928.2	24.0	12.0	222.8	267330.6	0.178E+06
Wearplate	680.0	16.0	32.0	108.8	348160.0	0.748E+05
Web	14.0	615.0	347.5	86.1	2991973.5	0.296E+05
BasePlate	450.0	50.0	680.0	225.0	15299993.0	0.335E+06
Totals	...	...	...	642.7	18907458.0	0.617E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18907458/643$$

$$= 294.199 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 58091$$

$$= 11822.8389 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 2.7612 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1055 \cdot \sin(120/2) / 1.05$$

$$= 887.522 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10493.2354 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.9044 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(58091 + 1091)450 / (4 \cdot 2070 \cdot 173))}$$

$$= 23.365 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1055 + 24 + 16)$$

$$= 1896.596 \text{ mm}$$

Distance between Ribs [e]:



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$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 1897 / (5 - 1) \\
 &= 474.149 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 474 \cdot 450 / 2 \\
 &= 1066.835 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 59182 / (2070 \cdot 450) \\
 &= 6.353 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 1067 \cdot 6.35 \\
 &= 6777.724 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 57.5 + 33.2 \\
 &= 90.730 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 6778 / 90.7 \\
 &= 7.326 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	425.0	...	59.5	...	0.896E+04

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 425 - 14 \\
 &= 411.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0 / 90.7 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 425 / 2 \\
 &= 212.500 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 8956 / 90.7 } \\
 &= 99.353 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1760 - \cos(120 / 2) (1055 + 24 + 16) - 50 \\
 &= 1162.500 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199943008 / 260 }
 \end{aligned}$$



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$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1162/99.4$$

$$= 11.701$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 10068 / (2 \cdot 2070) \cdot 474 \cdot 1162 / 2$$

$$= 670.173 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 152.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 7.33 / 152 + (670 \cdot 213 / 89559920) / 173$$

$$= 0.138$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	411.0	0.0	57.5	0.0	0.896E+04
Web	474.1	14.0	0.0	66.4	0.0	10.8
Totals	...	...	...	123.9	...	0.897E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/124$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1760 - \cos(1095/2) (50 + 0 + 0) - 0$$

$$= 615.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (8967/124) }$$

$$= 85.063 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 615/85.1$$

$$= 7.230$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 10068 / (2 \cdot 2070)$$

$$= 2.432 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.43 \cdot 474 \cdot 615$$



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= 709.087 Kg-m.

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

= Bending Moment / Section Modulus

= 709/421954

= 16.480 N./mm²

Compressive Allowable, $KL/r < C_c$ (7.23 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(7.23)²/(2 • 123²))260/

(5/3+3•(7.23)/(8• 123)-(7.23³)/(8•123³)

= 153.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 10.7/154 + (709 • 213/8967)/173

= 0.165

Review notes about nozzle loadings and supports in the Warnings report.

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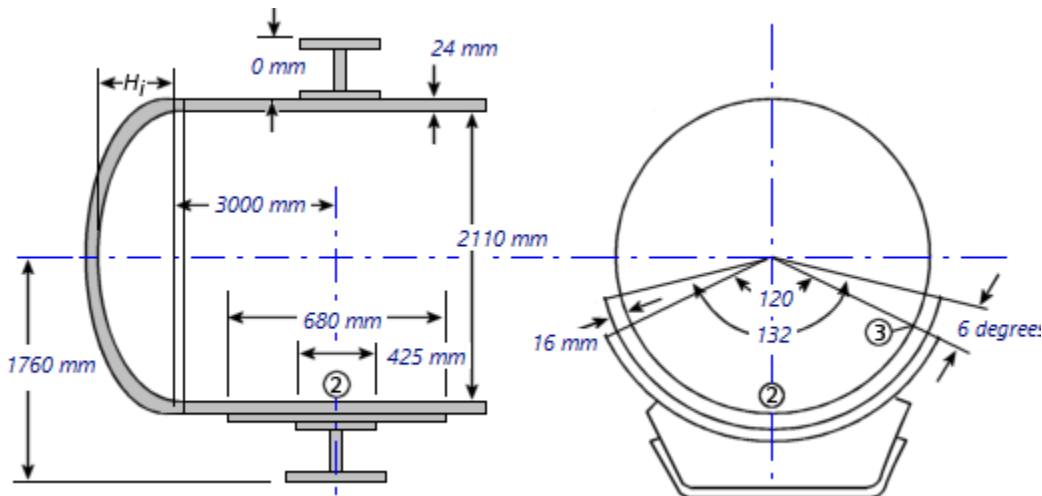
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1068.50	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	15260.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	680.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	103481.26	Kgf
Pressure used in Analysis	P	2220.341 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	45.01	138.00
Long. Stress at Bottom of Midspan	67.97	138.00
Long. Stress at Top of Saddles	126.30	138.00
Long. Stress at Bottom of Saddles	17.80	138.00
Tangential Shear in Shell	32.13	110.40
Circ. Stress at Horn of Saddle	69.18	172.50
Circ. Compressive Stress in Shell	3.17	138.00



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Saddle Calcs: Operating Case: Step: 22 9:26am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (2791/2 + 0) \cdot 1760/1897$$

$$= 3885.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(453, 0) \cdot 1760/9240$$

$$= 86.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(10230, 0) \cdot 1760/9240$$

$$= 1948.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (10230/2 + 0) \cdot 1760/1897$$

$$= 14239.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 89242 + \max(86.2, 3886, 1948, 14239)$$

$$= 103481.3 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.16 \cdot 1.2)^2)$$

$$= 4438.389 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	104572.08	Kgf
Transverse Shear Load	5114.81	Kgf
Longitudinal Shear Load	10229.61	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2110/2 + 24 + 16) \sin(120/2), 2 \cdot 1069)$$

$$= 1896.596 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -103481 \cdot 3000 [1 - (1 - 3000/15260 + (1069^2 - 0^2) / (2 \cdot 3000 \cdot 15260)) / (1 + (4 \cdot 0) / (3 \cdot 15260))]$$

$$= -57161.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 103481 \cdot 15260 / 4 ((1 + 2(1069^2 - 0^2)/(15260^2)) / (1 + (4 \cdot 0) / (3 \cdot 15260)) - 4 \cdot 3000 / 15260)$$



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$$(3 \cdot 15260) - 4 \cdot 3000 / 15260 \\ = 88210.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t) \\ = 2220 \cdot 1069 / (2 \cdot 21) - 88210 / (\pi \cdot 1069^2 \cdot 21) \\ = 45.01 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 2220 \cdot 1069 / (2 \cdot 21) + 88210 / (\pi \cdot 1069^2 \cdot 21) \\ = 67.97 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t) \\ = 2220 \cdot 1069 / (2 \cdot 21) - 57161 / (0.11 \cdot \pi \cdot 1069^2 \cdot 21) \\ = 126.30 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 2220 \cdot 1069 / (2 \cdot 21) + 57161 / (0.19 \cdot \pi \cdot 1069^2 \cdot 21) \\ = 17.80 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 103481 (15260 - 2 \cdot 3000) / (15260 + (4 \cdot 0 / 3)) \\ = 62794.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t) \\ = 1.17 \cdot 62794 / (1069 \cdot 21) \\ = 32.13 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{1069 \cdot 21} \\ = 116.840 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(425 + 1.56 \cdot \sqrt{1069 \cdot 21}, 2 \cdot 3000) \\ = 658.68 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1) \\ = \min(138 / 138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr)) \\ = -0.76 \cdot 103481 \cdot 0.1 / (659(21 + 1 \cdot 16)) \\ = -3.17 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -103481 / (4(21 + 1 \cdot 16)659) - \\ 3 \cdot 0.053 \cdot 103481 / (2(21 + 1 \cdot 16)^2) \\ = -69.18 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 9240.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.12426 \text{E-04} \cdot 9240 (150 - 21.1)$$



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= 14.798 mm

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	695.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1058 \cdot 21)} + 680)21^3)/12$$

$$= 70.424 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 70.4 + 192(306 - 10.5)^2$$

$$= 167449.703 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	912.5	21.0	10.5	191.6	201212.6	0.167E+06
Wearplate	680.0	16.0	29.0	108.8	315520.0	0.835E+05
Web	14.0	615.0	344.5	86.1	2966143.2	0.284E+05
BasePlate	450.0	50.0	677.0	225.0	15232495.0	0.310E+06
Totals	...	...	...	611.5	18715370.0	0.589E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18715370/612$$

$$= 306.041 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 103481$$

$$= 21060.6816 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.9188 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1760 - 1058 \cdot \sin(120/2) / 1.05$$

$$= 885.041 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 18639.9336 \text{ Kg-m.}$$



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Bending Stress, $S_b = (M \cdot C_1 / I) = 9.4902 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(103481 + 1091) \cdot 450) / (4 \cdot 2070 \cdot 175)}$
 $= 30.939 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (1055 + 24 + 16)$
 $= 1896.596 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1897 / (5 - 1)$
 $= 474.149 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 474 \cdot 450 / 2$
 $= 1066.835 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 104572 / (2070 \cdot 450)$
 $= 11.226 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1067 \cdot 11.2$
 $= 11975.927 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 57.5 + 33.2$
 $= 90.730 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 11976 / 90.7$
 $= 12.944 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	425.0	...	59.5	...	0.896E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 425 - 14$
 $= 411.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 90.7$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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Saddle Calcs: Operating Case: Step: 22 9:26am Sep 9,2026

= Saddle Width / 2
= 425/2
= 212.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{8956/90.7}$
= 99.353 mm

Length of Outer Rib [L]:
= Saddle Height - $\cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
= 1760 - $\cos(120/2) (1055 + 24 + 16) - 50$
= 1162.500 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
= 122.739

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1162/99.4
= 11.701

Bending Moment [Rm]:
= $FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
= 10230 / $(2 \cdot 2070) \cdot 474 \cdot 1162/2$
= 680.964 Kg-m.

Compressive Allowable, $KL/r < Cc$ (11.7 < 123) per AISC E2-1 [Sca]:
= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
= $(1 - (11.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7)^3 / (8 \cdot 123^3))$
= 153.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
= 12.9/153 + $(681 \cdot 213/89559920) / 175$
= 0.175

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	411.0	0.0	57.5	0.0	0.896E+04
Web	474.1	14.0	0.0	66.4	0.0	10.8
Totals	...	...	...	123.9	...	0.897E+04

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/124
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 425/2
= 212.500 mm

Length of Inner Rib [L]:
= Saddle Height - Outside Radius - Bpthk
= 1760 - $\cos(1095/2) (50 + 0 + 0) - 0$
= 615.000 mm

Radius of Gyration [r]:



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$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{8967/124} \\
 &= 85.063 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 615/85.1 \\
 &= 7.230
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 10230 / (2 \cdot 2070) \\
 &= 2.471 \text{ Kg/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 2.47 \cdot 474 \cdot 615 \\
 &= 720.504 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 721/421954 \\
 &= 16.745 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($7.23 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (7.23)^2 / (2 \cdot 123^2)) 262 / \\
 &\quad (5/3 + 3 \cdot (7.23) / (8 \cdot 123) - (7.23)^3 / (8 \cdot 123^3)) \\
 &= 154.9 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba} \\
 &= 19/155 + (721 \cdot 213/8967) / 175 \\
 &= 0.218
 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	90332.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1948.5 Kg
Maximum Transverse Force	Ft	5114.8 Kg
Maximum Longitudinal Force	F1	10229.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$\begin{aligned}
 &= FI / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 10230 / (4.87 \cdot 8) \\
 &= 25.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.
 \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$\begin{aligned}
 &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\
 &= 5115 / (4.87 \cdot 8) \\
 &= 12.9 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.
 \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $10230 / (82.8 \cdot 8)$
= 1.5145 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $1760 \cdot 5115 + 0$
= 9002.24 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $9002/90333$
= $99.65 \text{ mm} < B_{\text{plen}}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 1.51, 0]$
= 1.5145 cm^2

MDMT Calculations for: Left Saddle flo, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 16$, $c = 3 \text{ mm}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.81$, Temp. Reduction = $10 \text{ }^\circ\text{C}$
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-24 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1068.50	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	15260.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	680.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:



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Saddle Force Q, Operating Case 104426.08 Kgf
Pressure used in Analysis P 2220.341 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	44.90	138.00
Long. Stress at Bottom of Midspan	68.08	138.00
Long. Stress at Top of Saddles	126.93	138.00
Long. Stress at Bottom of Saddles	17.45	138.00
Tangential Shear in Shell	32.42	110.40
Circ. Stress at Horn of Saddle	69.81	172.50
Circ. Compressive Stress in Shell	3.19	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (2791/2 + 0) \cdot 1760/1897$$

$$= 3885.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(453, 0) \cdot 1760/9240$$

$$= 86.2 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl,eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(10230, 0) \cdot 1760/9240$$

$$= 1948.5 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (10230/2 + 0) \cdot 1760/1897$$

$$= 14239.3 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 90187 + \max(86.2, 3886, 1948, 14239)$$

$$= 104426.1 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.16 \cdot 1.2)^2)$$

$$= 4438.389 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 105516.90 Kgf

Transverse Shear Load 5114.81 Kgf

Longitudinal Shear Load 10229.61 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2110/2 + 24 + 16) \sin(120/2), 2 \cdot 1069)$$

$$= 1896.596 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434



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$$K7p = 0.0434$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\ &= -104426 \cdot 3000 [1 - (1 - 3000/15260 + (1069^2 - 0^2)/(2 \cdot 3000 \cdot 15260))/(1 + (4 \cdot 0)/(3 \cdot 15260))] \\ &= -57682.9 \text{ Kg-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L] \\ &= 104426 \cdot 15260/4 [(1 + 2(1069^2 - 0^2)/(15260^2))/(1 + (4 \cdot 0)/(3 \cdot 15260)) - 4 \cdot 3000/15260] \\ &= 89015.4 \text{ Kg-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t) \\ &= 2220 \cdot 1069/(2 \cdot 21) - 89015/(\pi \cdot 1069^2 \cdot 21) \\ &= 44.90 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t) \\ &= 2220 \cdot 1069/(2 \cdot 21) + 89015/(\pi \cdot 1069^2 \cdot 21) \\ &= 68.08 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned} &= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t) \\ &= 2220 \cdot 1069/(2 \cdot 21) - 57683/(0.11 \cdot \pi \cdot 1069^2 \cdot 21) \\ &= 126.93 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned} &= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t) \\ &= 2220 \cdot 1069/(2 \cdot 21) + 57683/(0.19 \cdot \pi \cdot 1069^2 \cdot 21) \\ &= 17.45 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} &= Q(L - 2a)/(L + (4 \cdot h_m/3)) \\ &= 104426 (15260 - 2 \cdot 3000)/(15260 + (4 \cdot 0/3)) \\ &= 63367.3 \text{ Kg} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned} &= K2 \cdot T / (R_m \cdot t) \\ &= 1.17 \cdot 63367/(1069 \cdot 21) \\ &= 32.42 \text{ N./mm}^2 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned} &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\ &= 0.78 \cdot \sqrt{(1069 \cdot 21)} \\ &= 116.840 \text{ mm} \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\ &= \min(425 + 1.56 \cdot \sqrt{(1069 \cdot 21)}, 2 \cdot 3000) \\ &= 658.68 \text{ mm} \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned} &= \min(S_r/S, 1) \\ &= \min(138/138, 1) \\ &= 1.0000 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot t_r))$$



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$$= -0.76 \cdot 104426 \cdot 0.1 / (659 (21 + 1 \cdot 16))$$

$$= -3.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_7, r]$:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -104426 / (4(21 + 1 \cdot 16)659) -$$

$$3 \cdot 0.053 \cdot 104426 / (2(21 + 1 \cdot 16)^2)$$

$$= -69.81 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw, c	695.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1058 \cdot 21) + 680})^3) / 12$$

$$= 70.424 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 70.4 + 192 (306 - 10.5)^2$$

$$= 167449.703 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	912.5	21.0	10.5	191.6	201212.6	0.167E+06
Wearplate	680.0	16.0	29.0	108.8	315520.0	0.835E+05
Web	14.0	615.0	344.5	86.1	2966143.2	0.284E+05
BasePlate	450.0	50.0	677.0	225.0	15232495.0	0.310E+06
Totals	...	...	...	611.5	18715370.0	0.589E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18715370 / 612$$

$$= 306.041 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 104426$$

$$= 21252.9727 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.9637 \text{ N./mm}^2$$



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Allowed Stress, $S_a = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$

$= 1760 - 1058 \cdot \sin(120/2) / 1.05$

$= 885.041 \text{ mm}$

Bending Moment, $M = F_h \cdot d = 18810.1211 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 9.5768 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(104426 + 1091) \cdot 450) / (4 \cdot 2070 \cdot 175)}$

$= 31.079 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120/2) (1055 + 24 + 16)$

$= 1896.596 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 1897 / (5 - 1)$

$= 474.149 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 474 \cdot 450 / 2$

$= 1066.835 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 105517 / (2070 \cdot 450)$

$= 11.327 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 1067 \cdot 11.3$

$= 12084.131 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 57.5 + 33.2$

$= 90.730 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 12084 / 90.7$

$= 13.061 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	425.0	...	59.5	...	0.896E+04

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 425 - 14$



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= 411.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/90.7

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 425/2

= 212.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{(8956/90.7)}$

= 99.353 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1760 - cos(120/2)(1055 + 24 + 16) - 50

= 1162.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/262)}$

= 122.739

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1162/99.4

= 11.701

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 10230 / (2 • 2070) • 474 • 1162/2

= 680.964 Kg-m.

Compressive Allowable, KL/r < Cc (11.7 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.7)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$

= 153.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 13.1/153 + (681 • 213/89559920) / 175

= 0.176

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	411.0	0.0	57.5	0.0	0.896E+04
Web	474.1	14.0	0.0	66.4	0.0	10.8
Totals	...	...	...	123.9	...	0.897E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/124

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 425/2



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Saddle Calcs: Operating Case: Step: 22 9:26am Sep 9,2026

= 212.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1760 - cos(1095/2)(50 + 0 + 0) - 0

= 615.000 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(8967/124)}$

= 85.063 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 615/85.1

= 7.230

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 10230/(2 • 2070)

= 2.471 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 2.47 • 474 • 615

= 720.504 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 721/421954

= 16.745 N./mm²

Compressive Allowable, KL/r < Cc (7.23 < 123) per AISC E2-1 [Sca]:

= $(1 - (Kl/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Kl/r) / (8 \cdot Cc) - (Kl/r)^3 / (8 \cdot Cc^3))$

= $(1 - (7.23)^2 / (2 \cdot 123^2)) 262 /$

$(5/3 + 3 \cdot (7.23) / (8 \cdot 123) - (7.23^3) / (8 \cdot 123^3))$

= 154.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 19.1/155 + (721 • 213/8967)/175

= 0.219

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	91277.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	1948.5 Kg
Maximum Transverse Force	Ft	5114.8 Kg
Maximum Longitudinal Force	F1	10229.6 Kg
Saddle Bolted to Steel Foundation	No	No

Friction force due to thermal growth, [Ff]:

= 27099.8 Kg

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:



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= max(Ff; F_I) / (Bolt Area • Number of Bolts)
 = 27100 / (4.87 • 8)
 = 68.3 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= F_t / (Bolt Area • Number of Bolts)
 = 5115 / (4.87 • 8)
 = 12.9 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= F_I / (BoltShearAllowable • Nbolts)
 = 10230 / (82.8 • 8)
 = 1.5145 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • F_t + Sum of X Moments
 = 1760 • 5115 + 0
 = 9002.24 Kg-m.

Eccentricity (e):

= Rmom / QO
 = 9002/91278
 = 98.62 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]
 = max[0, 1.51, 0]
 = 1.5145 cm²

MDMT Calculations for: RH fixed saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, t_g = 16, c = 3 mm, E* = 1

Thickness Ratio = t_r • E*/(t_g - c) = 0.81, Temp. Reduction = 10 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

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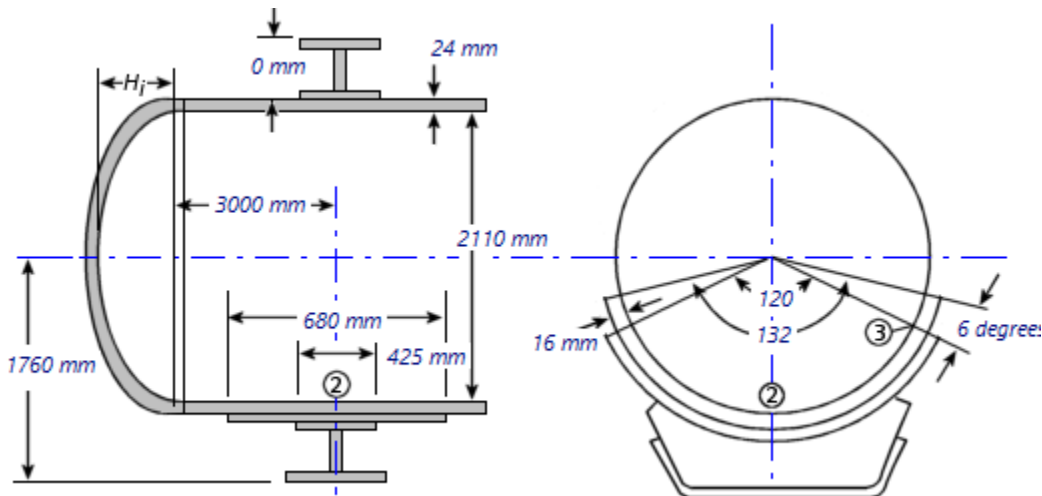
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1067.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	15260.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	680.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		86553.94	Kgf
Pressure used in Analysis	P	2883.458	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	55.67	234.00
Long. Stress at Bottom of Midspan	72.53	234.00
Long. Stress at Top of Saddles	115.34	234.00
Long. Stress at Bottom of Saddles	35.70	234.00
Tangential Shear in Shell	23.55	138.58
Circ. Stress at Horn of Saddle	49.92	350.99
Circ. Compressive Stress in Shell	2.39	234.00



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Saddle Calcs: Test Case:

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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1396/2 + 0) \cdot 1760/1897$$

$$= 1942.8 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(226, 0) \cdot 1760/9240$$

$$= 43.1 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 84611 + \max(43.1, 1943, 0, 0)$$

$$= 86553.9 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.16 \cdot 1.2)^2)$$

$$= 4438.389 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 87644.75 Kg

Transverse Shear Load 697.87 Kg

Longitudinal Shear Load 226.30 Kg

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2110/2 + 24 + 16) \sin(120/2), 2 \cdot 1067)$$

$$= 1896.596 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0529 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923 \quad K6p = 0.0434$$

$$K7p = 0.0434$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -86554 \cdot 3000 [1 - (1 - 3000/15260 + (1067^2 - 0^2)/(2 \cdot 3000 \cdot 15260))/(1 + (4 \cdot 0)/(3 \cdot 15260))]$$

$$= -47819.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 86554 \cdot 15260/4 ((1 + 2(1067^2 - 0^2)/(15260^2))/(1 + (4 \cdot 0)/(3 \cdot 15260)) - 4 \cdot 3000/15260)$$

$$= 73771.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 2883 \cdot 1067/(2 \cdot 24) - 73772/(\pi \cdot 1067^2 \cdot 24)$$

$$= 55.67 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$



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Saddle Calcs: Test Case:

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$$= 2883 \cdot 1067 / (2 \cdot 24) + 73772 / (\pi \cdot 1067^2 \cdot 24)$$

$$= 72.53 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 2883 \cdot 1067 / (2 \cdot 24) - 47820 / (0.11 \cdot \pi \cdot 1067^2 \cdot 24)$$

$$= 115.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 2883 \cdot 1067 / (2 \cdot 24) + 47820 / (0.19 \cdot \pi \cdot 1067^2 \cdot 24)$$

$$= 35.70 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 86554 \cdot (15260 - 2 \cdot 3000) / (15260 + (4 \cdot 0 / 3))$$

$$= 52522.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 52522 / (1067 \cdot 24)$$

$$= 23.55 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1067 \cdot 24)}$$

$$= 124.819 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(1067 \cdot 24)}, 2 \cdot 3000)$$

$$= 674.64 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 86554 \cdot 0.1 / (675(24 + 1 \cdot 16))$$

$$= -2.39 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -86554 / (4(24 + 1 \cdot 16)675) -$$

$$3 \cdot 0.053 \cdot 86554 / (2(24 + 1 \cdot 16)^2)$$

$$= -49.92 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	695.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1055 \cdot 24)} + 680)^3) / 12$$

$$= 106.932 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 107 + 223 (294 - 12)^2$$

$$= 177517.078 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	928.2	24.0	12.0	222.8	267330.6	0.178E+06
Wearplate	680.0	16.0	32.0	108.8	348160.0	0.748E+05
Web	14.0	615.0	347.5	86.1	2991973.5	0.296E+05
BasePlate	450.0	50.0	680.0	225.0	15299993.0	0.335E+06
Totals	...	...	...	642.7	18907458.0	0.617E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 18907458/643$$

$$= 294.199 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 86554$$

$$= 17615.6035 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.1142 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1760 - 1055 \cdot \sin(120/2) / 1.05$$

$$= 887.522 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 15634.5420 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 7.3074 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(86554 + 1091) \cdot 450) / (4 \cdot 2070 \cdot 173)}$$

$$= 28.434 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1055 + 24 + 16)$$

$$= 1896.596 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1897 / (5 - 1)$$

$$= 474.149 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 474 \cdot 450 / 2$$

$$= 1066.835 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 87645 / (2070 \cdot 450)$$

$$= 9.409 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1067 \cdot 9.41$$

$$= 10037.356 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 57.5 + 33.2$$

$$= 90.730 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 10037 / 90.7$$

$$= 10.849 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	425.0	...	59.5	...	0.896E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 425 - 14$$

$$= 411.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 90.7$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 425 / 2$$

$$= 212.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(Total \text{ Inertia} / Total \text{ Area})}$$

$$= \sqrt{(8956 / 90.7)}$$

$$= 99.353 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1760 - \cos(120 / 2) (1055 + 24 + 16) - 50$$

$$= 1162.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1162/99.4$$

$$= 11.701$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 226 / (2 \cdot 2070) \cdot 474 \cdot 1162/2$$

$$= 15.064 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$$

$$= 152.1 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 10.8/152 + (15.1 \cdot 213/89559920) / 173$$

$$= 0.073$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	411.0	0.0	57.5	0.0	0.896E+04
Web	474.1	14.0	0.0	66.4	0.0	10.8
Totals	...	...	...	123.9	...	0.897E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/124$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1760 - \cos(1095/2) (50 + 0 + 0) - 0$$

$$= 615.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (8967/124) }$$

$$= 85.063 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 615/85.1$$

$$= 7.230$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 226 / (2 \cdot 2070)$$

$$= 0.055 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.055 \cdot 474 \cdot 615$$

$$= 15.939 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 15.9/421954$$

$$= 0.370 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.23 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.23)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (7.23) / (8 \cdot 123) - (7.23^3) / (8 \cdot 123^3))$$

$$= 153.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 15.9/154 + (15.9 \cdot 213/8967) / 173$$

$$= 0.106$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load QO (Weight)	QO	85701.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	43.1 Kg
Maximum Transverse Force	Ft	697.9 Kg
Maximum Longitudinal Force	F1	226.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 226 / (4.87 \cdot 8)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 698 / (4.87 \cdot 8)$$

$$= 1.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 226 / (29 \cdot 8)$$

$$= 0.0957 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1760 \cdot 698 + 0$$

$$= 1228.28 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1228/85702$$

$$= 14.33 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.096, 0]

= 0.0957 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1067.00	mm
Shell Thickness used in this Case	t	24.000	mm
Stiffened Vessel Length per 4.15.6	L	15260.00	mm
Distance from Saddle to Vessel tangent	a1 or a	3000.00	mm
Saddle Width	b1 or b	425.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	680.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1760.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		87489.87	Kgf
Pressure used in Analysis	P	2883.458	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	55.58	234.00
Long. Stress at Bottom of Midspan	72.62	234.00
Long. Stress at Top of Saddles	115.90	234.00
Long. Stress at Bottom of Saddles	35.39	234.00
Tangential Shear in Shell	23.80	138.58
Circ. Stress at Horn of Saddle	50.46	350.99
Circ. Compressive Stress in Shell	2.42	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

= Ftr(Ft/Num of Saddles + Z Force Load) • B / E

= 3 (1396/2 + 0) • 1760/1897

= 1942.8 Kgf

Longitudinal Saddle Reaction Force [Fwl]:

= max(Fl,wind; Sum of X Forces) • B / Ls

= max(226, 0) • 1760/9240

= 43.1 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)

= 85547 + max(43.1, 1943, 0, 0)



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= 87489.9 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))

= 1.1 • 766 (0 + $\pi/4$ (2.16 • 1.2)²)

= 4438.389 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 88580.69 Kgf

Transverse Shear Load 697.87 Kgf

Longitudinal Shear Load 226.30 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)

= min(2(2110/2 + 24 + 16)sin(120/2), 2•1067)

= 1896.596 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405

K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434

K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]

= -87490•3000[1 - (1 - 3000/15260 + (1067²-0²)/

(2•3000•15260)/(1+(4•0)/(3•15260))]

= -48336.8 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²)/(1+(4hm²)/(3L))-4a/L)

= 87490•15260/4((1+2(1067²-0²)/(15260²)/(1+(4•0)/

(3•15260))-4•3000/15260)

= 74569.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)

= 2883 • 1067/(2•24) - 74569/(π •1067²•24)

= 55.58 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)

= 2883 • 1067/(2 • 24) + 74569/(π • 1067² • 24)

= 72.62 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)

= 2883•1067/(2•24)--48337/(0.11• π •1067²•24)

= 115.90 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)

= 2883•1067/(2•24)+-48337/(0.19• π •1067²•24)

= 35.39 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))

= 87490 (15260 - 2 • 3000)/(15260 + (4 • 0/3))

= 53090.2 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 53090 / (1067 \cdot 24)$$

$$= 23.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1067 \cdot 24)}$$

$$= 124.819 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(425 + 1.56 \cdot \sqrt{(1067 \cdot 24)}, 2 \cdot 3000)$$

$$= 674.64 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 87490 \cdot 0.1 / (675(24 + 1 \cdot 16))$$

$$= -2.42 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -87490 / (4(24 + 1 \cdot 16)675) -$$

$$3 \cdot 0.053 \cdot 87490 / (2(24 + 1 \cdot 16)^2)$$

$$= -50.46 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2070.1001	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	450.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	14.0000	mm
Web Thickness	Webtk	14.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	695.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1055 \cdot 24)} + 680)24^3) / 12$$

$$= 106.932 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 107 + 223(294 - 12)^2$$

$$= 177517.078 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	928.2	24.0	12.0	222.8	267330.6	0.178E+06
Wearplate	680.0	16.0	32.0	108.8	348160.0	0.748E+05
Web	14.0	615.0	347.5	86.1	2991973.5	0.296E+05
BasePlate	450.0	50.0	680.0	225.0	15299993.0	0.335E+06
Totals	...	...	...	642.7	18907458.0	0.617E+06

Distance to Centroid [C1]:



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$$\begin{aligned}
 &= AY / A \\
 &= 18907458/643 \\
 &= 294.199 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120)) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.2 \cdot 87490 \\
 &= 17806.0859 \text{ Kg}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 4.1586 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1760 - 1055 \cdot \sin(120/2) / 1.05 \\
 &= 887.522 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 15803.6025 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 7.3864 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(87490 + 1091) \cdot 450 / (4 \cdot 2070 \cdot 173)} \\
 &= 28.585 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 120/2) (1055 + 24 + 16) \\
 &= 1896.596 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1897 / (5 - 1) \\
 &= 474.149 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot B_{\text{pwid}} / 2 \\
 &= 474 \cdot 450/2 \\
 &= 1066.835 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 88581 / (2070 \cdot 450) \\
 &= 9.509 \text{ Kg/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 1067 \cdot 9.51 \\
 &= 10144.541 \text{ Kg}
 \end{aligned}$$

Area of the Rib and Web [Ar]:



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Saddle Calcs: Test Case:

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= Rib Area + Web Area

= 57.5 + 33.2

= 90.730 cm²

Compressive Stress [Sc]:

= P/Ar

= 10145/90.7

= 10.965 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	14.0	425.0	...	59.5	...	0.896E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 425 - 14

= 411.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/90.7

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 425/2

= 212.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{8956/90.7}$

= 99.353 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk

= 1760 - cos(120/2)(1055 + 24 + 16) - 50

= 1162.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1162/99.4

= 11.701

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 226 / (2 • 2070) • 474 • 1162/2

= 15.064 Kg-m.

Compressive Allowable, KL/r < Cc (11.7 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.7)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (11.7) / (8 \cdot 123) - (11.7^3) / (8 \cdot 123^3))$

= 152.1 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba



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Saddle Calcs: Test Case:

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$$= 11/152 + (15.1 \cdot 213/89559920)/173$$

$$= 0.074$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	14.0	411.0	0.0	57.5	0.0	0.896E+04
Web	474.1	14.0	0.0	66.4	0.0	10.8
Totals	...	...	...	123.9	...	0.897E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/124$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 425/2$$

$$= 212.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1760 - \cos(1095/2)(50 + 0 + 0) - 0$$

$$= 615.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(8967/124)}$$

$$= 85.063 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 615/85.1$$

$$= 7.230$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 226/(2 \cdot 2070)$$

$$= 0.055 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.055 \cdot 474 \cdot 615$$

$$= 15.939 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 15.9/421954$$

$$= 0.370 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($7.23 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.23)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (7.23) / (8 \cdot 123) - (7.23^3) / (8 \cdot 123^3))$$

$$= 153.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 16.1/154 + (15.9 \cdot 213/8967)/173$$

$$= 0.107$$

Input Data for Base Plate Bolting Calculations:



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Saddle Calcs: Test Case: Step: 23 9:26am Sep 9, 2026

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	6.1100 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.8662 cm ²
Saddle Load Q0 (Weight)	Q0	86637.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	43.1 Kg
Maximum Transverse Force	Ft	697.9 Kg
Maximum Longitudinal Force	F1	226.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 226 / (4.87 \cdot 8)$$

$$= 0.6 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 698 / (4.87 \cdot 8)$$

$$= 1.8 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 226 / (29 \cdot 8)$$

$$= 0.0957 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1760 \cdot 698 + 0$$

$$= 1228.28 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 1228 / 86638$$

$$= 14.18 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 0.096, 0]$$

$$= 0.0957 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N3-Tube Inlet

Nozl: 9

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Input, Nozzle Desc: N3-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2110.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	2500.0000	mm
Distance from Bottom/Left Tangent		1885.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		62.9921	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	2300.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		350.0004	mm
Flange Class		150	
Flange Grade		GR 1.1	



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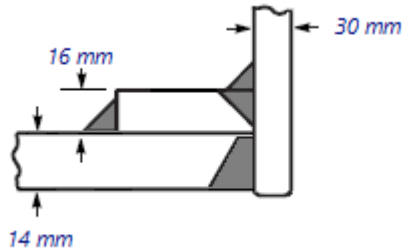
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Nozzle Calcs.: N3-Tube Inlet

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The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N3-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	62.992 in.
Actual Thickness Used in Calculation	1.181 in.

Messages regarding the Nozzle Input Data:

This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 1056 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 7.6862 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1000 \cdot 800 / (138 \cdot 1 + 0.4 \cdot 1000)$
 $= 5.7808 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 3.4092 mm

UG-40, Limits of Reinforcement: [External Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2500.0000 mm
Parallel to Vessel Wall, opening length	d	1250.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 6.7808 mm
Wall Thickness per UG16(b),	tr16b	= 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.6862 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.6862 mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7918 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000 mm
Wall Thickness per table UG-45	tb3	= 9.3312 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.33, \max(8.69, 3.5)]$
 $= 8.6862 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(6.78, 8.69)$
 $= 8.6862 \text{ mm}$



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Nozzle Calcs.: N3-Tube Inlet

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Available Nozzle Neck Thickness = 30.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.4,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	323.6 N./mm ²	Passed
Occasional	:	13.0,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	6.3,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 30, tr = 5.78, c = 1 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 16, c = 1 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.59$, Temp. Reduction = 23 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16, c = 1 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.59$, Temp. Reduction = 23 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14, tr = 7.69, c = 1 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.59$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14, tr = 7.69, c = 1 mm, E* = 1
Thickness Ratio = $tr * E* / (tg - c) = 0.59$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C



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Nozzle Calcs.: N3-Tube Inlet

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Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1000.00/1960.01 = 0.510

Warning: $Rn/R(0.73) > 0.7$ - Performing analysis per Appendix 46-3(a)(1).

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 29 / \ln[1 + 29/771]$$

$$= 785.410 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 29 / \ln[1 + 29/1056]$$

$$= 1070.434 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 138$$

$$= 207.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 103 \cdot 785 \cdot 179$$

$$= 1478.8 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 103 \cdot 1070 (176 + 29)$$

$$= 2315.1 \text{ Kg}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 103 \cdot 1070 \cdot 771$$

$$= 8703.8 \text{ Kg}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2315 + 8704 + 1479) / 103$$

$$= 11850.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 207 / (2 \cdot 11850 / 108 - 1070 / 29)$$

$$= 1138.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 138 [13 / 1070]$$

$$= 1675.8 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(1138, 1676)$$



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Nozzle Calcs.: N3-Tube Inlet

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= 1137.959 KPa.

Average Primary Membrane Stress [σ_{Avg}]:

= ($f_N + f_S + f_Y$) / AT

= (1479 + 2315 + 8704) / 108

= 11.315 N./mm²

General Primary Membrane Stress [σ_{Circ}]:

= P • Rxs / teff

= 103 • 1070 / 29

= 3.8 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})

= max(2 • 11.3 - 3.82, 3.82)

= 18.8 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	18.81	N./mm ²
Maximum Allowable External Pressure	Pmax	1137.96	KPa.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

= min[Sn/S, 1]

= min[138/138, 1]

= 1.000

Pad Material Factor [frp]:

= min[Sp/S, 1]

= min[138/138, 1]

= 1.000

Thickness of Nozzle [tn]:

= thickness - corrosion allowance

= 30 - 1

= 29.000 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t

= 2112/13

= 162.462 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance

= 2110/2 + 1

= 1056.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t) + W}$

= $\sqrt{(1056 \cdot 13) + 350}$

= 467.167 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$

= $\sqrt{((1056 + 13) \cdot (13 + 16))}$

= 176.071 mm



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Nozzle Calcs.: N3-Tube Inlet

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Limit [LR3]:

$$\begin{aligned}
 &= 2 \cdot R_n \\
 &= 2 \cdot 771 \\
 &= 1541.999 \text{ mm}
 \end{aligned}$$

Final value of [LR]:

$$\begin{aligned}
 &= \min(LR1, LR2, LR3) \\
 &= \min(467, 176, 1542) \\
 &= 176.071 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 13, 16] + \sqrt{(771 \cdot 29)} \\
 &= 165.529 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= L_{pr1} \\
 &= 600 \\
 &= 600.000 \text{ mm}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + t_e) \\
 &= 8(13 + 16) \\
 &= 232.000 \text{ mm}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[166, 600, 232] + 13 \\
 &= 178.529 \text{ mm}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 13 + ((28.2 \cdot 1) / 176) \\
 &= 29.000 \text{ mm}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 771 + 29) / (\sqrt{(2112 + 29) \cdot 29})) \\
 &= 6.305
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((N/5)^{0.85}, 1) \\
 &= 13 \cdot 176 \cdot \max((6.3/5)^{0.85}, 1) \\
 &= 27.876 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 29 \cdot 179 \\
 &= 51.773 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned}
 &= \min(W \cdot t_e, LR \cdot t_e) \\
 &= \min(350 \cdot 16, 176 \cdot 16) \\
 &= 28.171 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5)$$



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Nozzle Calcs.: N3-Tube Inlet

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$$= 27.9 + 1(51.8 + 0) + 0.5 + 0 + 0 + 1(28.2)$$

$$= 108.321 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 29 / \ln[1 + 29/771]$$

$$= 785.410 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 29 / \ln[1 + 29/1056]$$

$$= 1070.434 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 138 \cdot 1$$

$$= 207.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1000 \cdot 785 \cdot 179$$

$$= 14298.9 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1000 \cdot 1070 (176 + 29)$$

$$= 22385.2 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1000 \cdot 1070 \cdot 771$$

$$= 84161.0 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (22385 + 84161 + 14299) / 1000$$

$$= 11850.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 207 / (2 \cdot 11850 / 108 - 1070 / 29)$$

$$= 1138.0 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$

$$= 138 [13/1070]$$

$$= 1675.8 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(1138, 1676)$$

$$= 1137.959 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (14299 + 22385 + 84161) / 108$$

$$= 109.407 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1000 \cdot 1070 / 29$$

$$= 36.9 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$



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$$= \max(2 \cdot 109 - 36.9, 36.9)$$

$$= 181.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	181.90	N./mm ²
Maximum Allowable Working Pressure	Pmax	1137.96	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (771 + 29) / 771$$

$$= 1.038 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (771 + 29)$$

$$= 1256.637 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_{TP}]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (771 + 29 + 350)$$

$$= 1806.416 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 29.000, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 13.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(84161 \cdot 1.04, 1.5 \cdot 138(51.8 + 0), \pi/4 \cdot 1000 \cdot 771^2 \cdot 1.04^2)$$

$$= 51258.516 \text{ Kg}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 51259 \cdot 13 \cdot 138 / (13 \cdot 138 + 16 \cdot 138)$$

$$= 22977.957 \text{ Kg}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 51259 \cdot 16 \cdot 138 / (13 \cdot 138 + 16 \cdot 138)$$

$$= 28280.561 \text{ Kg}$$

Shear Stress [τ₁]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 22978 / (1257 \cdot (0.6 \cdot 13 + 0.49 \cdot 0))$$

$$= 22.990 \text{ N./mm}^2$$

Shear Stress [τ₂]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L41T))$$

$$= 28281 / (1257 \cdot (0.6 \cdot 16 + 0.49 \cdot 7.07))$$

$$= 16.893 \text{ N./mm}^2$$

Shear Stress [τ₃]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L42T))$$

$$= 28281 / (1806 \cdot (0.49 \cdot 8.49))$$

$$= 36.927 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(23, 16.9, 36.9)$$



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Nozzle Calcs.: N3-Tube Inlet Nozl: 9 9:26am Sep 9,2026

= 36.9 must be less than or equal to 138.0 N./mm²

Weld Size Calculations, Description: N3-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000	mm
Intermediate Calc. for pad/shell Welds	TminPad	15.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness	
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo	mm
Pad Weld	7.5000 = 0.5*TminPad	8.4840 = 0.7 * Wp	mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1000 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 367.2314 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 981.2314 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.911 %

Input Echo, WRC297 Item 1, Description: N3-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2110.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	1599.999	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	2300.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-12237.40	Kgf
Circumferential Shear	Vc	8158.27	Kgf
Longitudinal Shear	Vl	8158.27	Kgf
Circumferential Moment	Mc	36534.93	Kg-m.
Longitudinal Moment	Ml	36534.93	Kg-m.



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Nozzle Calcs.: N3-Tube Inlet

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Torsional Moment Mt 52192.88 Kg-m.

Include Axial Pressure Thrust No

Include Pressure Stress Indices per Div. 2 No

Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Dv (.748) is > 0.5
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 1600.0 + 2 \cdot 1.65 \cdot \sqrt{1070.5(14.0 - 1.0)}$$

$$= 1989.294 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2141.000 mm

Nozzle Outside Diameter (d) = 1599.999 mm

Vessel Thickness used (T) = 29.000 mm

Nozzle Thickness used (t) = 29.000 mm

T / t = 1.000

d / t = 55.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.421$$

Nr/P = 0.026

Mr/P = 0.071

M0/P = 0.022

N0/P = 0.136

MrD/Mc = 0.194

NrDL/Mc = 0.053

M0d/Mc = 0.067

N0DL/Mc = 0.131

MrD/Ml = 0.051

NrDL/Ml = 0.019

M0D/Ml = 0.016

N0DL/Ml = 0.100

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	19	19	19	19
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-26	-26	26	26
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	35	36	35	36
Outplane Stress Summary	21	36	126	38

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5



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Inplane Bending (ML)	-81	81	81	-81
Inplane Pressure Stress	17	17	17	17

Inplane Stress Summary	-4	37	168	-115
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	5	5	3	3
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	28	42	169	153
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	19	19	19	19
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-34	-34	34	34
Outplane Bending (Mc)	-107	107	107	-107
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17

Outplane Stress Summary	-85	90	197	-53
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-309	309	309	-309
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	35	36	35	36

Inplane Stress Summary	-223	275	423	-314
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	3	3	5	5
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Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	223	275	423	315
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	19	19	19	19
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-26	-26	26	26
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	32	31	32

Hoop Stress Summary	24	25	77	78
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-66	66	66	-66
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	-6	26	138	-92
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4

Shear Stress Summary	5	5	3	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	32	31	138	171
------------------------	----	----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	19	19	19	19
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-34	-34	34	34
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	32	31	32



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Hoop Stress Summary	15	17	85	86
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	49	-49	49	-49
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-266	266	266	-266
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	-207	227	339	-293
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4

Shear Stress Summary	3	3	5	5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	223	227	339	379

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 1600.0 + 2 \cdot 1.65 \cdot \sqrt{(1062.5(14.0 - 1.0))}$$

$$= 1987.837 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2125.000 mm
Nozzle Outside Diameter	(d) =	2300.000 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	379.000 mm
T / t	=	0.034
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 13.838$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.071
M0/P	=	0.018
N0/P	=	0.020
MrD/Mc	=	0.237
NrDL/Mc	=	0.093
M0d/Mc	=	0.065
N0DL/Mc	=	0.037
MrD/Ml	=	0.034
NrDL/Ml	=	0.034
M0D/Ml	=	0.011



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NØDL/Ml = 0.015

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	78	-78	78	-78
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-61	61	61	-61
Normal Pressure Stress	80	81	80	81
Outplane Stress Summary	97	64	248	-29

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	32	32	32	32
Inplane Bending (P)	304	-304	304	-304
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-31	-31	31	31
Inplane Bending (ML)	-186	186	186	-186
Inplane Pressure Stress	40	40	40	40
Inplane Stress Summary	159	-76	595	-385

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	160	141	595	385

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	78	-78	78	-78
Outplane Membrane (Mc)	-34	-34	34	34
Outplane Bending (Mc)	-361	361	361	-361
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	40	40	40	40
Outplane Stress Summary	-262	303	528	-351



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	32	32	32	32
Inplane Bending (P)	304	-304	304	-304
Inplane Membrane (Mc)	-85	-85	85	85
Inplane Bending (Mc)	-1309	1309	1309	-1309
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	80	81	80	81
Inplane Stress Summary	-977	1034	1813	-1414

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	977	1034	1813	1414

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	35	36	35	36	35	36	35	36
Circ. Pl (SUS)	-7	-7	45	45	-11	-11	17	17
Circ. Q (SUS)	-7	7	43	-43	-249	249	369	-369
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-2	-2	8	8	-15	-15	53	53
Long. Q (SUS)	-21	21	141	-141	-89	89	125	-125
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	35.0	36.0	35.0	36.0	35.0	36.0	35.0	36.0
Pm+Pl (SUS)	28.1	29.1	80.0	81.0	24.0	25.0	70.1	70.1
Pm+Pl+Q (Total)	28.8	41.0	166.2	154.1	225.1	274.0	421.1	316.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Nozzle Calcs.: N3-Tube Inlet

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Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	81.02	207.00	Passed
Pm+Pl+Q (TOTAL)	421.11	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	80	81	80	81	80	81	80	81
Circ. Pl (SUS)	0	0	28	28	-53	-53	117	117
Circ. Q (SUS)	17	-17	139	-139	-1005	1005	1613	-1613
Long. Pm (SUS)	40	40	40	40	40	40	40	40
Long. Pl (SUS)	1	1	63	63	-20	-20	48	48
Long. Q (SUS)	118	-118	490	-490	-283	283	439	-439
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	80.0	81.0	80.0	81.0	80.0	81.0	80.0	81.0
Pm+Pl (SUS)	80.0	81.0	108.2	109.2	27.1	28.1	197.0	198.0
Pm+Pl+Q (Total)	159	141	593	387	978	1033	1810	1415

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	81.00	138.00	Passed
Pm+Pl (SUS)	198.01	207.00	Passed
Pm+Pl+Q (TOTAL)	1810.02	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	-7	-7	45	45	-15	-15	53	53
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-17	17	115	-115	-217	217	315	-315
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0



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Nozzle Calcs.: N3-Tube Inlet

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Pm+Pl (SUS)	24.1	25.1	76.0	77.0	16.1	17.1	84.0	85.0
Pm+Pl+Q (Total)	33.5	30.5	136.1	171.1	224.1	226.0	336.1	379.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	138.00	Passed
Pm+Pl (SUS)	85.02	207.00	Passed
Pm+Pl+Q (TOTAL)	379.13	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Nozzle Calcs.: N1-Shell Inlet

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Input, Nozzle Desc: N1-Shell Inlet

From: 60

Pressure for Reinforcement Calculations	P	2210.000	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2110.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5771.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0157	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	55.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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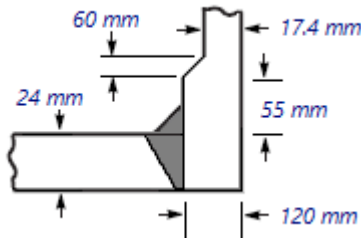
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Nozzle Calcs.: N1-Shell Inlet

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Reinforcement CALCULATION, Description: N1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.016 in.
Actual Thickness Used in Calculation	0.687 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2210 \cdot 1058 / (138 \cdot 1 - 0.6 \cdot 2210)$
 $= 17.1090 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2210 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 2210)$
 $= 4.8537 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 103 / 0.5773$
 $= 177.6223 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1162.1984	mm
Parallel to Vessel Wall, opening length	d	581.0992	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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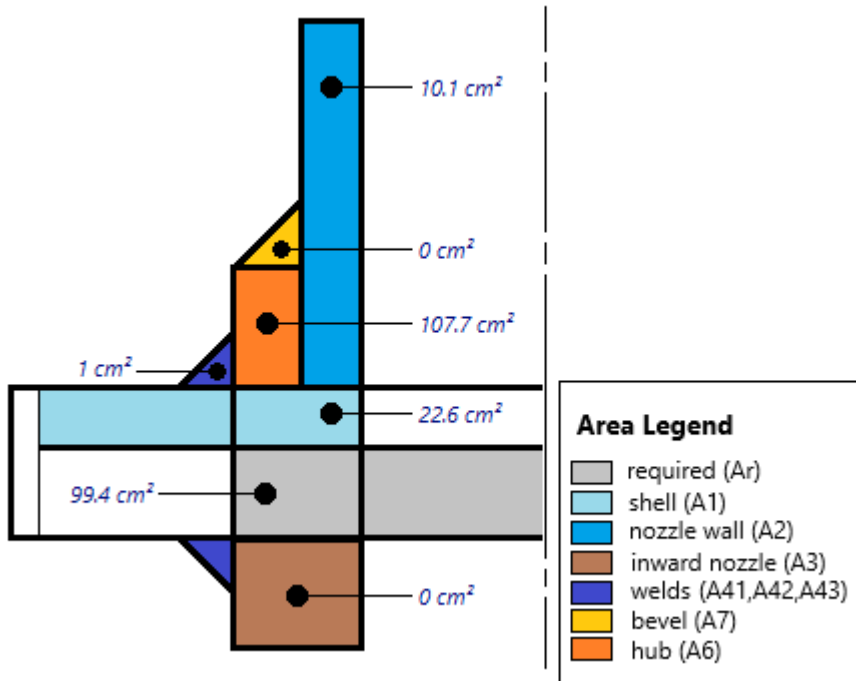
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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	99.420	NA	NA
Area in Shell	A1	22.610	NA	NA
Area in Nozzle Wall	A2	10.076	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	107.678	NA	NA
TOTAL AREA AVAILABLE	Atot	141.364	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (581 \cdot 17.1 \cdot 1 + 2 \cdot 117 \cdot 17.1 \cdot 1 \cdot (1 - 1))$$

$$= 99.420 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 581(1 \cdot 21 - 1 \cdot 17.1) - 2 \cdot 117$$

$$(1 \cdot 21 - 1 \cdot 17.1) \cdot (1 - 1)$$

$$= 22.610 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 52.5)(14.4 - 4.85)1$$

$$= 10.076 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr2 + (W_i - c_n / 0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(52.5, 350.55)) \cdot (120 - 17.4) \cdot 1$$

$$= 107.678 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 7.8537 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 20.1090 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 20.1090 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(20.1, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(7.85, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	38.1,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	306.9 N./mm ²	Passed
Occasional	:	21.7,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	12.6,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 17.4$, $t_r = 4.85$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.34$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 24$, $t_r = 17.1$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.81$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 2210.00/5100.00 = 0.433

Weld Size Calculations, Description: N1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (99.4 - 22.6 + 2 \cdot 117 \cdot 1 \cdot (1 \cdot 21 - 17.1)) \cdot 138)$
 $= 120895.78 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (118 + 1 - 0 \cdot 1) \cdot 138$
 $= 167104.98 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (118 + 0 + 1 + (49.1)) \cdot 138$
 $= 236252.69 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (118 + 0 + 1 + 0 + (49.1)) \cdot 138$
 $= 236252.69 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 66067. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 298) \cdot (120 - 3) \cdot 0.7 \cdot 138$
 $= 1078112. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (24 - 3) \cdot 0.74 \cdot 138$
 $= 209528. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (66067 + 1078111) = 1144179 \text{ Kgf}$



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Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (66067 + 0 + 209528 + 0) = 275596 Kgf

Failure Path [Path 3-3]:

= (Sonw + Tngw + Sinw)

= (66067 + 209528 + 0) = 275596 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 1144179 Kgf, must exceed W = 120895 Kgf or W1 = 167104 Kgf

Path 2-2 = 275595 Kgf, must exceed W = 120895 Kgf or W2 = 236252 Kgf

Path 3-3 = 275595 Kgf, must exceed W = 120895 Kgf or W3 = 236252 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2210 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 45.0493 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 419.0493 mm

Input Echo, WRC297 Item 1, Description: N1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2110.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	815.099	mm
Nozzle Thickness	120.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2210.00	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.792) is < 10



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check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 815.099 + 2 \cdot 1.65 \cdot \sqrt{(1068.5(24.0 - 3.0))}$$

$$= 1309.422 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2137.000 mm
 Nozzle Outside Diameter (d) = 815.099 mm
 Vessel Thickness used (T) = 21.000 mm
 Nozzle Thickness used (t) = 117.000 mm
 T / t = 0.179
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.848$$

Nr/P = 0.050
 Mr/P = 0.085
 M0/P = 0.026
 N0/P = 0.050
 MrD/Mc = 0.256
 NrDL/Mc = 0.093
 M0d/Mc = 0.077
 N0DL/Mc = 0.080
 MrD/M1 = 0.068
 NrDL/M1 = 0.052
 M0D/M1 = 0.020
 N0DL/M1 = 0.054

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	110	112	110	112
Outplane Stress Summary	106	111	155	92

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	55	55	55	55
Inplane Stress Summary	48	57	176	-41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress		106	111	177	133

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)		5	5	5	5
Outplane Bending (P)		15	-15	15	-15
Outplane Membrane (Mc)		-11	-11	11	11
Outplane Bending (Mc)		-64	64	64	-64
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		55	55	55	55
Outplane Stress Summary		0	97	151	-8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		5	5	5	5
Inplane Bending (P)		52	-52	52	-52
Inplane Membrane (Mc)		-12	-12	12	12
Inplane Bending (Mc)		-215	215	215	-215
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		110	112	110	112
Inplane Stress Summary		-60	267	395	-136

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3
Shear Stress Summary		2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		61	267	395	137

Nozzle Stresses (N./mm²):



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LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8
Hoop Stress Summary	2	5	18	20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	1	1	6	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	5	18	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8
Hoop Stress Summary	0	2	21	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-2	6	11	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2	6	21	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	110	112	110	112	110	112	110	112
Circ. Pl (SUS)	-2	-2	12	12	-7	-7	17	17
Circ. Q (SUS)	-2	2	32	-32	-163	163	267	-267
Long. Pm (SUS)	55	55	55	55	55	55	55	55
Long. Pl (SUS)	-2	-2	12	12	-6	-6	16	16
Long. Q (SUS)	-5	5	109	-109	-49	49	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	110.0	112.0	110.0	112.0	110.0	112.0	110.0	112.0
Pm+Pl (SUS)	108.0	110.0	122.0	124.0	103.0	105.0	127.0	129.0
Pm+Pl+Q (Total)	106.3	112.3	176.2	134.1	60.1	268.0	394.1	138.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	112.00	138.00	Passed
Pm+Pl (SUS)	129.02	207.00	Passed
Pm+Pl+Q (TOTAL)	394.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	8	5	8	5	8	5	8
Circ. Pl (SUS)	-2	-2	12	12	-6	-6	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	8.0	5.0	8.0	5.0	8.0	5.0	8.0
Pm+Pl (SUS)	3.0	6.0	17.0	20.0	2.0	2.0	21.0	24.0
Pm+Pl+Q (Total)	3.0	6.0	17.0	20.0	3.0	5.0	21.0	27.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	24.00	207.00	Passed
Pm+Pl+Q (TOTAL)	27.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N2-Shell Outlet

From: 60

Pressure for Reinforcement Calculations	P	2230.682	KPa.
Temperature for Internal Pressure	Temp	150	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2110.00	mm
Shell Finished (Minimum) Thickness	t	24.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		17041.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0157	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	55.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		120.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		22.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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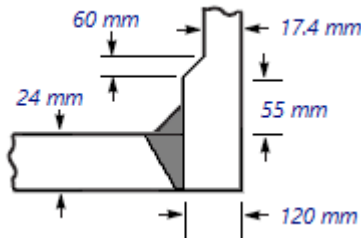
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Reinforcement CALCULATION, Description: N2-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.016 in.
Actual Thickness Used in Calculation	0.687 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2231 \cdot 1058 / (138 \cdot 1 - 0.6 \cdot 2231)$
 $= 17.2707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2231 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 2231)$
 $= 4.8988 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 103 / 0.5773$
 $= 177.6223 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1162.1984	mm
Parallel to Vessel Wall, opening length	d	581.0992	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	52.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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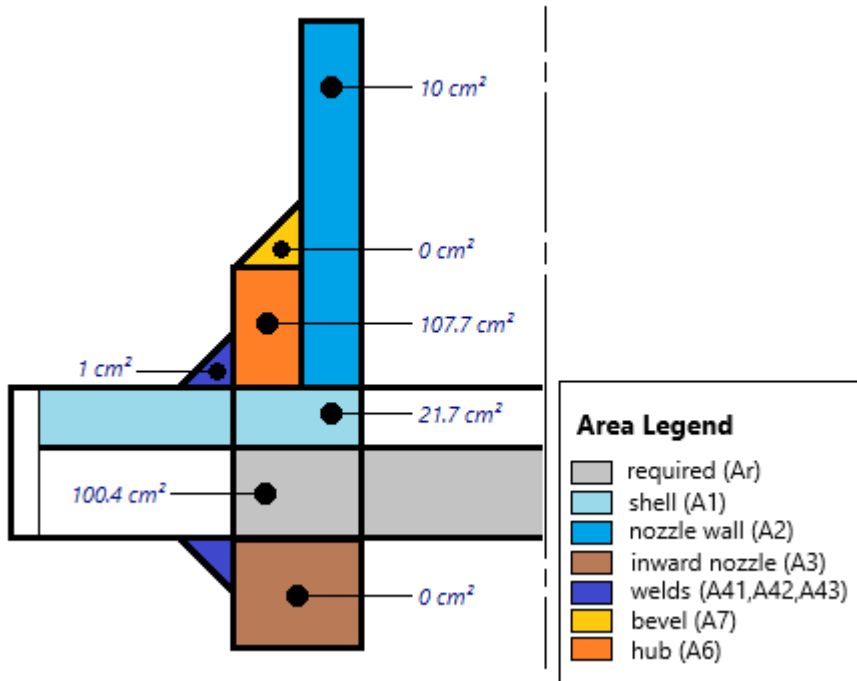
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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	100.360	NA	NA
Area in Shell	A1	21.671	NA	NA
Area in Nozzle Wall	A2	10.029	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	107.678	NA	NA
TOTAL AREA AVAILABLE	Atot	140.377	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (581 \cdot 17.3 \cdot 1 + 2 \cdot 117 \cdot 17.3 \cdot 1 \cdot (1 - 1))$$

$$= 100.360 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 581(1 \cdot 21 - 1 \cdot 17.3) - 2 \cdot 117$$

$$(1 \cdot 21 - 1 \cdot 17.3) \cdot (1 - 1)$$

$$= 21.671 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 52.5)(14.4 - 4.9)1$$

$$= 10.029 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - c_n / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(52.5, 350.55)) \cdot (120 - 17.4) \cdot 1$$

$$= 107.678 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 7.8988 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 20.2707 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 20.2707 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(20.3, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(7.9, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	38.3,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	306.7 N./mm ²	Passed
Occasional	:	21.9,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	12.6,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 17.4$, $t_r = 4.9$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.34$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 24$, $t_r = 17.3$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.82$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -2 °C

Min Metal Temp. at Required thickness (UCS 66.1) -12 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 2230.68/5100.00 = 0.437

Weld Size Calculations, Description: N2-Shell Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (100 - 21.7 + 2 • 117 • 1 •
(1 • 21 - 17.3))138)
= 123007.57 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (118 + 1 - 0 • 1) • 138
= 167038.31 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (118 + 0 + 1 + (49.1)) • 138
= 236186.02 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (118 + 0 + 1 + 0 + (49.1)) • 138
= 236186.02 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 610 • 10 • 0.49 • 138
= 66067. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 298) • (120 - 3) • 0.7 • 138
= 1078112. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 610 • (24 - 3) • 0.74 • 138
= 209528. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (66067 + 1078111) = 1144179 Kgf



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Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (66067 + 0 + 209528 + 0) = 275596 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (66067 + 209528 + 0) = 275596 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 1144179 Kgf, must exceed W = 123007 Kgf or W1 = 167038 Kgf
 Path 2-2 = 275595 Kgf, must exceed W = 123007 Kgf or W2 = 236186 Kgf
 Path 3-3 = 275595 Kgf, must exceed W = 123007 Kgf or W3 = 236186 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2210 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 45.0493 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 419.0493 mm

Input Echo, WRC297 Item 1, Description: N2-Shell Outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2110.000	mm
Shell Thickness	24.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	815.099	mm
Nozzle Thickness	120.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2230.68	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (6.792) is < 10



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check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 815.099 + 2 \cdot 1.65 \cdot \sqrt{(1068.5(24.0 - 3.0))}$$

$$= 1309.422 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2137.000 mm
 Nozzle Outside Diameter (d) = 815.099 mm
 Vessel Thickness used (T) = 21.000 mm
 Nozzle Thickness used (t) = 117.000 mm
 T / t = 0.179
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.848$$

Nr/P = 0.050
 Mr/P = 0.085
 M0/P = 0.026
 N0/P = 0.050
 MrD/Mc = 0.256
 NrDL/Mc = 0.093
 M0d/Mc = 0.077
 N0DL/Mc = 0.080
 MrD/Ml = 0.068
 NrDL/Ml = 0.052
 M0D/Ml = 0.020
 N0DL/Ml = 0.054

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	111	113	111	113
Outplane Stress Summary	107	112	157	93

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	55	55	55	55
Inplane Stress Summary	48	58	177	-41

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress		107	112	177	134

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)		5	5	5	5
Outplane Bending (P)		15	-15	15	-15
Outplane Membrane (Mc)		-11	-11	11	11
Outplane Bending (Mc)		-64	64	64	-64
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		55	55	55	55
Outplane Stress Summary		1	98	152	-8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		5	5	5	5
Inplane Bending (P)		52	-52	52	-52
Inplane Membrane (Mc)		-12	-12	12	12
Inplane Bending (Mc)		-215	215	215	-215
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		111	113	111	113
Inplane Stress Summary		-59	268	396	-135

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3
Shear Stress Summary		2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		60	268	396	135

Nozzle Stresses (N./mm²):



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LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8
Hoop Stress Summary	3	5	18	20

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	1	1	6	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	3	5	18	20

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	8	5	8
Hoop Stress Summary	0	2	21	24

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Nozzle Calcs.: N2-Shell Outlet Nozl: 11 9:26am Sep 9,2026

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-2	6	11	-2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	2	6	21	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	111	113	111	113	111	113	111	113
Circ. Pl (SUS)	-2	-2	12	12	-7	-7	17	17
Circ. Q (SUS)	-2	2	32	-32	-163	163	267	-267
Long. Pm (SUS)	55	55	55	55	55	55	55	55
Long. Pl (SUS)	-2	-2	12	12	-6	-6	16	16
Long. Q (SUS)	-5	5	109	-109	-49	49	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	111.0	113.0	111.0	113.0	111.0	113.0	111.0	113.0
Pm+Pl (SUS)	109.0	111.0	123.0	125.0	104.0	106.0	128.0	130.0
Pm+Pl+Q (Total)	107.3	113.3	176.2	135.1	59.1	269.0	395.1	137.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	113.00	138.00	Passed
Pm+Pl (SUS)	130.02	207.00	Passed
Pm+Pl+Q (TOTAL)	395.07	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



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Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	8	5	8	5	8	5	8
Circ. Pl (SUS)	-2	-2	12	12	-6	-6	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	0	0	2	-2	-4	4	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	8.0	5.0	8.0	5.0	8.0	5.0	8.0
Pm+Pl (SUS)	3.0	6.0	17.0	20.0	2.0	2.0	21.0	24.0
Pm+Pl+Q (Total)	3.0	6.0	17.0	20.0	3.0	5.0	21.0	27.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	8.00	138.00	Passed
Pm+Pl (SUS)	24.00	207.00	Passed
Pm+Pl+Q (TOTAL)	27.00	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N4-Tube Outlet

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Input, Nozzle Desc: N4-Tube Outlet

From: 80

Pressure for Reinforcement Calculations	P	1020.682	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2110.00	mm
Design Length of Section	L	3000.0000	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	2500.0000	mm
Distance from Bottom/Left Tangent		20917.52	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		62.9921	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	30.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	2300.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		350.0004	mm
Flange Class		150	
Flange Grade		GR 1.1	



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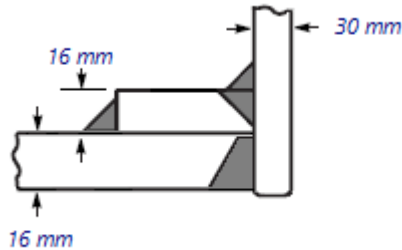
Nozzle Calcs.: N4-Tube Outlet

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The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N4-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	62.992 in.
Actual Thickness Used in Calculation	1.181 in.

Messages regarding the Nozzle Input Data:

This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1021 \cdot 1056 / (138 \cdot 1 - 0.6 \cdot 1021)$
 $= 7.8459 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1021 \cdot 800 / (138 \cdot 1 + 0.4 \cdot 1021)$
 $= 5.9000 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 3.4092 mm

UG-40, Limits of Reinforcement: [External Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	2500.0000 mm
Parallel to Vessel Wall, opening length	d	1250.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 6.9000 mm
Wall Thickness per UG16(b),	tr16b	= 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.8459 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.8459 mm
Wall Thickness, shell/head, external pressure	trb2	= 1.7918 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000 mm
Wall Thickness per table UG-45	tb3	= 9.3312 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.33, \max(8.85, 3.5)]$
 $= 8.8459 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(6.9, 8.85)$
 $= 8.8459 \text{ mm}$



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Nozzle Calcs.: N4-Tube Outlet

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Available Nozzle Neck Thickness = 30.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	21.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	323.3 N./mm ²	Passed
Occasional	:	13.3,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	6.3,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 30, tr = 5.9, c = 1 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.2$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	24 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 16, c = 1 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16, c = 1 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16, tr = 7.85, c = 1 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16, tr = 7.85, c = 1 mm, E* = 1

Thickness Ratio = $tr * E^* / (tg - c) = 0.52$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C



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Nozzle Calcs.: N4-Tube Outlet

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Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1020.68/1960.01 = 0.521$

Warning: $Rn/R(0.73) > 0.7$ - Performing analysis per Appendix 46-3(a)(1).

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$= t_n / \ln[1 + t_n/R_n]$

$= 29 / \ln[1 + 29/771]$

$= 785.410 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:

$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$

$= 31 / \ln[1 + 31/1056]$

$= 1071.427 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:

$= 1.5 \cdot S$

$= 1.5 \cdot 138$

$= 207.0 \text{ N./mm}^2$

Force acting on the Nozzle [fN]:

$= P \cdot R_{xn} \cdot L_H$

$= 103 \cdot 785 \cdot 181$

$= 1495.3 \text{ Kg}$

Force acting on the Shell [fS]:

$= P \cdot R_{xs} (L_R + t_n)$

$= 103 \cdot 1071 (182 + 29)$

$= 2386.6 \text{ Kg}$

Discontinuity Force from External Pressure [fY]:

$= P \cdot R_{xs} \cdot R_{nc}$

$= 103 \cdot 1071 \cdot 771$

$= 8711.9 \text{ Kg}$

Area Resisting External Pressure [Ap]:

$= (f_S + f_Y + f_N)/P$

$= (2387 + 8712 + 1495)/103$

$= 11941.6 \text{ cm}^2$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$

$= 207 / (2 \cdot 11942/114 - 1071/31)$

$= 1187.5 \text{ KPa.}$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$= S / (t / R_{xs})$

$= 138 [15/1071]$

$= 1931.9 \text{ KPa.}$

Maximum Allowable Working Pressure [Pmax]:

$= \min(P_{max1}, P_{max2})$

$= \min(1188, 1932)$



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Nozzle Calcs.: N4-Tube Outlet

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= 1187.534 KPa.

Average Primary Membrane Stress [σ_{Avg}]:

= ($f_N + f_S + f_Y$) / AT

= (1495 + 2387 + 8712) / 114

= 10.801 N./mm²

General Primary Membrane Stress [σ_{Circ}]:

= P • Rxs / teff

= 103 • 1071 / 31

= 3.6 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})

= max(2 • 10.8 - 3.57, 3.57)

= 18.0 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	18.03	N./mm ²
Maximum Allowable External Pressure	Pmax	1187.53	KPa.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

= min[Sn/S, 1]

= min[138/138, 1]

= 1.000

Pad Material Factor [frp]:

= min[Sp/S, 1]

= min[138/138, 1]

= 1.000

Thickness of Nozzle [tn]:

= thickness - corrosion allowance

= 30 - 1

= 29.000 mm

Shell Diameter to Thickness ratio [D/t]:

= Di/t

= 2112/15

= 140.800 must be less than 400.

Effective Pressure Radius [Reff]:

= Di/2 + corrosion allowance

= 2110/2 + 1

= 1056.000 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

= $\sqrt{(Reff \cdot t) + W}$

= $\sqrt{(1056 \cdot 15) + 350}$

= 475.857 mm

Limit [LR2]:

= $\sqrt{((Reff + t) \cdot (t + te))}$

= $\sqrt{((1056 + 15) \cdot (15 + 16))}$

= 182.211 mm



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Nozzle Calcs.: N4-Tube Outlet

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Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 771$$

$$= 1541.999 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(476, 182, 1542)$$

$$= 182.211 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 15, 16] + \sqrt{(771 \cdot 29)}$$

$$= 165.529 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 600$$

$$= 600.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(15 + 16)$$

$$= 248.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[166, 600, 248] + 15$$

$$= 180.529 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 15 + ((29.2 \cdot 1) / 182)$$

$$= 31.000 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff} }))$$

$$= \min(12, (2 \cdot 771 + 29) / (\sqrt{(2112 + 31) \cdot 31 }))$$

$$= 6.095$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((N/5)^{0.85}, 1)$$

$$= 15 \cdot 182 \cdot \max((6.1/5)^{0.85}, 1)$$

$$= 32.343 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 29 \cdot 181$$

$$= 52.353 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10^2$$

$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, LR \cdot t_e)$$

$$= \min(350 \cdot 16, 182 \cdot 16)$$

$$= 29.154 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43 + frp(A5)$$



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Nozzle Calcs.: N4-Tube Outlet

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$$= 32.3 + 1(52.4 + 0) + 0.5 + 0 + 0 + 1(29.2)$$

$$= 114.350 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 29 / \ln[1 + 29/771]$$

$$= 785.410 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 31 / \ln[1 + 31/1056]$$

$$= 1071.427 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 138 \cdot 1$$

$$= 207.0 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1021 \cdot 785 \cdot 181$$

$$= 14758.1 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1021 \cdot 1071 (182 + 29)$$

$$= 23554.1 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1021 \cdot 1071 \cdot 771$$

$$= 85981.3 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (23554 + 85981 + 14758) / 1021$$

$$= 11941.6 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 207 / (2 \cdot 11942 / 114 - 1071 / 31)$$

$$= 1187.5 \text{ KPa.}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$

$$= 138 [15/1071]$$

$$= 1931.9 \text{ KPa.}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(1188, 1932)$$

$$= 1187.534 \text{ KPa.}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (14758 + 23554 + 85981) / 114$$

$$= 106.596 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 1021 \cdot 1071 / 31$$

$$= 35.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$



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Nozzle Calcs.: N4-Tube Outlet

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$$= \max(2 \cdot 107 - 35.3, 35.3)$$

$$= 177.9 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	N./mm ²
Local Primary Membrane Stress	PL	177.91	N./mm ²
Maximum Allowable Working Pressure	Pmax	1187.53	KPa.

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (771 + 29) / 771$$

$$= 1.038 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (771 + 29)$$

$$= 1256.637 \text{ mm}$$

Weld Length of Pad to Shell Weld [L_TP]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (771 + 29 + 350)$$

$$= 1806.416 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 29.000, 6.000) \text{ Passed}$$

$$t_{f1}, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 16.000, 0.6 \cdot 15.000) \text{ Failed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(85981 \cdot 1.04, 1.5 \cdot 138(52.4 + 0), \pi/4 \cdot 1021 \cdot 771^2 \cdot 1.04^2)$$

$$= 52318.660 \text{ Kgf}$$

Discontinuity Force [fws]:

$$= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 52319 \cdot 15 \cdot 138 / (15 \cdot 138 + 16 \cdot 138)$$

$$= 25315.479 \text{ Kgf}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 52319 \cdot 16 \cdot 138 / (15 \cdot 138 + 16 \cdot 138)$$

$$= 27003.180 \text{ Kgf}$$

Shear Stress [τ₁]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 25315 / (1257 \cdot (0.6 \cdot 15 + 0.49 \cdot 0))$$

$$= 21.951 \text{ N./mm}^2$$

Shear Stress [τ₂]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L41T))$$

$$= 27003 / (1257 \cdot (0.6 \cdot 16 + 0.49 \cdot 7.07))$$

$$= 16.130 \text{ N./mm}^2$$

Shear Stress [τ₃]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L42T))$$

$$= 27003 / (1806 \cdot (0.49 \cdot 8.49))$$

$$= 35.259 \text{ N./mm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(22, 16.1, 35.3)$$



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Nozzle Calcs.: N4-Tube Outlet Nozl: 12 9:26am Sep 9,2026

= 35.3 must be less than or equal to 138.0 N./mm²

Weld Size Calculations, Description: N4-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000	mm
Intermediate Calc. for pad/shell Welds	TminPad	15.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness	
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo	mm
Pad Weld	7.5000 = 0.5*TminPad	8.4840 = 0.7 * Wp	mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1000 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 367.2314 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 983.2314 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.911 %

Input Echo, WRC297 Item 1, Description: N4-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2110.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	1599.999	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	2300.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1020.68	KPa.
Radial Load	P	-12237.40	Kgf
Circumferential Shear	Vc	8158.27	Kgf
Longitudinal Shear	Vl	8158.27	Kgf
Circumferential Moment	Mc	36534.93	Kg-m.
Longitudinal Moment	Ml	36534.93	Kg-m.



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Nozzle Calcs.: N4-Tube Outlet

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Torsional Moment Mt 52192.88 Kg-m.

Include Axial Pressure Thrust No

Include Pressure Stress Indices per Div. 2 No

Local Loads applied at end of Nozzle/Attachment No

*Warning - The ratio, Dn/Dv (.747) is > 0.5
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 1600.0 + 2 \cdot 1.65 \cdot \sqrt{1071.5(16.0 - 1.0)}$$

$$= 2018.365 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2143.000 mm

Nozzle Outside Diameter (d) = 1599.999 mm

Vessel Thickness used (T) = 31.000 mm

Nozzle Thickness used (t) = 29.000 mm

T / t = 1.069

d / t = 55.172

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.208$$

Nr/P = 0.027

Mr/P = 0.070

M0/P = 0.023

N0/P = 0.137

MrD/Mc = 0.191

NrDL/Mc = 0.055

M0d/Mc = 0.069

N0DL/Mc = 0.137

MrD/Ml = 0.052

NrDL/Ml = 0.020

M0D/Ml = 0.017

N0DL/Ml = 0.110

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	34	35	34	35
Outplane Stress Summary	18	34	117	36

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4



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Nozzle Calcs.: N4-Tube Outlet Nozl: 12 9:26am Sep 9,2026

Inplane Bending (ML)	-72	72	72	-72
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	-4	35	150	-99

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	25	40	150	136

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-31	-31	31	31
Outplane Bending (Mc)	-96	96	96	-96
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-77	82	179	-47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	52	-52	52	-52
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-267	267	267	-267
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	34	35	34	35
Inplane Stress Summary	-190	240	370	-268

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N4-Tube Outlet

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CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	190	240	370	268
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	33	31	33

Hoop Stress Summary	23	24	74	75
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	48	-48	48	-48
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-6	-6	6	6
Axial Bending (ML)	-66	66	66	-66
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	-7	28	138	-91
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4

Shear Stress Summary	5	5	3	3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	33	32	138	167
------------------------	----	----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-31	-31	31	31
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	33	31	33



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Nozzle Calcs.: N4-Tube Outlet Nozl: 12 9:26am Sep 9, 2026

Hoop Stress Summary	17	18	80	82
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	48	-48	48	-48
Axial Membrane (Mc)	-6	-6	6	6
Axial Bending (Mc)	-261	261	261	-261
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15

Axial Stress Summary	-203	223	333	-286
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4

Shear Stress Summary	3	3	5	5
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	220	223	333	369

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 1600.0 + 2 \cdot 1.65 \cdot \sqrt{(1063.5(16.0 - 1.0))}$$

$$= 2016.800 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2127.000 mm
Nozzle Outside Diameter	(d) =	2300.000 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	379.000 mm
T / t	=	0.040
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 12.877$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.021
MrD/Mc	=	0.237
NrDL/Mc	=	0.092
M0d/Mc	=	0.066
N0DL/Mc	=	0.039
MrD/Ml	=	0.034
NrDL/Ml	=	0.035
M0D/Ml	=	0.011



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Nozzle Calcs.: N4-Tube Outlet

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NØDL/ML = 0.016

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	59	-59	59	-59
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-46	46	46	-46
Normal Pressure Stress	71	72	71	72
Outplane Stress Summary	83	59	199	-11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	24	24	24	24
Inplane Bending (P)	228	-228	228	-228
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-23	-23	23	23
Inplane Bending (ML)	-141	141	141	-141
Inplane Pressure Stress	35	35	35	35
Inplane Stress Summary	122	-50	454	-285

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	123	110	454	285

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	10	10	10	10
Outplane Bending (P)	59	-59	59	-59
Outplane Membrane (Mc)	-26	-26	26	26
Outplane Bending (Mc)	-272	272	272	-272
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	35	35	35	35
Outplane Stress Summary	-192	232	404	-258



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Nozzle Calcs.: N4-Tube Outlet

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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	24	24	24	24
Inplane Bending (P)	228	-228	228	-228
Inplane Membrane (Mc)	-63	-63	63	63
Inplane Bending (Mc)	-984	984	984	-984
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	71	72	71	72
Inplane Stress Summary	-724	789	1373	-1052

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	724	789	1373	1052

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	34	35	34	35	34	35	34	35
Circ. Pl (SUS)	-8	-8	42	42	-9	-9	15	15
Circ. Q (SUS)	-8	8	40	-40	-215	215	319	-319
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-1	-1	7	7	-14	-14	48	48
Long. Q (SUS)	-20	20	124	-124	-80	80	112	-112
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	34.0	35.0	34.0	35.0	34.0	35.0	34.0	35.0
Pm+Pl (SUS)	26.1	27.1	76.0	77.0	25.0	26.0	65.1	65.1
Pm+Pl+Q (Total)	24.2	40.5	148.3	137.1	190.1	241.1	368.1	269.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Nozzle Calcs.: N4-Tube Outlet

Nozl: 12

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Pm (SUS)	35.00	138.00	Passed
Pm+Pl (SUS)	77.02	207.00	Passed
Pm+Pl+Q (TOTAL)	368.13	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	71	72	71	72	71	72	71	72
Circ. Pl (SUS)	-1	-1	21	21	-39	-39	87	87
Circ. Q (SUS)	13	-13	105	-105	-756	756	1212	-1212
Long. Pm (SUS)	35	35	35	35	35	35	35	35
Long. Pl (SUS)	1	1	47	47	-16	-16	36	36
Long. Q (SUS)	87	-87	369	-369	-213	213	331	-331
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	71.0	72.0	71.0	72.0	71.0	72.0	71.0	72.0
Pm+Pl (SUS)	70.0	71.0	92.1	93.1	32.1	33.1	158.0	159.0
Pm+Pl+Q (Total)	123	109	451	287	724	789	1370	1053

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	72.00	138.00	Passed
Pm+Pl (SUS)	159.01	207.00	Passed
Pm+Pl+Q (TOTAL)	1370.03	413.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	33	31	33	31	33	31	33
Circ. Pl (SUS)	-8	-8	42	42	-14	-14	48	48
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-6	-6	6	6	-6	-6	6	6
Long. Q (SUS)	-18	18	114	-114	-213	213	309	-309
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	31.0	33.0	31.0	33.0	31.0	33.0	31.0	33.0



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Nozzle Calcs.: N4-Tube Outlet

Nozl: 12

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Pm+Pl (SUS)	23.1	25.1	73.0	75.0	17.1	19.1	79.0	81.0
Pm+Pl+Q (Total)	33.5	31.1	135.1	168.1	221.1	222.0	330.1	369.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	81.02	207.00	Passed
Pm+Pl+Q (TOTAL)	369.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

Case: 1 9:26a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	2210.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2210.00 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	24.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2110.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	150.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	1.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2110.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	3704
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	15500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		15260.00 mm
Design Temperature of the Tubes		150.00 °C
Tube Material		SB-338 1
Tube Material UNS Number		R50250
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		45.30 N./mm ²
Tube Allowable Stress At Ambient		68.90 N./mm ²
Tube Yield Stress At design Temperature	Syt	74.10 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1029.310 mm



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ASME TS Calc:

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	31602.9 cm ²
Perimeter of Tube Layout	Cp	6619.8 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	150.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	120.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2385.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	2194.00 mm
Step 1 Depth on the Tubeside	ht1	9.00 mm
Step 2 Diameter on the Tubeside	dt2	2127.00 mm
Step 2 Depth on the Tubeside	ht2	7.00 mm
Is the Tubesheet Flat on Shellside?		Yes
Calculated Tubesheet Diameter per Div 2, 4.18.4		2385.00 mm

Note: Tubesheet diameter is now: 2385.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	520.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Analyze as Simply Supported

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2385.000 mm
Flange Inside Diameter	B	2110.000 mm
Flange Face Outside Diameter	Fod	2197.000 mm
Flange Face Inside Diameter	Fid	2110.000 mm
Gasket Outside Diameter	Go	2186.000 mm
Gasket Inside Diameter	Gi	2142.000 mm
Small end Hub thk.	g0	16.0000 mm
Large end Hub thk.	g1	24.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects



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ASME TS Calc:

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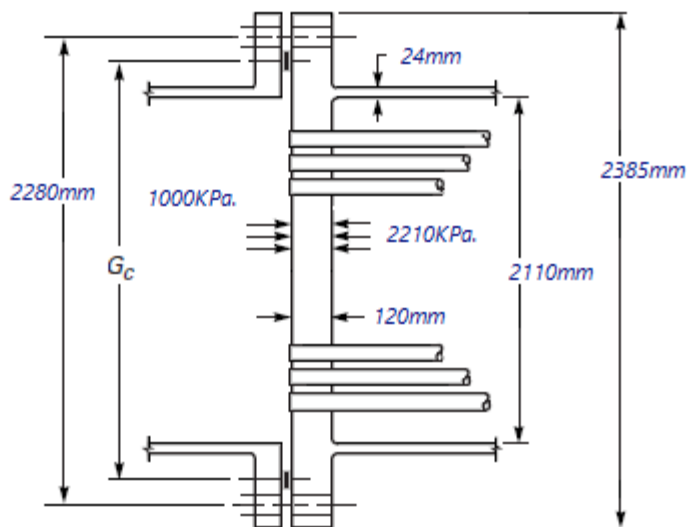
Bolting Information:

Diameter of Bolt Circle	C	2280.000 mm
Nominal Bolt Diameter	dB	41.2750 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	56
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with		Shell
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	104.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.



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ASME TS Calc:

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Tubes - TE-5 Titanium Gr 1,2,3,7,11,12,16 and 17
Tubes - TM-5 R50250

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000082793 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.10058E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.10687E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	150.0 °C	0.10058E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.10687E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(1000 \cdot (25.4/2 - 0)) / (45.3 \cdot 1 + 0.4 \cdot 1000)$
= $0.2779 + 0.0000 = 0.2779$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart NFT-3 at 150.00 °C
Elastic Modulus for Material 98975232.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	15260.00	12.04	50.0000	0.0075908	38.74

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 38.7) / (3 \cdot 12) = 4290$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.1977	25.40	15260.00	21.21	50.0000	0.0024457	35.16

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 35.2) / (3 \cdot 21.2) = 2210$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.1977 mm
Allowable Internal Pressure at Corroded thickness	8061.85 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	4290.28 KPa.
Required External Design Pressure	2210.00 KPa.
Required Thickness due to Shell Side pressure	1.1977 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	22.000 mm
Basic Gasket Width,	$b0 = N / 2$	11.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	8.355 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2169.291 mm

Note:

The Tubesheet stress reduction option was Simply Supported. Performing



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ASME TS Calc:

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the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

= l_{tx} / h (modified for corrosion if present)

= 0/120

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(25.4 - 2 \cdot 2.11 (100579424/195062688) \cdot$

$(45.3/138) \cdot (0); 25.4 - 2 \cdot 2.11)$

= 25.4000 mm

Equivalent outer tube limit circle diameter $[Do]$:

= $2 \cdot r_o + dt$

= $2 \cdot 1029 + 25.4$

= 2084.020 mm

Determine the basic ligament efficiency for shear $[\mu]$:

= $(p - dt) / p$

= $(31.8 - 25.4) / 31.8$

= 0.200

Equivalent outer tube limit radius $[ao]$:

= $Do / 2$

= 2084/2

= 1042.0100 mm

Effective tube pitch $[p^*]$:

= $p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$

= $31.8 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2084 \cdot 31.8) / (\pi \cdot 2084^2)) }$

= 31.7500 mm

Effective ligament efficiency for bending $[\mu^*]$:

= $(p^* - d^*) / p^*$

= $(31.8 - 25.4) / 31.8$

= 0.2000

Ratio $[ps]$:

= as / ao

= 1055/1042

= 1.0125

Ratio $[pc]$:

= ac / ao

= 1085/1042

= 1.0409

Compute parameter $[xt]$:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$

= $1 - 3704 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1042))^2$

= 0.6174

Determine parameter $[xs]$:

= $1 - Nt(dt / (2 \cdot ao))^2$

= $1 - 3704 (25.4 / (2 \cdot 1042))^2$

= 0.4498



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ASME TS Calc:

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Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts)E_s/L$$

$$= \pi \cdot 24 (2110 + 24)195062688/15260$$

$$= 209735.4219 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(D_t - tt)E_t/L$$

$$= \pi \cdot 2.11 (25.4 - 2.11)100579424/15260$$

$$= 103.7655 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 209735 / (3704 \cdot 104)$$

$$= 0.54569$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_j)$$

$$= 1 / (1 + 209735/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2110 + 24) 24)$$

$$= 0.0080 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.008 \cdot 195062688 \cdot 24^3 / (6(1 - 0.3^2))$$

$$= 3967053312.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(2110 + 24) \cdot 3.97E+09/120^3 \cdot (1 + 120 \cdot 0.008 + (120 \cdot 0.008)^2/2)$$

$$= 71384024.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot \nu_s / (2(D_s + ts)))$$

$$= 2110^2 / (4 \cdot 195062688 \cdot 24) \cdot (1 - 2110 \cdot 0.3 / (2(2110 + 24)))$$

$$= 0.20248E+00 \text{ mm/N} \cdot \text{mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 3.779528 ; μ* = 0.200000 ; E* = 30302818 KPa.

E*/E = 0.155349 ; ν* = 0.458572

Note: As h/p (3.780) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^{*2})N_t \cdot E_t \cdot tt(d_t - tt)a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.46^2) 3704 \cdot 100579424 \cdot 2.11$$

$$(25.4 - 2.11) 1042^2) / (30302818 \cdot 15260 \cdot 120^3))^{1/4}$$

$$= 4.6598$$

Values from Table 4.18.3



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ASME TS Calc:

Case: 1

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$$\begin{aligned} Z_d &= 0.014860 ; Z_v = 0.045100 ; Z_m = 0.303012 \\ Z_a &= 0.157425E+02 ; Z_w = 0.045100 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 2385/2084 \\ &= 1.1444 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 30302818 \cdot (71384024 + 0 + 195062688 \cdot \ln(1.14)) \\ &= 1.7456 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*) F \\ &= (1 + 0.46) 1.75 \\ &= 2.5461 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.01 - 1 - 2.55 \cdot 0.045) / (1 + 2.55 \cdot 0.3) \\ &= -0.057783 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.015 + -0.058 \cdot 0.045) / 2 \cdot 4.66^4 \\ &= 2.8889 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.045 + -0.058 \cdot 0.3) / 2 \cdot 4.66^4 \\ &= 6.5045 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.045 + (1.01 - 1) 0.3) 4.66^4 / (1 + 2.55 \cdot 0.3) \\ &= 13.0090 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / D_o \text{ (config b)} \\ &= (2169 - 2280) / 2084 = -0.053 \end{aligned}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.01 \cdot 3.97E+09 \cdot 0.008 \cdot 0.20247635 (1 + 120 \cdot 0.008) \\ &= 12829.8418 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ &= 1042^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 12830 \\ &= -12744.9453 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)] \\ &= 1.04 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 120) - 0.000 (0/120 + 0 \cdot 0/2)] \end{aligned}$$



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ASME TS Calc:

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$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$\begin{aligned} &= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1)/4 - (\rho_s - 1)/2] - \omega c \\ &= 1042^2 [(1.04^2 + 1) \cdot (1.04 - 1)/4 - (1.01 - 1)/2] - 0 \\ &= 16372.8418 \text{ mm}^2 \end{aligned}$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$\begin{aligned} &= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - \\ &\quad [(\rho_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \} \\ &= 2210 \{ 0.45 + 2(1 - 0.45) 0.32 + [2/0.55 \cdot \\ &\quad (2110/2084)^2] 0.3 - [(1.01^2 - 1)/(1 \cdot 0.55)] - \\ &\quad [(1 - 1)/(2 \cdot 1 \cdot 0.55)] [(0^2 - (2110)^2)/2084^2] \} \\ &= 4161.5459 \text{ KPa.} \end{aligned}$$

Pressure [$P't$]:

$$\begin{aligned} &= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t \\ &= [0.62 + 2(1 - 0.62) 0.32 + 1/(1 \cdot 0.55)] 1000 \\ &= 2694.8066 \text{ KPa.} \end{aligned}$$

Pressure [P_y]:

$$\begin{aligned} &= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2) \\ &= 3704 \cdot 104 \cdot 0 / (3.14 \cdot 1042^2) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned} &= -\gamma_b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2) \\ &= -0.053 \cdot 13 \cdot 411733 / (2 \cdot \pi \cdot 1042^2) \\ &= 408.9989 \text{ KPa.} \end{aligned}$$

Calculate the pressure [P_{rim}]:

$$\begin{aligned} &= -(U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) \\ &= -(13/1042^2) \cdot (-12745 \cdot 2210 - 16373 \cdot 1000) \\ &= 533.6335 \text{ KPa.} \end{aligned}$$

Calculate the pressure [P_ω]:

$$\begin{aligned} &= (U/a_o^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c) \\ &= (13/1042^2) \cdot (12830 \cdot 0 - 0 \cdot 0) \\ &= 0.0000 \text{ KPa.} \end{aligned}$$

Determine the effective pressure [P_e]:

$$\begin{aligned} &= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho_s - 1) Qz2)) \cdot \\ &\quad (P's - P't + P_y + P_\omega + P_w + P_{rim}) \\ &= 0.1000E+01 \cdot 0.55 / (1 + 1 \cdot 0.55 (2.89 + (1.01 - 1) 6.5)) \cdot (4162 - 2695 + 0 + 0 + 409 + 534) \\ &= 501.6847 \text{ KPa.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$\begin{aligned} &= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + \\ &\quad W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\ &= [((-12745 \cdot 2210 - 16373 \cdot 1000) - \\ &\quad (12830 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + \\ &\quad 411733 \cdot -0.053 / (2 \cdot \pi)] / (1 + 2.55 \cdot 0.3) \\ &= -4528.9609 \text{ Kg} \end{aligned}$$



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Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\
 &= -0.058 + 2 \cdot -4529 / (502 \cdot 1042^2) \\
 &= -0.22085
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.110423

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.11 / 0.2) \cdot (2 \cdot 1042 / (120 - 0))^2 \cdot 502 \\
 &= 125.3197 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot Fm / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.092 / 0.2) \cdot (2 \cdot 1042 / (85.6 - 0))^2 \cdot 506 \\
 &= 206.9776 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 85.6 + 0 + 0 \\
 &= 85.5548 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.2 \cdot 120 / 1042) \cdot 110 \\
 &= 5.0855 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned}
 \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\
 \text{Tubesheet Shear Stress} &= 66.1119 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(85.6; 3.81) \\
 &= 85.5548 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.092, F_{tmax} = 1.47$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs) \\
 &= ((2210 \cdot 0.45 - 1000 \cdot 0.62) - (502) \cdot -0.092) / \\
 &\quad (0.62 - 0.45) \\
 &= 2.5214 \text{ N./mm}^2
 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmax}) / (Xt - Xs) \\
 &= ((2210 \cdot 0.45 - 1000 \cdot 0.62) - (502) \cdot 1.47) / \\
 &\quad (0.62 - 0.45) \\
 &= -2.1575 \text{ N./mm}^2
 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.52 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:



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ASME TS Calc:

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$$= Sot = 45.3018 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q3 \cdot Z_w)Xa^4); 1.25)$$

$$= \max((3.25 - 0.25(0.015 + -0.22 \cdot 0.045)4.66^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 100579424 / 74.1)}$$

$$= 163.6864$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 520 / 8.27$$

$$= 62.8932$$

Buckling Allowable Stress [Stb]:

$$= (Syt / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (74.1 / 2) \cdot (1 - 62.9 / (2 \cdot 164))$$

$$= 29.934 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.52 \cdot 1.54$$

$$= 39.69 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 45.3$$

$$= 713.163 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 45.3$$

$$= 697.9028 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 45.3 / 45.3$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 45.3$$

$$= 451.5842 \text{ Kgf}$$



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ASME TS Calc:

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Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 713 is $\geq$ tube strength 713 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1042^2 / ((2110 + 24) 24) \cdot [502 + (1.01^2 - 1)(2210 - 1000)] + 1055^2 \cdot 1000 / ((2110 + 24) 24)$$

$$= 33.0132 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1042^2 / ((2110 + 24) 24) [502 + (1.01^2 - 1)(2210 - 1000)] + 1055^2 \cdot 1000 / ((2110 + 24) 24)$$

$$= 33.0132 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 3.97E+09 / 24^2 \{ 0.008 [0.20247635 \cdot 2210 + 1055^2 \cdot 0 / (4.68E+09)] + 6(1 - 0.46^2) / (30302818) (1042 / 120)^3 (1 + 120 \cdot 0.008 / 2) [502 (0.045 + 0.3 \cdot -0.058) + 2 / 1042^2 \cdot 0.3 \cdot -4529] \}$$

$$= 79.9148 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore | \sigma_{s, m} | + | \sigma_{s, b} | \leq SPS_s$$

$$\therefore | 33 | + | 79.9 | \leq 464 \text{ N./mm}^2$$

$$\therefore 113, \text{ must be } \leq 464 \text{ N./mm}^2$$

Note: The Cylinder-to-Tubesheet Juncture stress is $<$ SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Intermediate Shell side parameters for the Simply Supported case:

betas, k_s , λ bdas, δ tas are all = 0



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ASME TS Calc:

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Intermediate Channel side parameters for the Simply Supported case:
betaC, kc, lambdaC, deltaC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.46) / 30302818 \cdot (0 + 0 + 195062688 \cdot \ln(1.14))$$

$$= 0.4702$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.46) 0.47$$

$$= 0.6858$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.01 - 1 - 0.69 \cdot 0.045) / (1 + 0.69 \cdot 0.3)$$

$$= -0.015285$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.015 + -0.015 \cdot 0.045) / 2 \cdot 4.66^4$$

$$= 3.3408$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.045 + -0.015 \cdot 0.3) / 2 \cdot 4.66^4$$

$$= 9.5403$$

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.045 + (1.01 - 1) 0.3) 4.66^4 / (1 + 0.69 \cdot 0.3)$$

$$= 19.0806$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / D_o \text{ (config b)}$$

$$= (2169 - 2280) / 2084 = -0.053$$

Calculate parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 0 \cdot 0 \cdot 0.000 (1 + 120 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 1042^2 (1.01^2 - 1) \cdot (1.01 - 1) / 4 - 0$$

$$= 84.8964 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.04 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 120) - 0.000 (0/120 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 1042^2 [(1.04^2 + 1) \cdot (1.04 - 1) / 4 - (1.01 - 1) / 2] - 0$$

$$= 16372.8418 \text{ mm}^2$$



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ASME TS Calc:

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Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega * s \cdot Ps - \omega * c \cdot Pt)$$

$$= -(19.1/1042^2) \cdot (84.9 \cdot 2210 - 16373 \cdot 1000)$$

$$= 284.4239 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega s \cdot P*s - \omega c \cdot P*c)$$

$$= (19.1/1042^2) \cdot (0 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + Py + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.55 / (1 + 1 \cdot 0.55(3.34 + (1.01 - 1)9.54)) \cdot (4162 - 2695 + 0 + 0 + 600 + 284)$$

$$= 444.2440 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s \cdot Ps - \omega * c \cdot Pt) - (\omega s \cdot P*s - \omega c \cdot P*c)) + W * \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((84.9 \cdot 2210 - 16373 \cdot 1000) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 411733 \cdot -0.053 / (2 \cdot \pi)] / (1 + 0.69 \cdot 0.3)$$

$$= -4248.7446 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.015 + 2 \cdot -4249 / (444 \cdot 1042^2)$$

$$= -0.18804$$

Fm Value from Table 4.18.3 = 0.094020

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.094 / 0.2) \cdot (2 \cdot 1042 / (120 - 0))^2 \cdot 444$$

$$= 94.4863 \text{ N./mm}^2$$

Allowable Tube sheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tube sheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu *) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.085 / 0.2) \cdot (2 \cdot 1042 / (80.5 - 0))^2 \cdot 483$$

$$= 206.9959 \text{ N./mm}^2$$

Reqd Tube sheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 80.5 + 0 + 0$$

$$= 80.4543 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy)$$

$$= (2 \cdot 0.2 \cdot 120 / 1042) \cdot 110$$

$$= 5.0855 \text{ N./mm}^2$$



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ASME TS Calc:

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Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [tf]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (31603 \cdot 970) / (0.2 \cdot 21.9 \cdot 6620)$$

$$= 105.8674 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 21.9 + 0 + 0$$

$$= 21.8634 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(80.5; 21.9)$$

$$= 80.4543 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -0.18, F_{tmax} = 1.61$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((2210 \cdot 0.45 - 1000 \cdot 0.62) - (444) \cdot -0.18) / (0.62 - 0.45)$$

$$= 2.7214 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((2210 \cdot 0.45 - 1000 \cdot 0.62) - (444) \cdot 1.61) / (0.62 - 0.45)$$

$$= -2.0105 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 2.72 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 45.3018 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.015 + -0.19 \cdot 0.045)4.66^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 100579424 / 74.1)}$$

$$= 163.6864$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 520 / 8.27$$

$$= 62.8932$$

Buckling Allowable Stress [Stb]:

$$= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$



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ASME TS Calc:

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$$= (74.1/2) \cdot (1 - 62.9/(2 \cdot 164))$$

$$= 29.934 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 2.72 \cdot 1.54$$

$$= 42.84 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 45.3$$

$$= 713.163 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 45.3$$

$$= 697.9028 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 45.3/45.3$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 45.3$$

$$= 451.5842 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 713 is $\geq$ tube strength 713 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [$\sigma_{s,m}$]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 1042^2 / ((2110 + 24) 24) \cdot [444 + (1.01^2 - 1)(2210 - 1000)] + 1055^2 \cdot 1000 / ((2110 + 24) 24)$$

$$= 31.7954 \text{ N./mm}^2$$



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ASME Section VIII-2 Part 4.18.8.4 Step 11:

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:

$$= \frac{a_o^2}{(D_s + t_s) t_s} [P_e + (p_s^2 - 1)(P_s - P_t)] + \frac{a_s^2 P_t}{(D_s + t_s) t_s}$$

$$= \frac{1042^2}{(2110 + 24) 24} [444 + (1.01^2 - 1)(2210 - 1000)] + \frac{1055^2 \cdot 1000}{((2110 + 24) 24)}$$

$$= 31.7954 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:

$$= 6 \cdot \frac{k_s}{t_s^2} \left\{ \beta_s [\delta_s P_s + \frac{a_s^2 P_s}{E_s t_s}] + \frac{6(1 - \nu^2)}{(E_s)(a_o/h)^3 (1 + h \beta_s/2)} [P_e (Z_v + Z_m Q_1) + 2 a_o^2 Z_m Q_2] \right\}$$

$$= 6 \cdot \frac{0}{24^2} \left\{ 0 [0.000 \cdot 2210 + 1055^2 \cdot 0 / (4.68E+09)] + \frac{6(1 - 0.46^2)}{(30302818)(1042/120)^3 (1 + 120 \cdot 0/2)} [444(0.045 + 0.3 \cdot -0.015) + 2/1042^2 \cdot 0.3 \cdot -4249] \right\}$$

$$= -0.0000 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_s, m| + |\sigma_s, b| \leq \text{SPS}, s$$

$$\therefore |31.8| + |-0| \leq 464 \text{ N./mm}^2$$

$$\therefore 31.8, \text{ must be } \leq 464 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	94.5 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	8.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	2.7 <=	45.3 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-2.0 <=	-29.9 N./mm ²	Pass
Maximum Force on any one Tube	42.8 <=	713.2 Kgf	Pass
Axial Membrane Stress in Shell	31.8 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	31.8 <=	464.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	80.454	120.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Tbsht	Thk. + CA Extnsn	---- Tubesheet Stresses	Case Type	Pass/Fail
			Bend Allwd Shear Allwd		
D1uc	81.231	50.430	106 207 7 110	Ps+Pt-Th	D1 Ok
D2uc	39.465	...	86 207 16 110	Ps+Pt-Th	D2 Ok
D3uc	80.454	...	94 207 9 110	Ps+Pt-Th	D3 Ok
O1uc	21.246	...	89 438 3 110	Ps+Pt+Th	O1 Ok
O2uc	68.278	...	191 438 26 110	Ps+Pt+Th	O2 Ok
O3uc	63.884	...	176 438 16 110	Ps+Pt+Th	O3 Ok
O4uc	40.405	...	96 438 7 110	Ps+Pt+Th	O4 Ok
D1c	96.849	50.430	134 207 7 110	Ps+Pt-Th-c	D1 Ok
D2c	58.539	...	100 207 17 110	Ps+Pt-Th-c	D2 Ok
D3c	94.506	...	119 207 10 110	Ps+Pt-Th-c	D3 Ok
O1c	54.538	...	121 438 3 110	Ps+Pt+Th-c	O1 Ok



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ASME TS Calc:

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02c	83.421	...	226	438	29	110	Ps+Pt+Th-c 02	Ok
03c	80.914	...	215	438	17	110	Ps+Pt+Th-c 03	Ok
04c	60.779	...	125	438	8	110	Ps+Pt+Th-c 04	Ok

Max: 96.8487 50.430 mm 0.650 0.261 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass : Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	14	137	...	...	...	...	...	...	...	Ok
D2uc	18	137	...	...	...	...	...	...	...	Ok
D3uc	32	137	...	...	...	...	...	...	...	Ok
O1uc	18	464	...	...	...	...	...	...	...	Ok
O2uc	29	464	...	...	...	...	...	...	...	Ok
O3uc	39	464	...	...	...	...	...	...	...	Ok
O4uc	8	464	...	...	...	...	...	...	...	Ok
D1c	17	137	...	...	...	...	...	...	...	Ok
D2c	20	137	...	...	...	...	...	...	...	Ok
D3c	36	137	...	...	...	...	...	...	...	Ok
O1c	21	464	...	...	...	...	...	...	...	Ok
O2c	32	464	...	...	...	...	...	...	...	Ok
O3c	44	464	...	...	...	...	...	...	...	Ok
O4c	9	464	...	...	...	...	...	...	...	Ok
Max RATIO	0.264	...	...	...	...	...	...	...	...	...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	8	45	-6	-45	126	713	14	464	...	...	Ok
D2uc	10	45	-10	-37	153	713	18	464	...	...	Ok
D3uc	3	45	-2	-29	43	713	32	464	...	...	Ok
O1uc	6	90	-6	-45	90	1426	197	464	...	...	Ok
O2uc	11	90	-11	-29	168	1426	227	464	...	...	Ok
O3uc	5	90	-5	-29	80	1426	115	464	...	...	Ok
O4uc	3	90	-3	-29	50	1426	65	464	...	...	Ok
D1c	9	45	-6	-45	144	713	17	464	...	...	Ok
D2c	11	45	-11	-44	169	713	20	464	...	...	Ok



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ASME TS Calc:

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D3c	3	45	-2	-29	46	713	36	464	...	...	Ok
01c	6	90	-6	-45	92	1426	243	464	...	...	Ok
02c	11	90	-11	-29	177	1426	246	464	...	...	Ok
03c	5	90	-5	-29	82	1426	116	464	...	...	Ok
04c	3	90	-3	-29	53	1426	90	464	...	...	Ok

Max RATIO 0.238 0.377 0.238 0.529 ...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	96.849	120.000	Ok
Tubesheet thickness flanged extension :	50.430	104.000	Ok
Tube thickness :	1.198	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (24.000) adj. to tubesheet: 405.060 mm

For simply supported geometries that meet stress and thickness requirements, the above minimum length calculations are for information only and are not an ASME requirement. See VIII-2 sections 4.18.7.5, 4.18.8.8 and 4.18.9.7 for more information.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.946 mm
 Due to Pressure Shell Compresses by : -0.495 mm
 Due to Pressure + Thermal Shell Extends by : +1.041 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	1539.21	1.000	4560.10	1.000
Tubesheet Shear Stress	11483.60	1.000	11072.25	1.000
Tube Tensile Stress	4964.92	1.000	9301.55	1.000
Tube Compressive Stress	5296.85	0.634	6913.63	1.000
Tube-Tubesheet Joint load	4964.92	1.000	9301.55	1.000
Shell Stress (Axial, Junction)	2077.96	1.000	3230.59	1.000
Tube Pressure Stress	8061.84	1.000	4290.27	1.000
Tubesheet Extension Stress	2585.47	...	No Calc	No Calc
Minimum MAWP	1539.21		3230.59	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	1950.88	1.000	5363.73	1.000
Tubesheet Shear Stress	16175.16	1.000	15592.13	1.000
Tube Tensile Stress	8463.92	1.000	15447.25	1.000
Tube Compressive Stress	10896.72	1.000	14714.86	1.000
Tube-Tubesheet Joint load	8463.92	1.000	15447.24	1.000
Shell Stress (Axial, Junction)	2785.45	1.000	4179.58	1.000
Tube Pressure Stress	12261.12	1.000	6902.52	1.000
Tubesheet Extension Stress	2585.47	...	No Calc	No Calc
Minimum MAPnc	1950.88		4179.58	

(*) All load cases were analyzed to compute the MAWP for determining the test pressure.



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ASME TS Calc:

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Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0950	0.0951	...	...	...	...
2	-0.4951	-0.4952	...	...	...	...
3	-0.4000	-0.4002	...	...	...	...
4	...	...	...	...	...	...
5	0.0950	0.0951	0.9460	0.9463	...	...
6	-0.4951	-0.4952	0.9460	0.9463	...	...
7	-0.4000	-0.4002	0.9460	0.9463	...	...
8	...	...	0.9460	0.9463	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, shell thickness))
 = max(120/4, min(120, 24))
 = 30.000 mm

Thickness Ratio = 0.53, Temperature Reduction per Fig. UCS 66.1 = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -25 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:
 = tubesheet thickness/4
 = 120/4
 = 30.000 mm

Thickness Ratio = 0.53, Temperature Reduction per Fig. UCS 66.1 = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -13 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(1000/1951, 2210/4180)
 = 0.529

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Problems/Failures Summary:

Step: 33

9:26am

Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Shell/Head in this vessel!**

**** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: